

DATA CONVERTER

Lecture 3

Building blocks

Li Fule

Summer, 2010

Equations to remember

$$I_{DS} = K' \frac{W}{L} (V_{GS} - V_T)^2 (1 + \lambda V_{DS})$$

**Strong
inversion**

$$r_{DS} = \frac{1}{\lambda I_{DS}} = \frac{V_E L}{I_{DS}}$$

$$g_m = \frac{2I_{DS}}{V_{GS} - V_T}$$

$$A_V = \frac{2V_E L}{V_{GS} - V_T}$$

$$f_T = \frac{3\mu(V_{GS} - V_T)}{4n\pi L^2}$$

VE为工艺参数

$$\overline{dV_{ieq}^2} = 4kT \left(\frac{2/3}{g_m} + R' \right) df$$

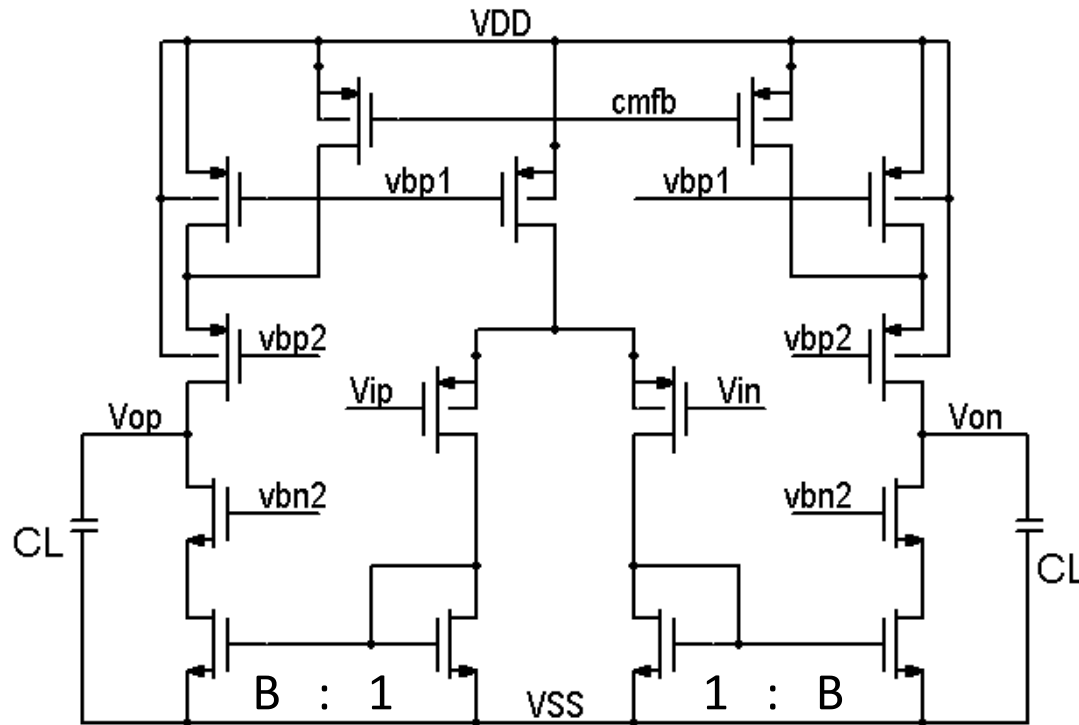
$$R' = R_G + R_S + R_B (n-1)^2$$

$$\overline{dV_{ieqf}^2} = \frac{KF_F}{WLC_{ox}^2} \frac{df}{f}$$

跨导运放

- Opamp-based converter中最重要的单元
- 主要指标
 - DC GAIN, GBW, SR, PM, GM, FOM, Output Swing
- 转换器中常用的运放结构
 - Current-Mirror based single stage
 - Folded-Cascode single Stage
 - Telescopic single stage
 - Two stage
- Gain boosting

Symmetrical OTA



考虑到PMOS比流过相同电流的NMOS要大得多，故 $f_{nd} < f_{tn}/(B+3)$

对称性好
输入共模范围比较大

$$V_{cmi} \in (V_{GSN} - V_{TP}, V_{DD} - V_{ov} - V_{GSP})$$

当B增大时，电流效率增加，
但噪声性能差

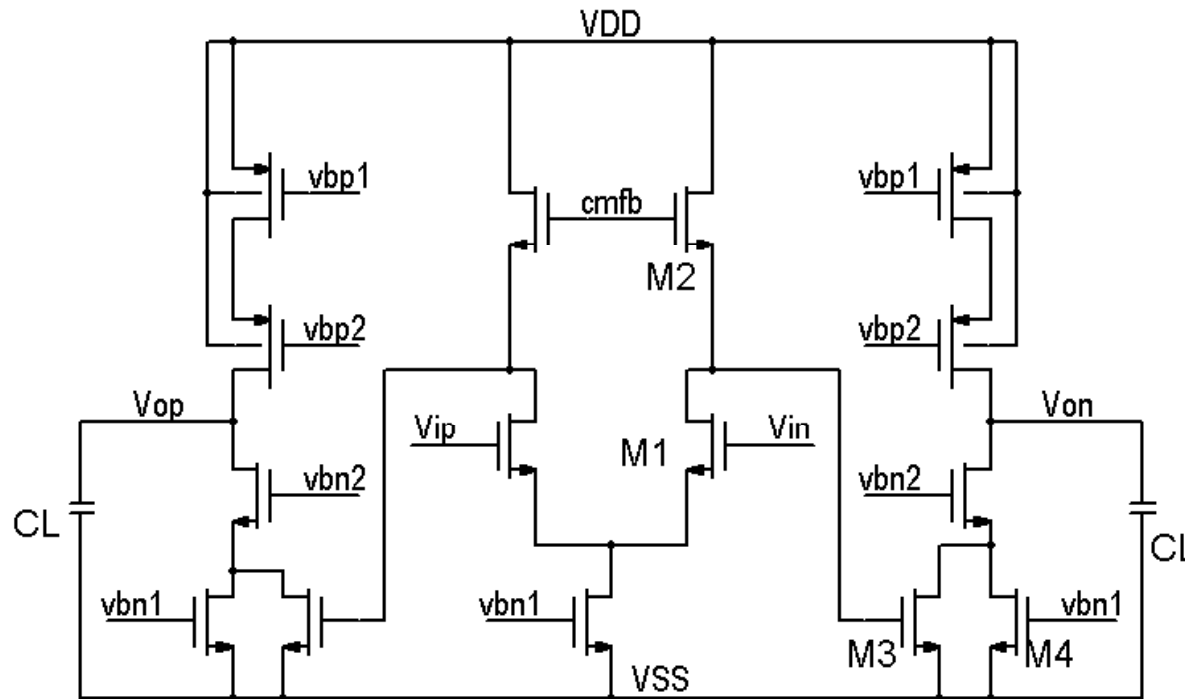
$$GBW = \frac{Bg_{mi}}{2\pi C_L}$$

fnd = ftn/(B+3) ???

Output Swing:

$$V_{OS} < V_{DD} - 4V_{ov}$$

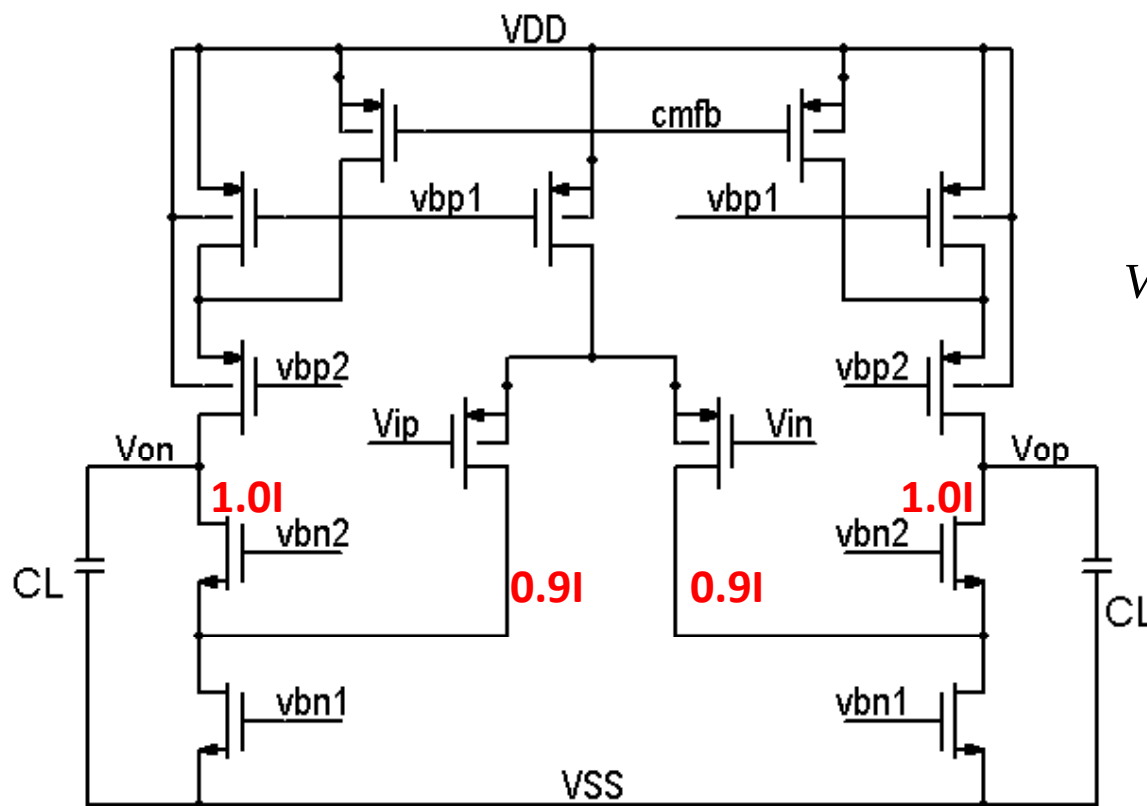
Symmetrical OTA



M4比较小，它主要用于避免M2完全cutoff，使得共模反馈环路失效

- 第一级：都是NMOS，速度快，也可提供一定的增益
 - 第二级：负载驱动、主要增益
 - 共模反馈：无须倒相
 - 第一级输出低阻，无须miller反馈
 - 相当于Symmetrical OTA，还是单级结构
 - 输入共模范围： $V_{cmi} \in (V_{GSN} + V_{ov}, V_{DD} - 2V_{ov})$
 - Output Swing:
- $$V_{OS} < V_{DD} - 4V_{ov}$$

Folded-Cascode OTA



在实际设计中，为避免cascode NMOS管脱离正常的饱和区工作状态，输入差分对的偏置电流应稍小于其他两个支路

对称性好

输入共模范围最大

$$V_{cmi} \in (V_{ov} - V_{TP}, V_{DD} - V_{ov} - V_{GSP})$$

电流效率比telescopic结构要低，只有一半的电流用于产生gm

$$GBW = \frac{g_{mi}}{2\pi C_L}$$

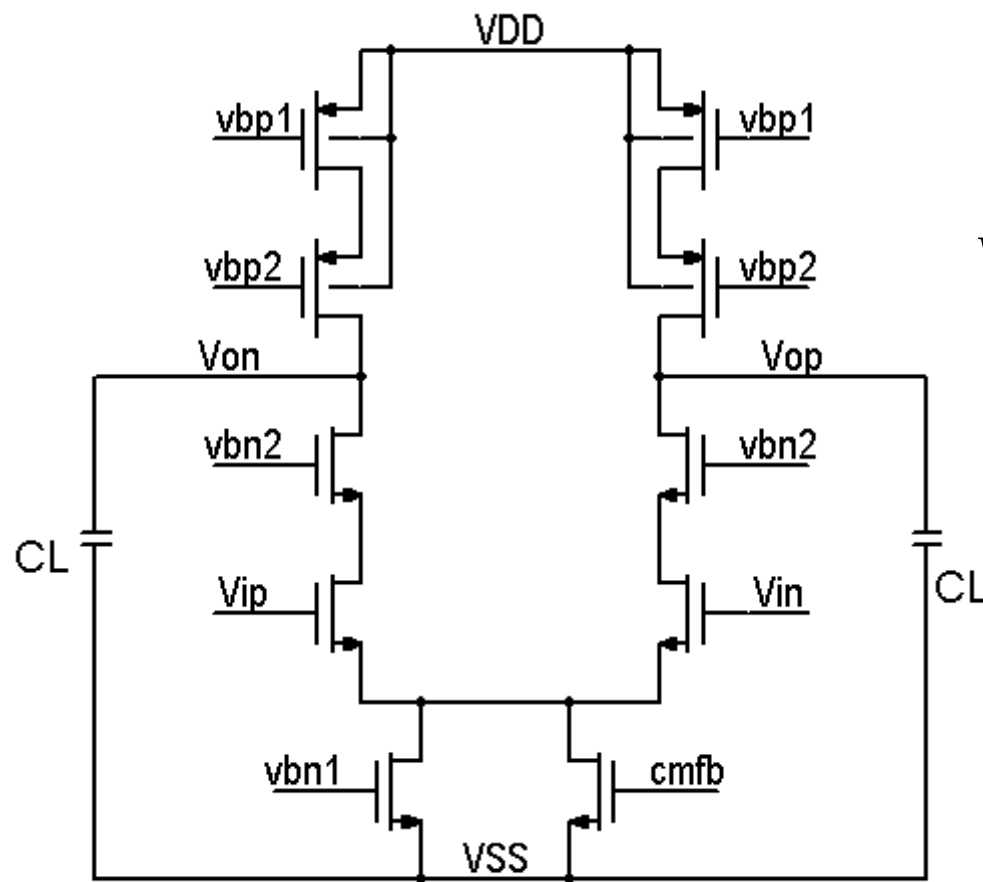
速度快：

$$f_{nd} = f_{tn}/3?$$

Output Swing比较大

$$V_{OS} < V_{DD} - 4V_{ov}$$

Telescopic OTA



在2.5V, 3V, 或更高的电源下，是一种优先选择的结构
在1.8V电源或以下，output swing 严重受限

对称性好

输入共模范围最小！

$$V_{cmi} \in (V_{ov} + V_{GSN}, V_{DD} - 3V_{ov} - V_{OS} + V_{TN})$$

电流效率最高，是folded结构的2倍，所有的电流用于产生gm

$$GBW = \frac{g_{mi}}{2\pi C_L}$$

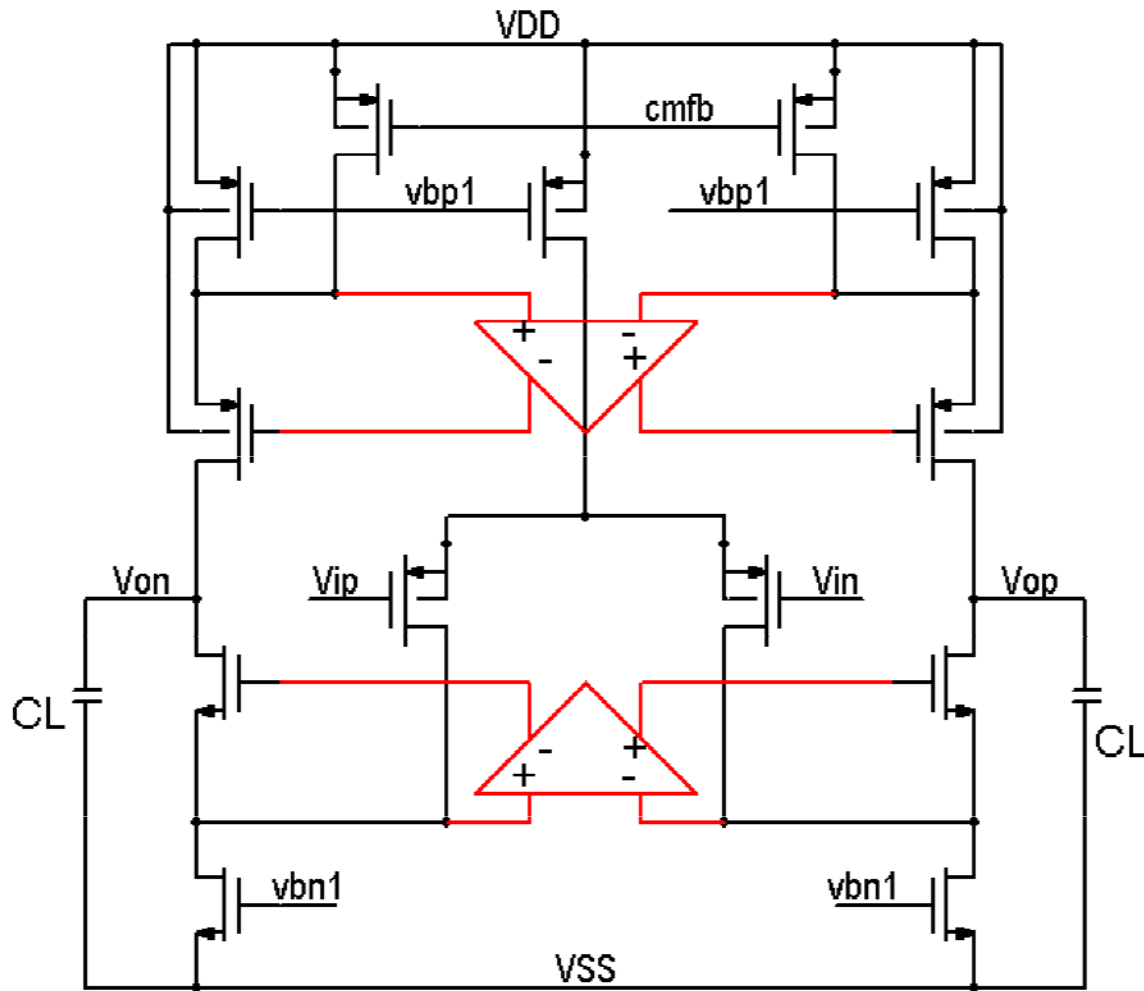
速度最快：

$$f_{nd} = f_{tn}/2$$

Output Swing受限

$$V_{OS} < V_{DD} - 5V_{ov}$$

Gain-boosting for higher gain



为了实现更高的增益，同时
不损害output swing，可采用
gain-boosting

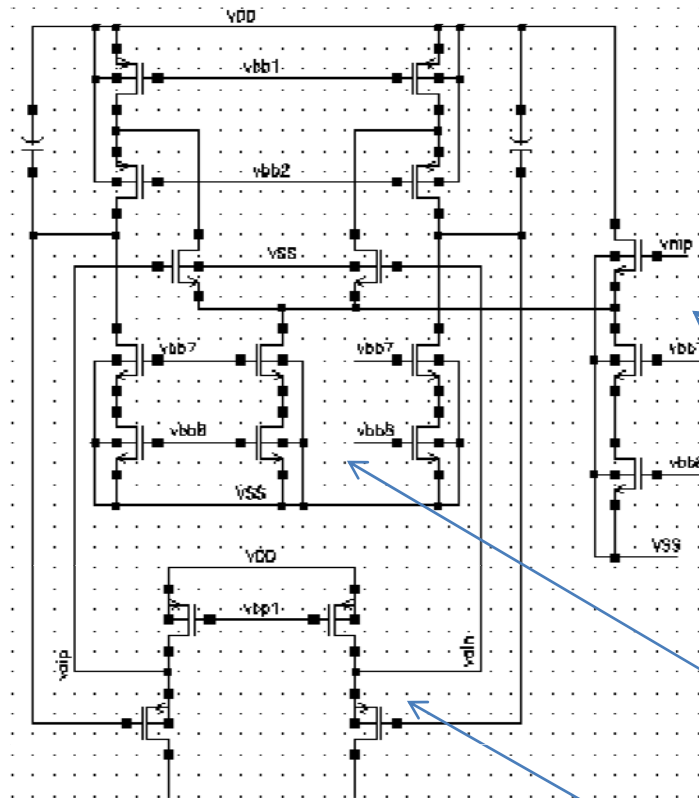
$$A = A_{main} \cdot A_{gb}$$

辅助的gain-boosting运放的
尺度通常是主运放1/10左右，
功耗增加的比例不大

但gain-boosting运放会引入
doublet, 设计中要注意它的位
置

通常加入gain-boosting后瞬态建
立相应更难控制，对高速设计
不利

Gain boosting amplifier



- Gain boosting会引入doublet
- 为不影响带内特性, 希望doublet在闭环带外
- 为了不影响稳定性, 希望doublet小于 f_{nd}

$$f \cdot GBW_{main} < GBW_{gb} < f_{nd}$$

Common-voltage control

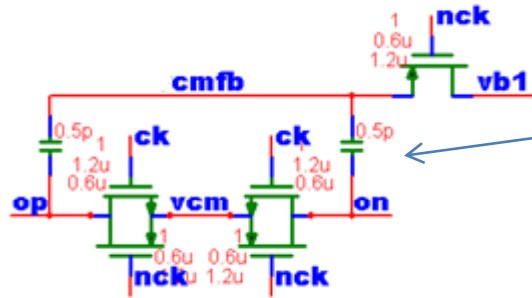
Gain-boosting OPA for PMOS SIDE

MAIN STAGE PMOS SIDE

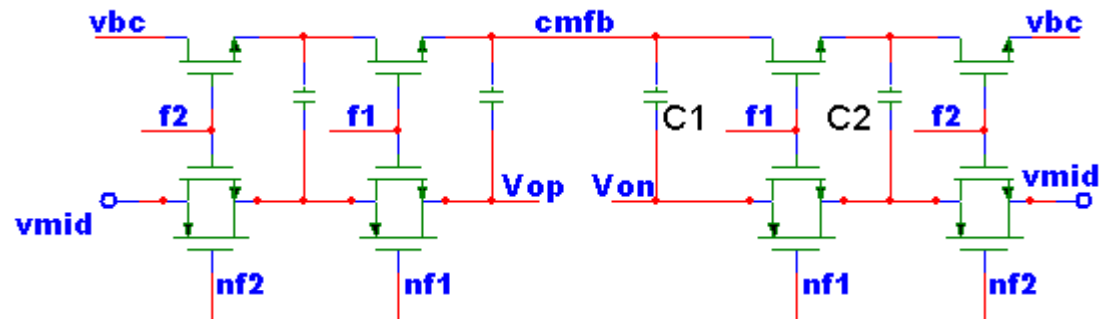
NMOS端有对偶的设计

Common Feedback Circuit

开关电容共模反馈：不限制差模output swing，广泛应用于开关电容式模数转换器电路中



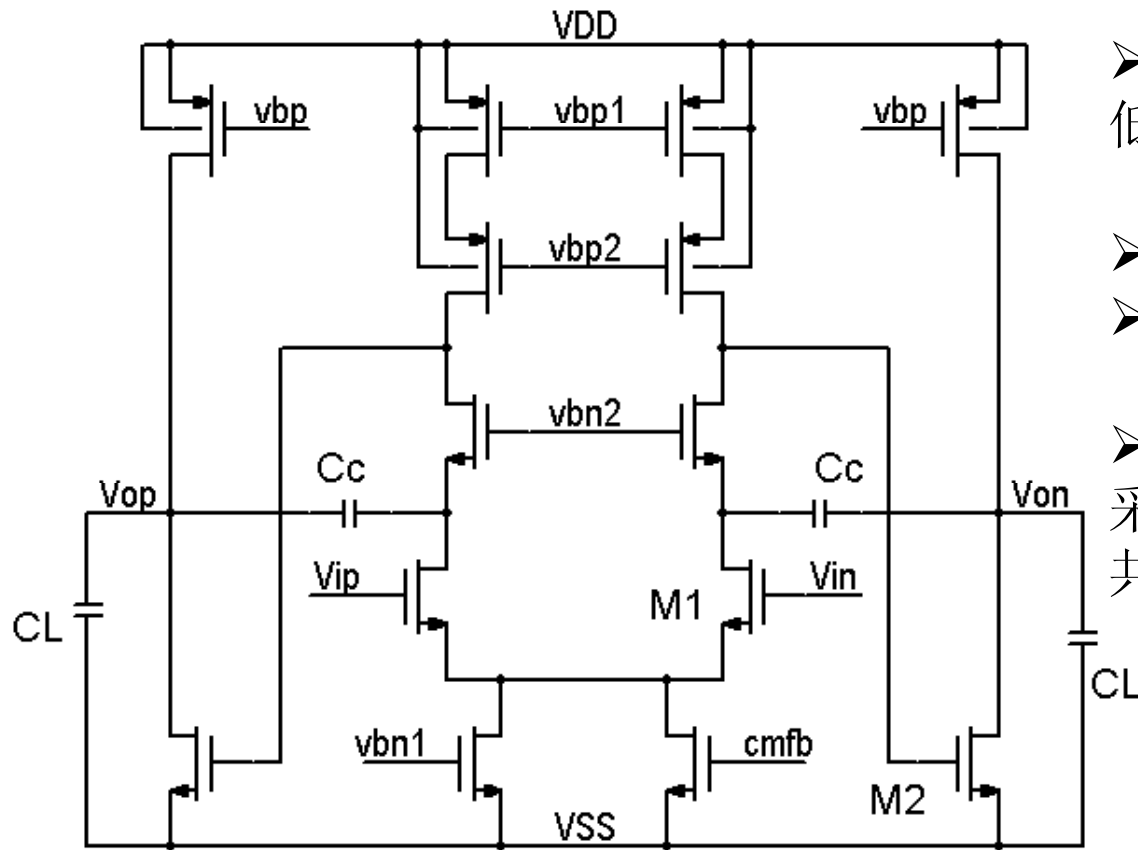
单相时钟，电路简单，但运放必须复位，不能连续工作



需要双相不交叠时钟，四个电容，但运放可连续工作

C1, C2的取值？

Two Stage OTA



共模反馈需要倒相，这增加了功耗，增加了共模回路的复杂性

➤增益大，Output Swing大，适合低电源电压

$$A = A_1 \cdot A_2$$

➤功耗大，速度不如单级快

➤噪声主要取决于第一级

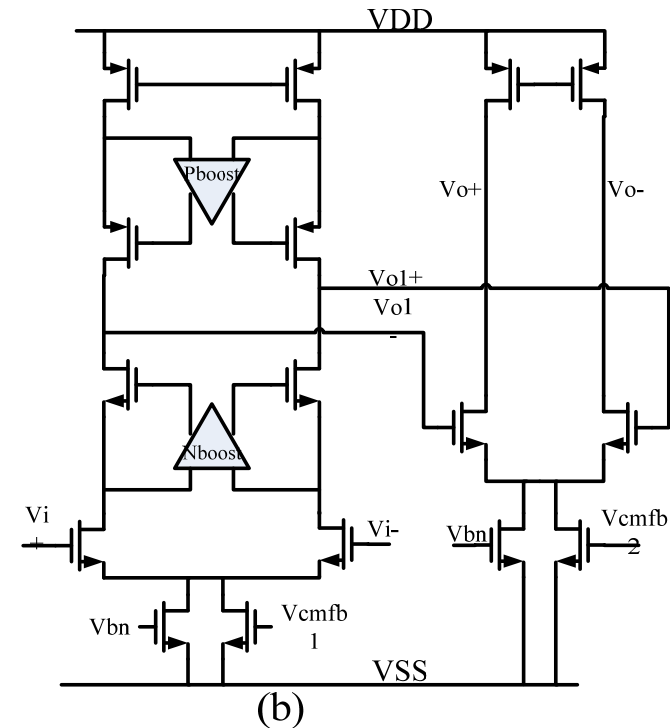
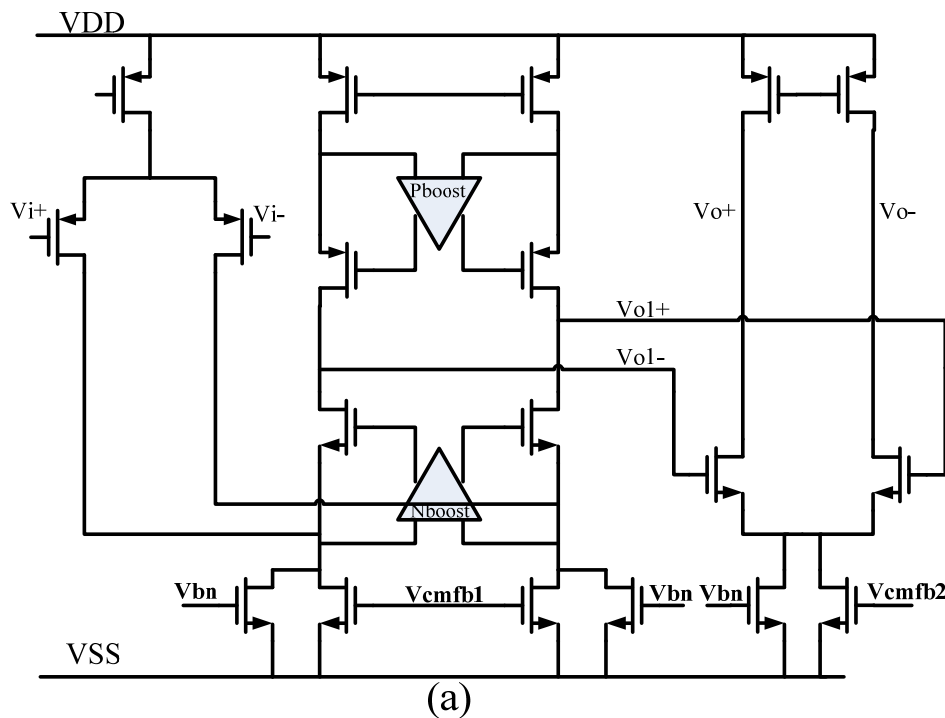
➤第一级输出output swing小，可采用省电的telescopic，不过此时共模输入范围较小

$$GBW = \frac{g_{m1}}{2\pi C_C}$$

$$f_{nd} = \frac{g_{m2}}{2\pi C_L} \cdot \frac{1}{1 + \frac{C_{n1}}{C_C}}$$

设计中注意第一级输出寄生电容Cn1要优化

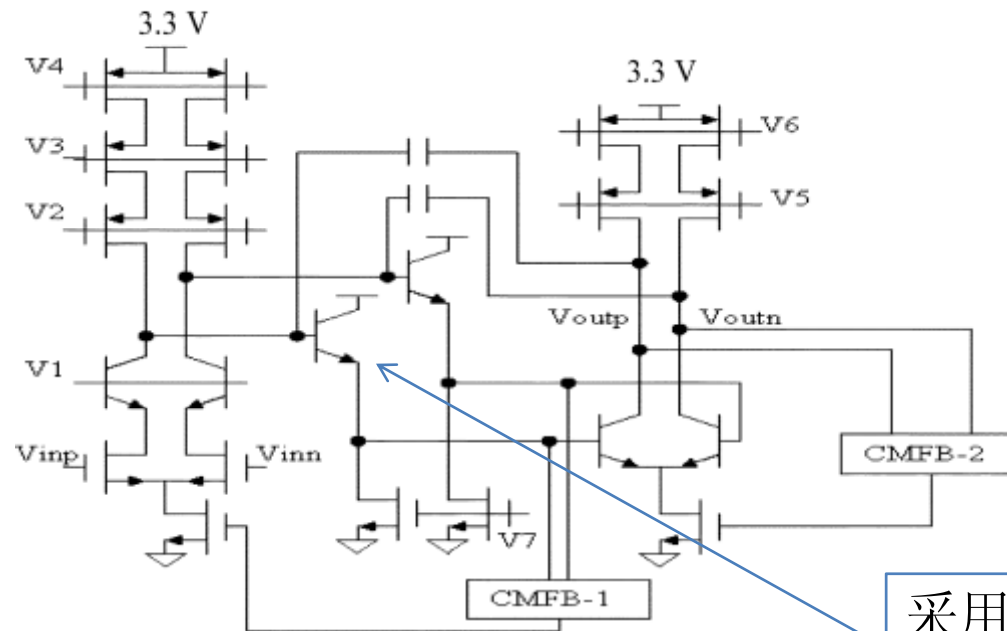
Two-Stage OTA with two cmfb



共模反馈倒相问题：可采用“两级各自共模反馈的办法”，
第一级和第二级均采用开关电容共模反馈电路

问题：第一级 C_n 会增加，因为多了 C_{cm1}
运放output swing减小了一个 V_{ov}

Two-Stage OTA with two cmfb



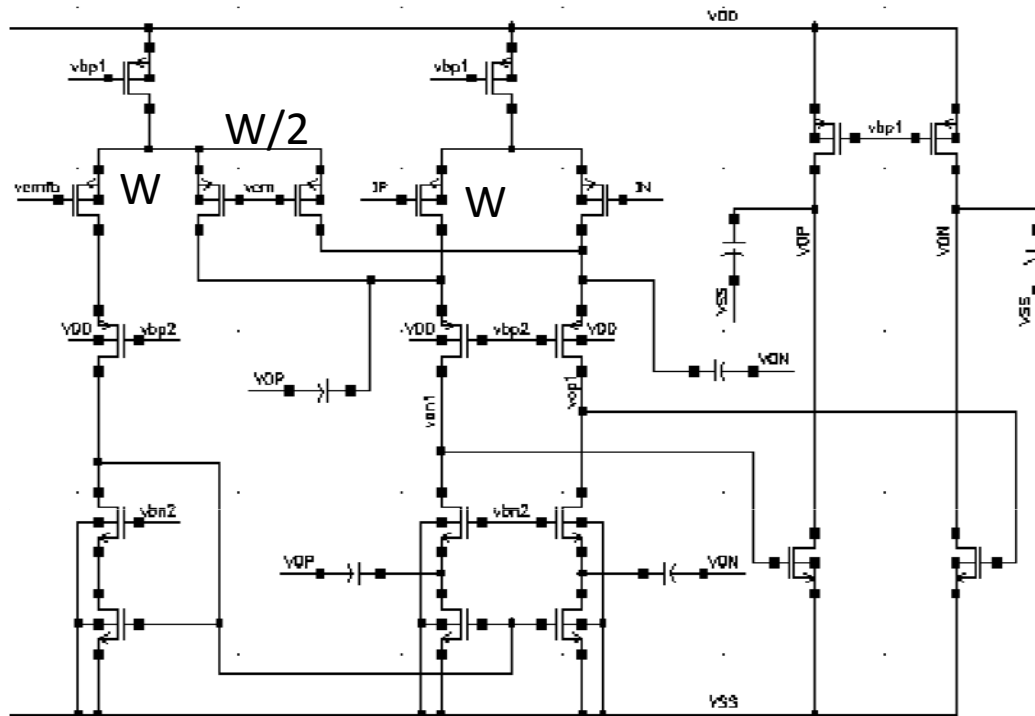
GAIN: 120dB
BW: 750MHz
Settling: 0.25LSB in 2ns
SC CMFB for stage 1&2

采用source follower隔离
Ccm1的影响

Buffering and level-shifting
between stage 1 and 2

Ref: Ahmed M. A. Ali, etc, "A 14-bit 125 MS/s IF/RF Sampling Pipelined ADC With 100 dB SFDR and 50 fs Jitter",
JSSC, 41.8

Two-Stage OTA without cmfb inverter



➤无须共模倒相，也不用两级分别共模反馈

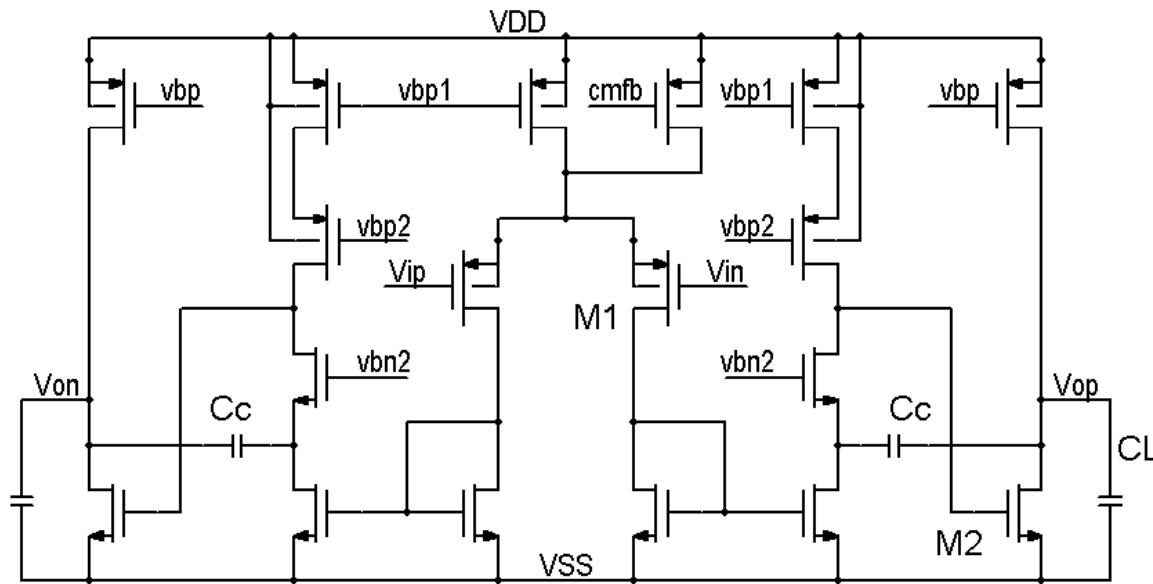
➤共模回路相对简单，Output Swing大

➤共模gm与差模gm相似，且共模与差模信号经过相似的路径

➤第一级增加了一半功耗

➤第一级是telescopic结构，输入共模范围小

Two-Stage OTA without cmfb inverter



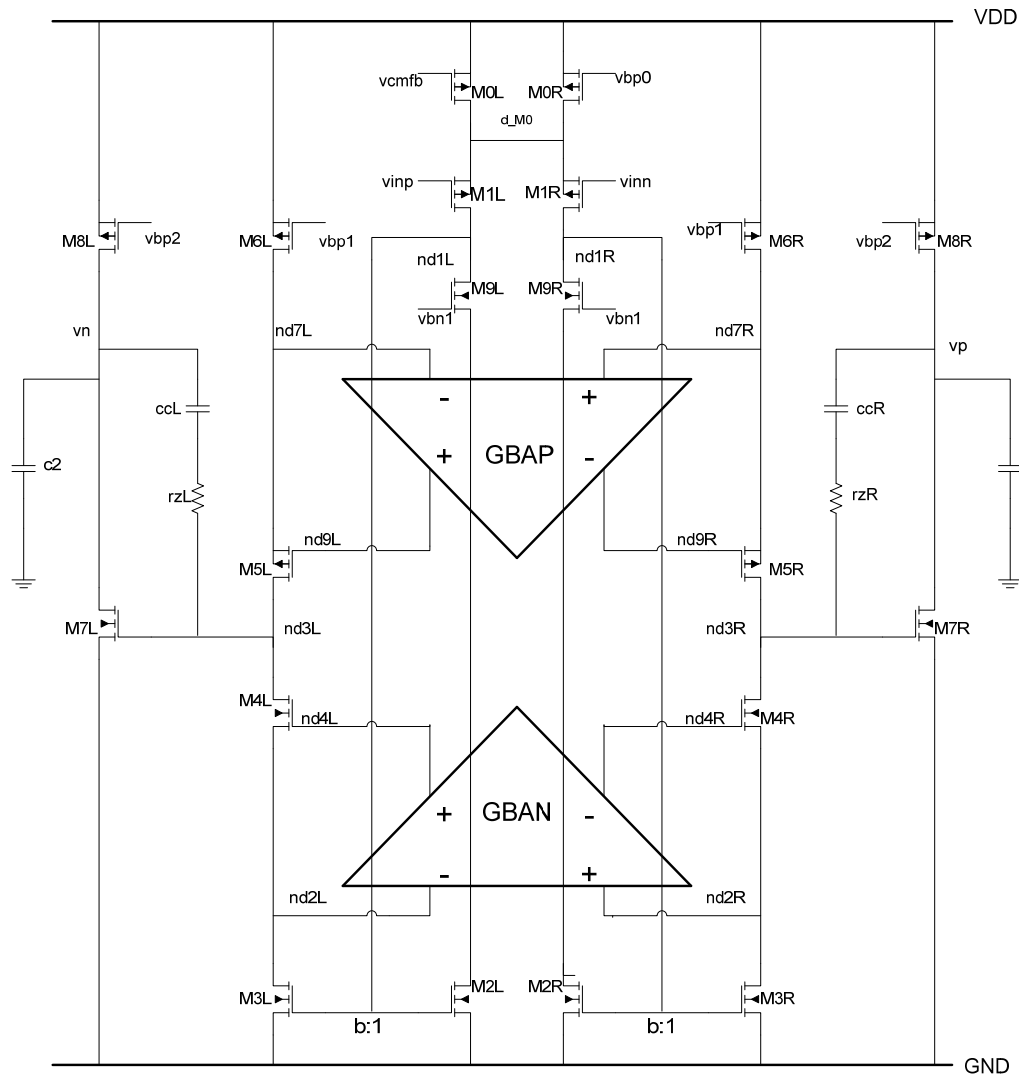
➤第一级采用 symmetrical 结构，则可采用单个共模反馈回路，无须共模倒相

➤Output Swing 大

➤输入共模范围宽

➤第一级的 cascode 上可叠加 gain-boosting，进一步提高增益

Add gain-boosting



在上一页所示运放的基础上增加gain-boosting, 进一步增加增益

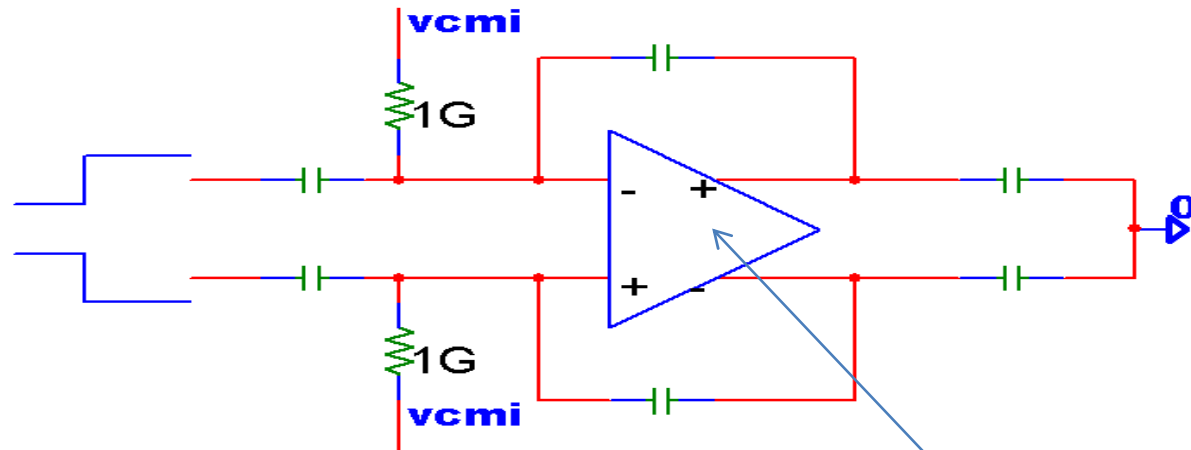
Smic 0.13um, 1.2V supply

Metrics	Specifications	TT	Worst
DC gain(dB)	90	107.5	97.2
Bandwidth (MHz)	1000	1499	1028
Phase Margin($^{\circ}$)	65	70.8	65.9
Output VS(V)	1.2	1.82	1.21
Power(mW)	/	71.0	83.4

Mostly used amplifiers

Num.	Structure	Power	GBW	Adc	Swing	noise
1	Simple OTA	1	Max.	$\sim A_T$	Avg.	4
2	Telescopic	1	Max.	$\sim A_T^2$	Small	4
3	Symmetrical (B=3)	1.33	Mid.	$\sim A_T$	Max.	16
4	Folded casc.	2	Larg.	$\sim A_T^2$	Avg.	4
5	Miller 2-stage	10	Mid.	$\sim A_T^3$	Max.	4
6	Gain boosting	~ 2	Larg.	$\sim A_T^3$ $\sim A_T^4$	Avg.	
7	Including 2- stage cascode			$\sim A_T^3$	Small	

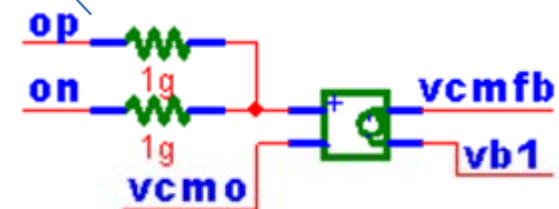
瞬态建立分析



对开关电容式ADC来说，由运放构成的开关电容放大器的建立特性对速度至关重要
运放，特别是结构较复杂的运放，AC分析满足指标，建立特性不一定就能满足指标，因此，需要对运放做详细的瞬态建立分析

1) 将运放的开关电容共模反馈替换成连续共模反馈电路模型

2) 按照上面的电容放大器来进行瞬态分析，其中，运放输入端可通过高阻连接到 v_{cmi} 以确定工作点

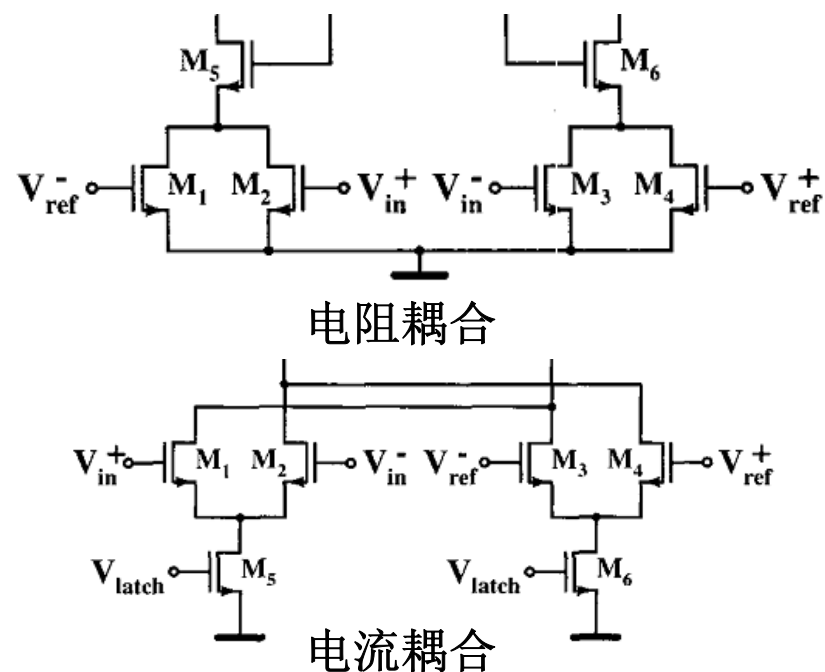
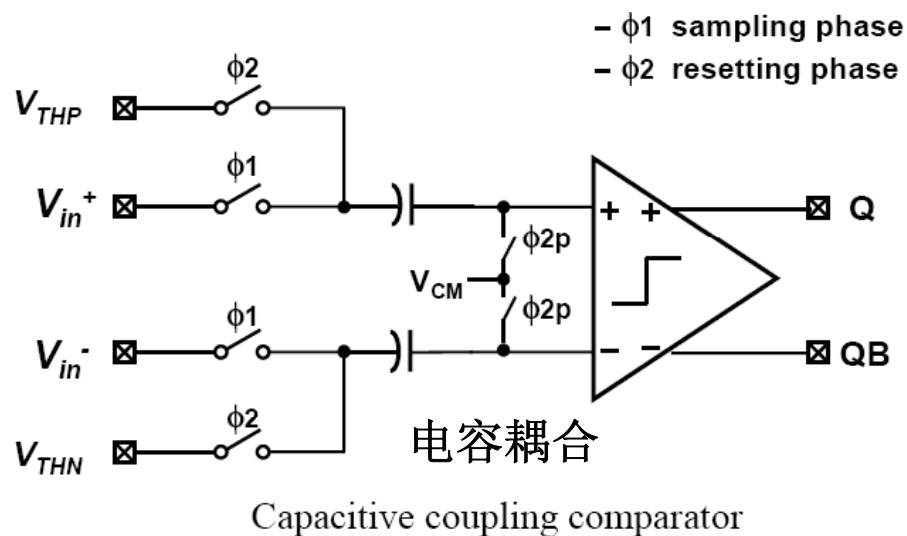


CMFB for sim.

Comparator Design

- 实现差分比较
- 所关心的性能
 - 失调电压（精度）
 - 速度（与输入信号差有关）
 - 回程噪声
 - 功耗
- 主要类型：
 - 动态和静态
- 选择标准
 - 根据转换器系统结构来选择最合适的类型

Differential Comparator



实现差分比较: $(V_{ip} - V_{in}) > (V_{THP} - V_{THN}) ? 0:1$

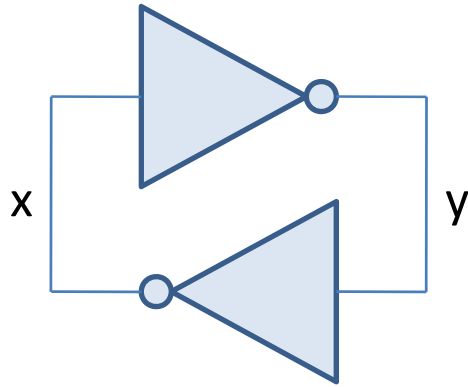
对于电容耦合, 电容不能太小, 可取比较器输入电容的3倍以上

$C = 0.1 \text{ pF} ?$

该电容并不直接构成对输入信号的负载

Offset Cancellation: 取决于级分辨率

速度？



$$T_{latch} = \tau_l \cdot \ln \left(\frac{\Delta V_{logic}}{\Delta V_0} \right)$$

$$\tau_l = \frac{C_n}{g_m}$$

C_n 包括了latch输出本身的寄生电容，以及负载电容

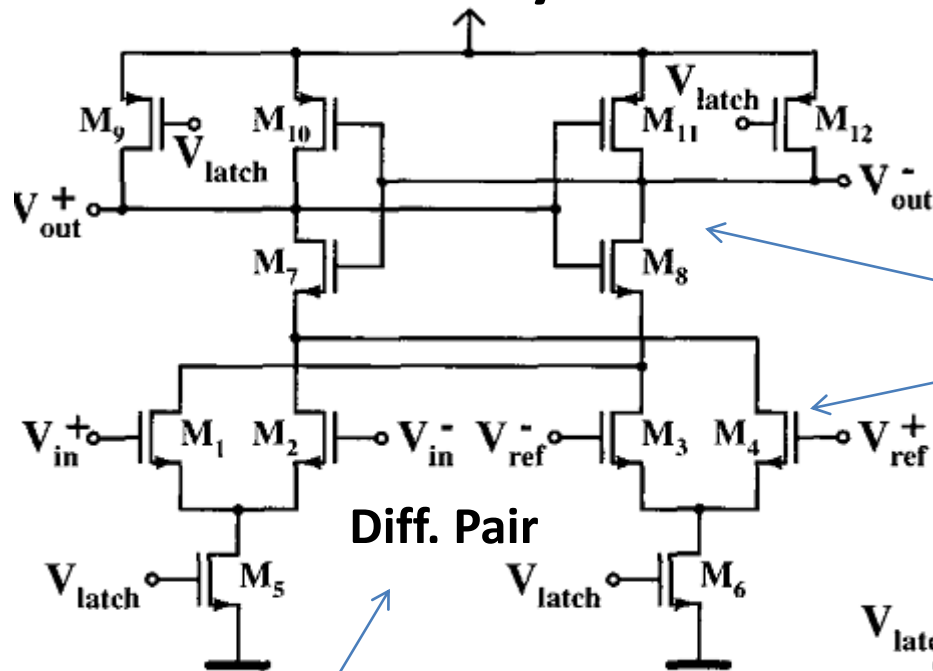
若 T_{latch} 超出了规定的时间，则会出现亚稳态(MetalStability)

在时间 t_0 内可分辨的最小输入电压为：
$$V_{min} = \Delta V_{logic} \cdot e^{-\frac{t_0}{\tau_l}}$$

若在前面增加pre-amp，增益为 A_v ，则：
$$V_{min} = \frac{1}{A_v} \Delta V_{logic} \cdot e^{-\frac{t_0}{\tau_l}}$$

显然，增加pre-amp可提高分辨率，降低误差出现的概率

Dynamic Comparator

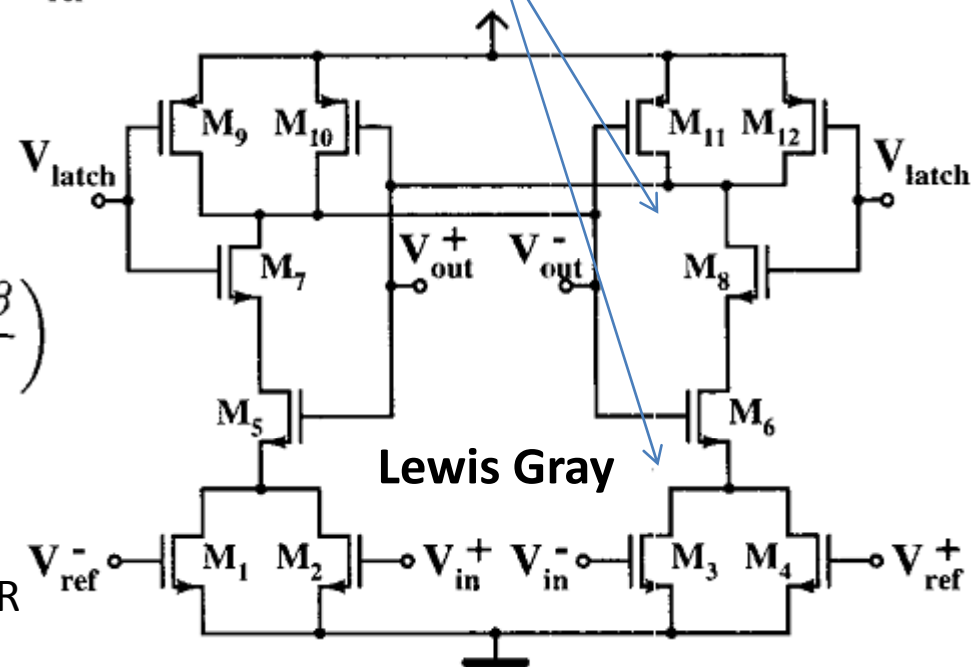


Diff. Pair

$$V_{os} = \Delta V_T + \frac{V_{gs} - V_T}{2} \left(\frac{\Delta R_L}{R_L} + \frac{\Delta \beta}{\beta} \right)$$

功耗小，失调大，
回程噪声大

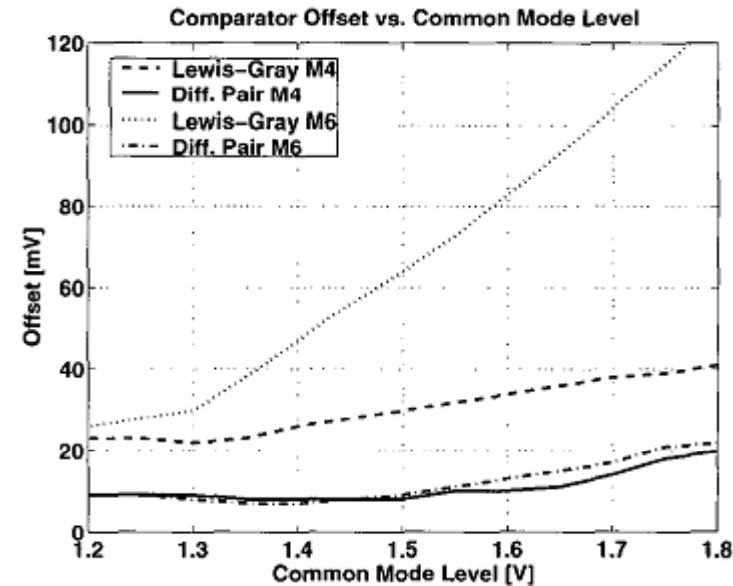
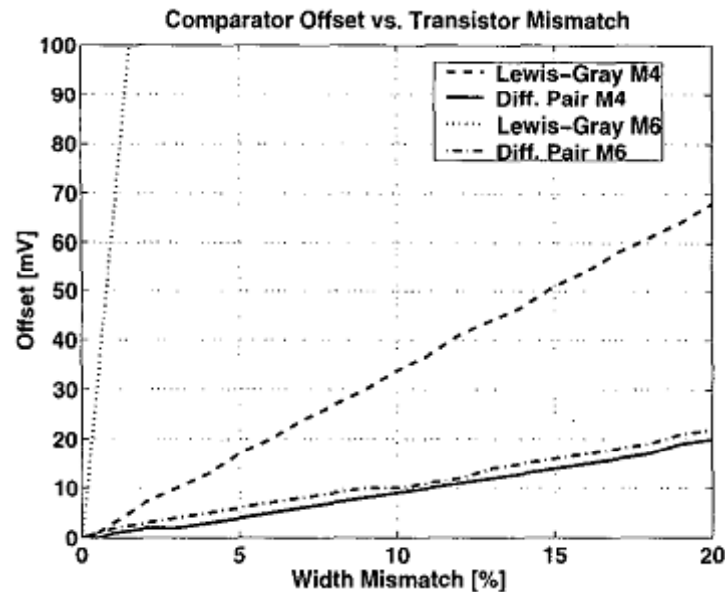
速度对比？
回程噪声？



Lewis Gray

Lauri Sumaner, et, al. AMISMATCH
INSENSITIVE CMOS DYNAMIC COMPARATOR
FOR PIPELINE A/D CONVERTERS, 2000

Dynamic Comparator



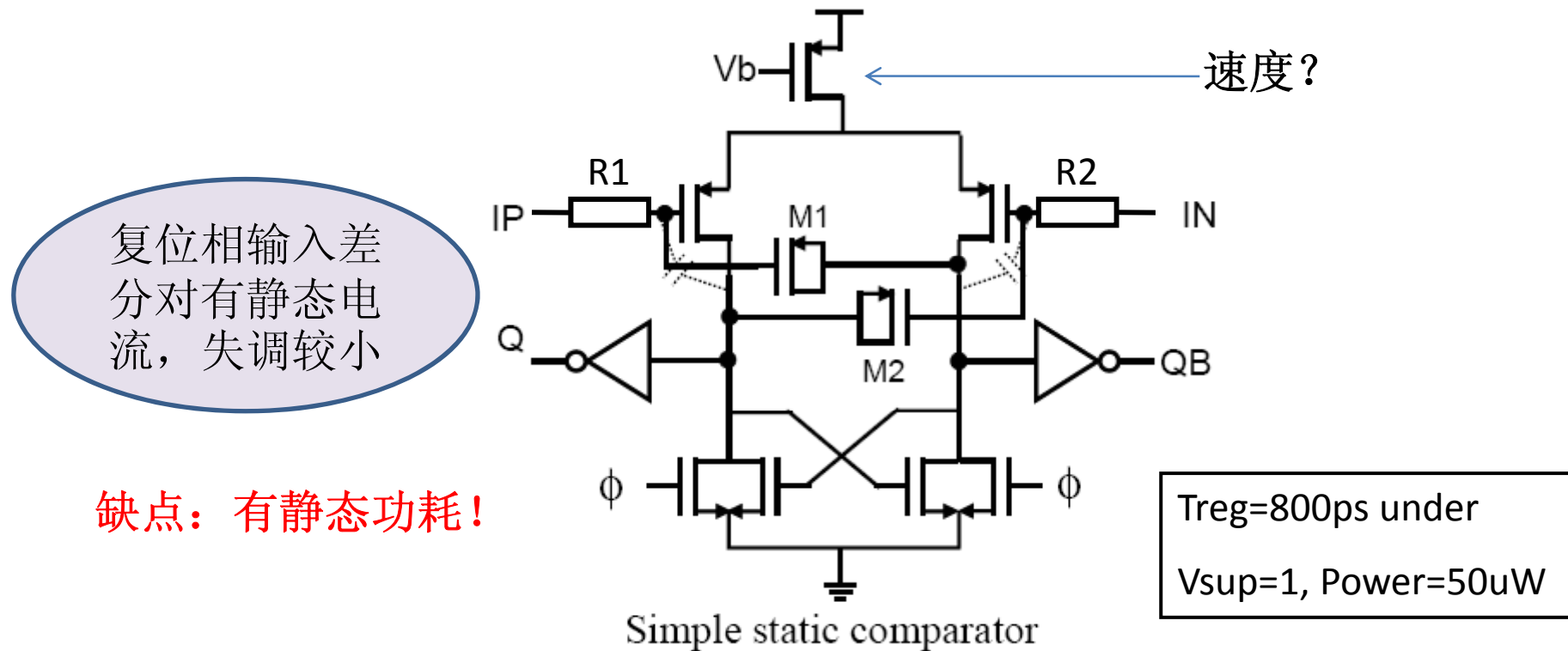
➤ Lewis-Gray: 失调大，速度慢，回程噪声小

➤ Diff. Pair: 失调小，速度快，回程噪声大

Lauri Sumaner, et, al. AMISMATCH
INSENSITIVE CMOS DYNAMIC COMPARATOR
FOR PIPELINE A/D CONVERTERS, 2000

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Static Comparator

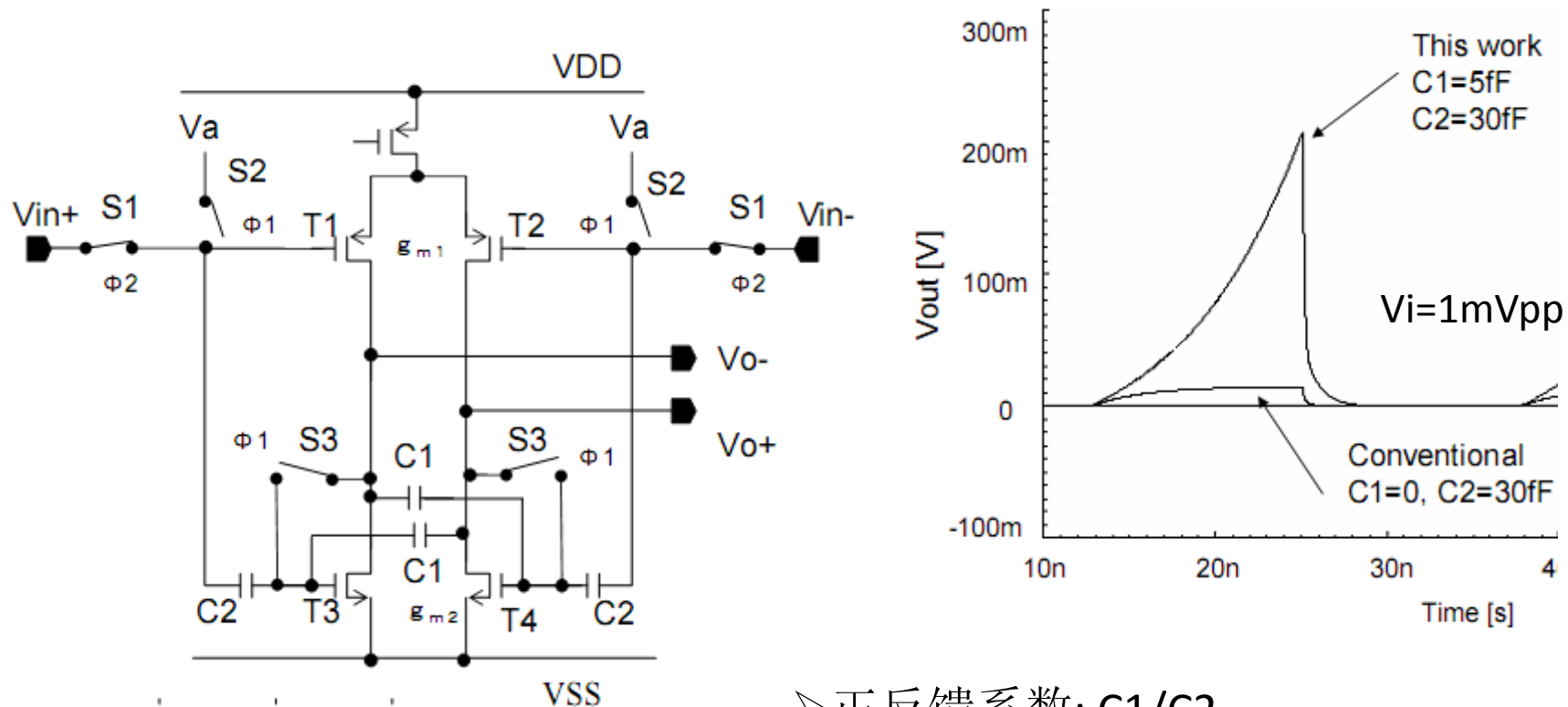


PMOS input and NMOS latch: low 1/f noise, fast latching

M1, M2: half size of the input pair, to reduce diff. kick-back noise

R1, R2: to reduce kick-back noise

Comparator with positive feedback



➤ 正反馈系数: $C1/C2$

利用正反馈来提升pre_amp增益, 降低offset, 用于sub-ranging ADC, 或multi-bit pipeline stage

[7] Y. Shimizu, S. Murayama, K. Kudoh, H. Yatsuda, and A. Ogawa, "A 30 mw 12b 40 MS/s sub-ranging ADC with a high-gain offset-canceling positive-feedback amplifier in 90 nm digital CMOS," IEEE, ISSCC 2006, Dig. of Tech. Papers, pp.222–225, Feb. 2006.

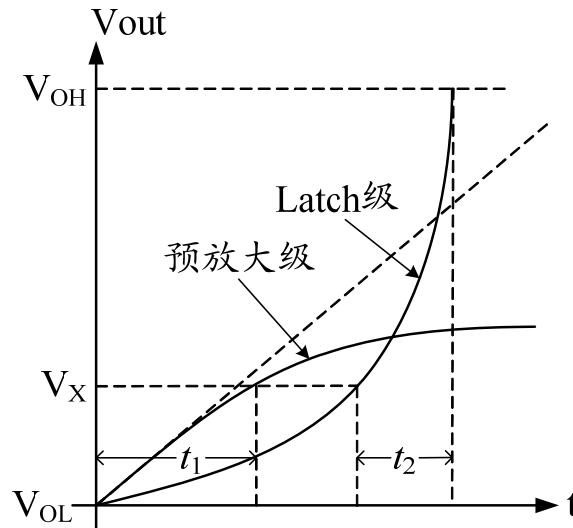
Comparator with pre-amp

M1采用交叉耦合构成负阻，用于提高增益：

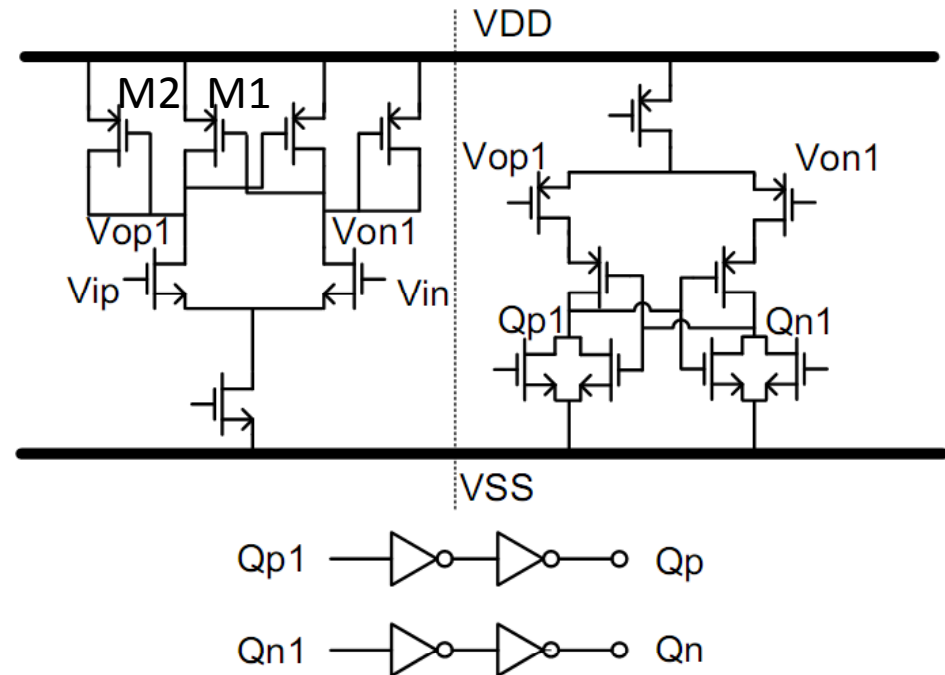
$$A_v = g_{m1} / (g_{m2} - g_{m1})$$

$W_2 > W_1$ ：放大器

$W_2 < W_1$ ：带迟滞效应的比较器



缺点：有静态功耗！



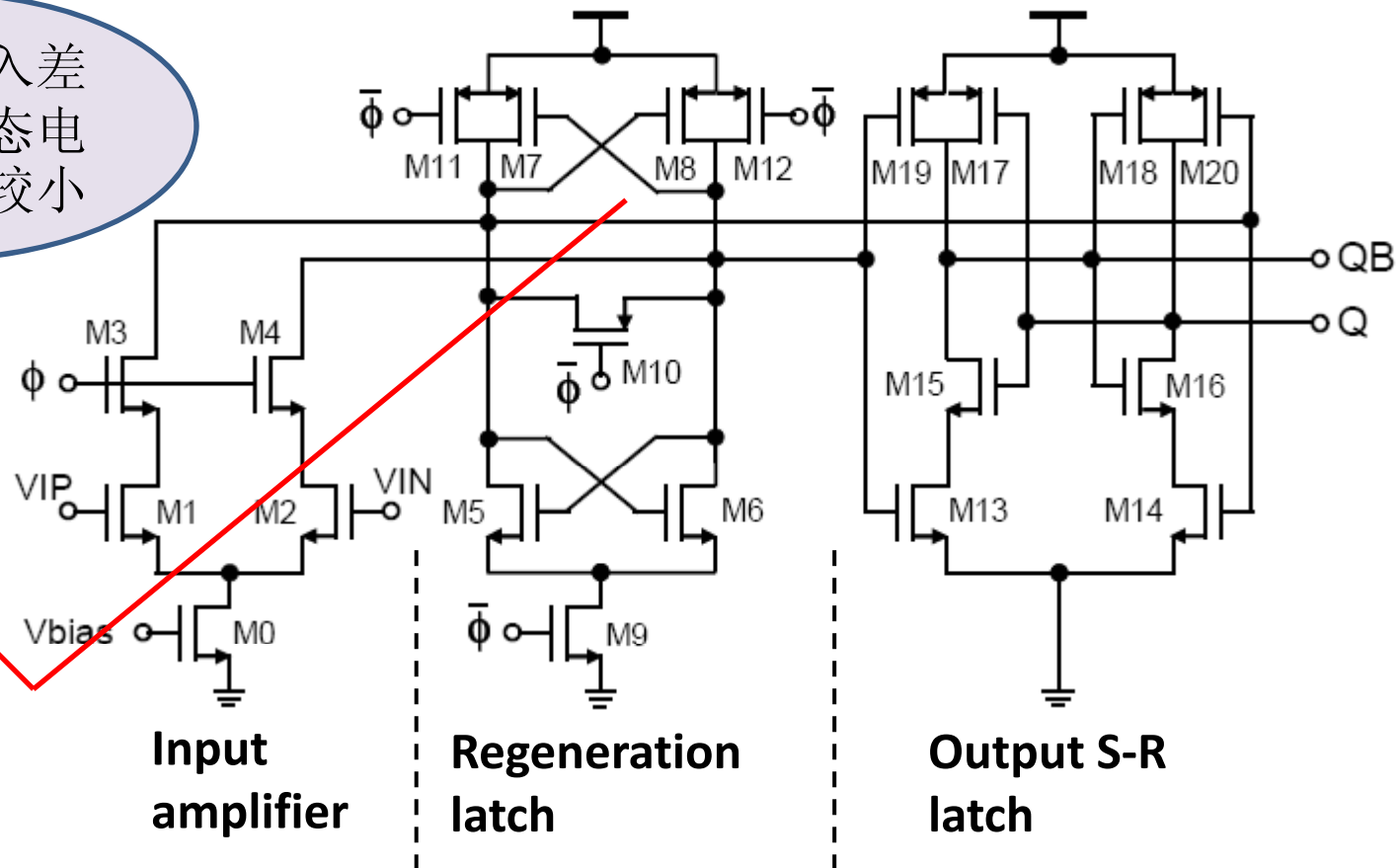
加前置放大器的好处：

- 1) 降低latch的offset
- 2) 提高比较器速度
- 3) 隔离kick-back Noise

High Speed Comparator

复位相输入差分对有静态电流，失调较小

失调小
速度快
回程噪声小



M3, M4: 节约功耗; 阻止kick-back noise

Φ : 下降沿与MDAC中的采样沿对齐

$T_{reg} < 250ps$ under
 $V_{sup} = 1.8$, Power = 0.3mW
 $V_{id} = 2mV$

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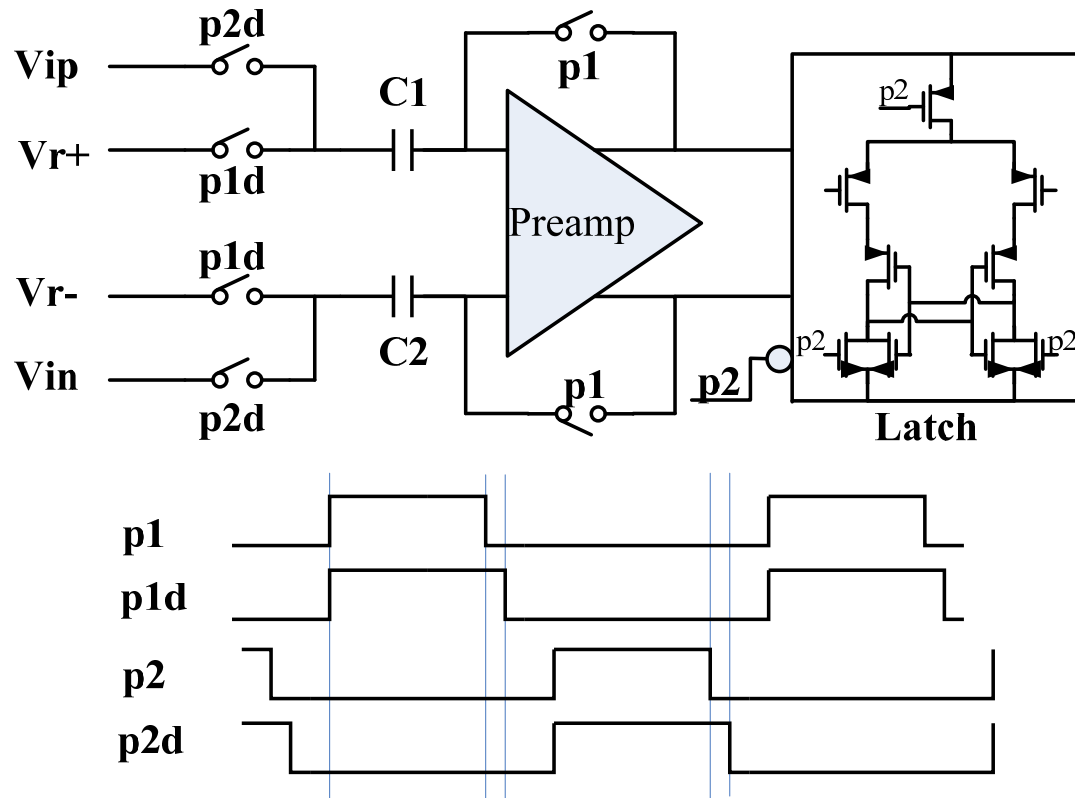
Offset cancellation comparator

N-bit级电路能容忍的
比较器的失调为：

$$\frac{V_{ref}}{2^{N+1}}$$

比较器采用
输入失调存储技术

$$V_{OS} = \frac{V_{os1}}{1 + A_1} + \frac{\Delta Q}{C} + \frac{V_{os2}}{A_1}$$



比较器结构和时序

Clock generator for high speed ADC

商用ADC的时钟指标

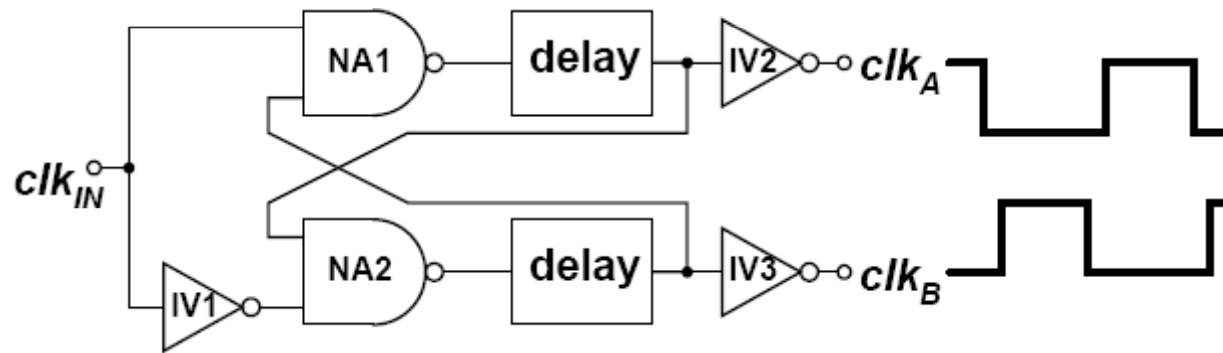
$$R_{SN} = -20 \lg (2\pi f_{\text{analog}} t_{\text{jitter}})$$

$$t_{\text{jitter}} = \frac{10^{\frac{-R_{SN}}{20}}}{2\pi f_{\text{analog}}}$$

对宽带转换器设计，
时钟抖动的控制至关重要，
并已经成为设计的瓶颈和难点

产品	采样率 (Msp/s) / 精度 (位)	时钟抖动/ ps	占空比可调整范围/ %
AD9480	250/ 8	0.25	30~70
AD9430	210/ 12	0.25	30~70
AD9445	125/ 14	0.06	30~70
AD9446	100/ 16	0.06	30~70
ADC081000	1 000/ 8	0.4	30~65
ADC10080	80/ 10	2	40~60
ADC12L080	80/ 12	0.7	20~80
ADS5520	125/ 12	0.3	40~60
ADS5550	125/ 14	0.3	40~60
LTC2220-1	185/ 12	0.15	30~70
LTC2208	130/ 16	0.05	30~70
MAX1214	210/ 12	0.2	40~60

Clock generator



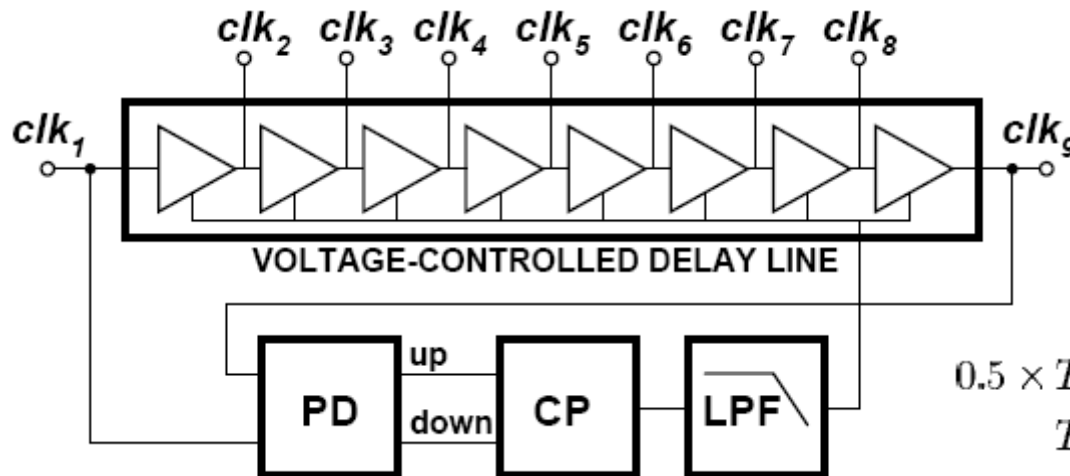
常用的双相不交叠时钟产生电路

为考虑worst corner
所导致的延时过剩
问题

占空比问题

Clock skew问题

DLL 还可用于time-
interleaved结构

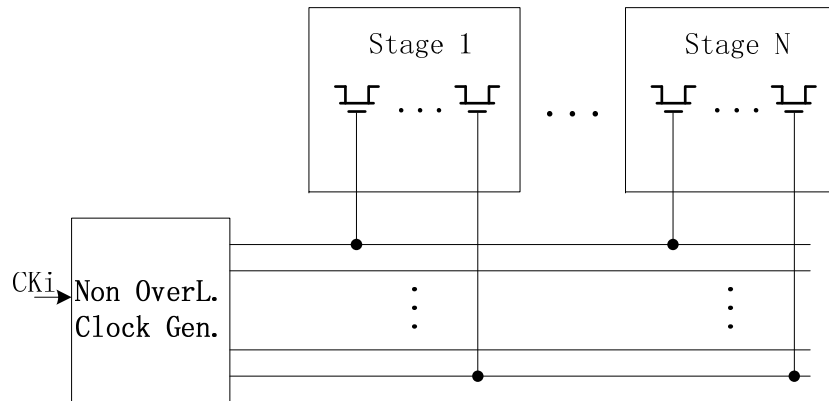


基于DLL的时钟相产生电路

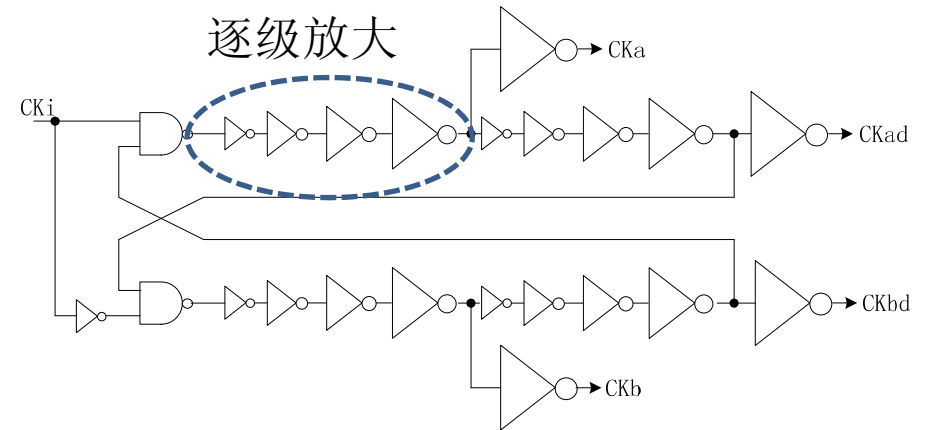
$$0.5 \times T_{CLK} < T_{VCDL-min} < T_{CLK}$$

$$T_{CLK} < T_{VCDL-max} < 1.5 \times T_{CLK}$$

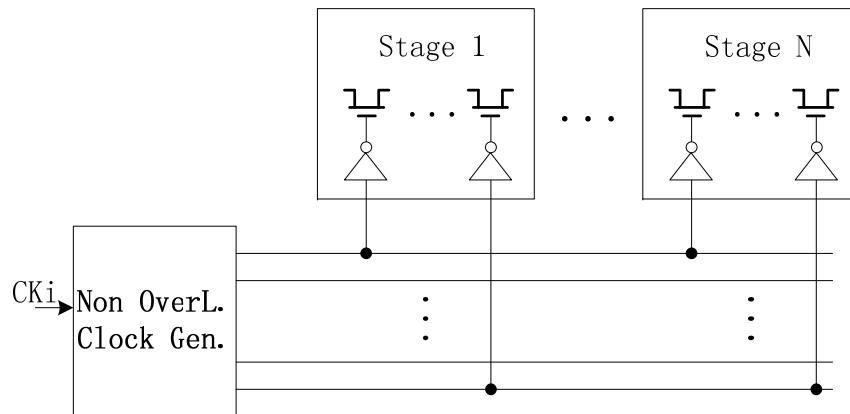
Global & Local Driver



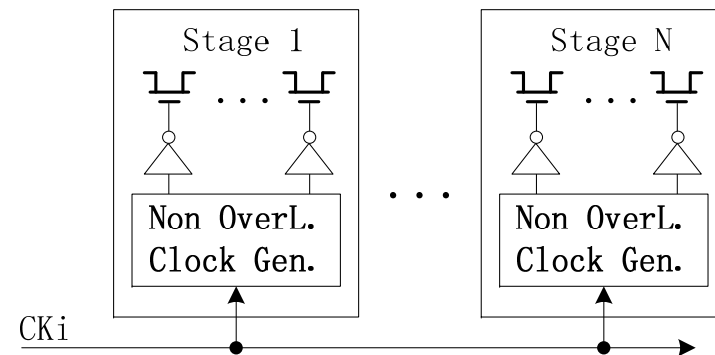
Global Generator & Driver



Non-Overlapping Clock Generator

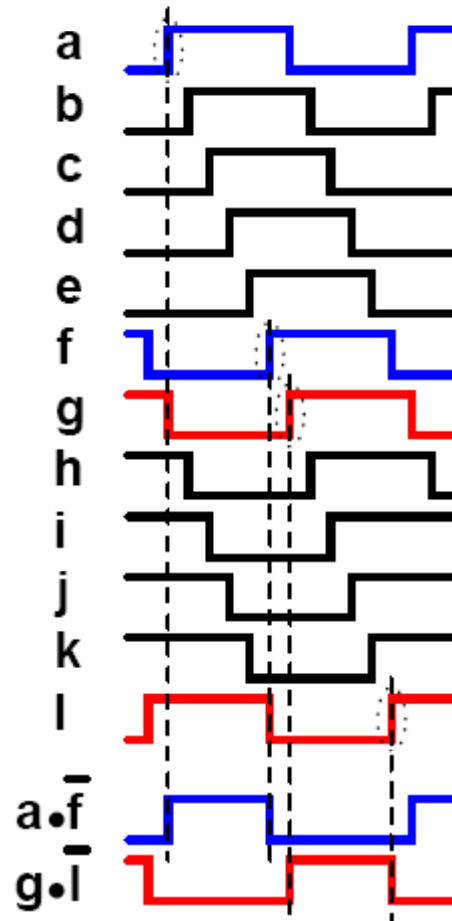


Global Generator & Local Driver



Local Generator & Driver

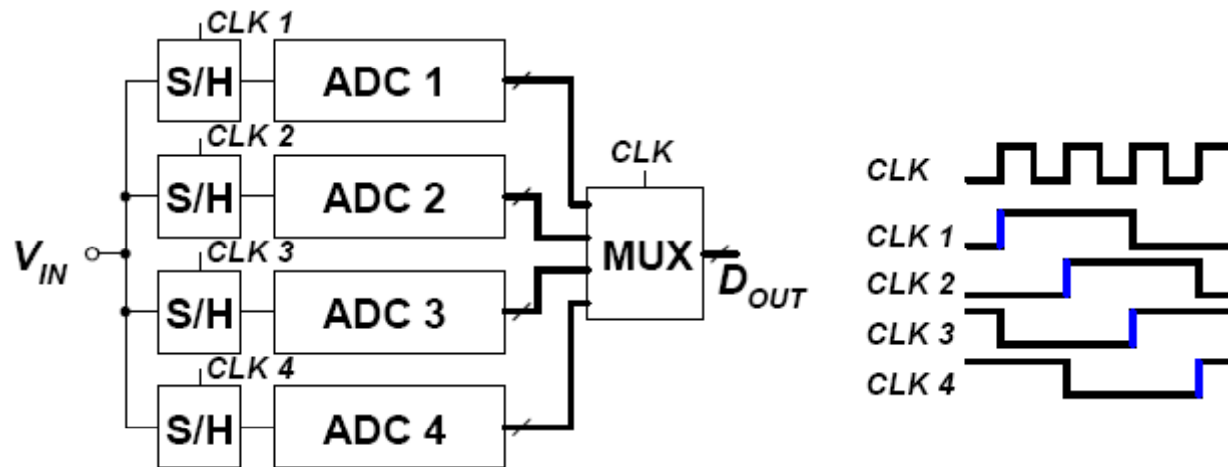
DLL-based Clock generator



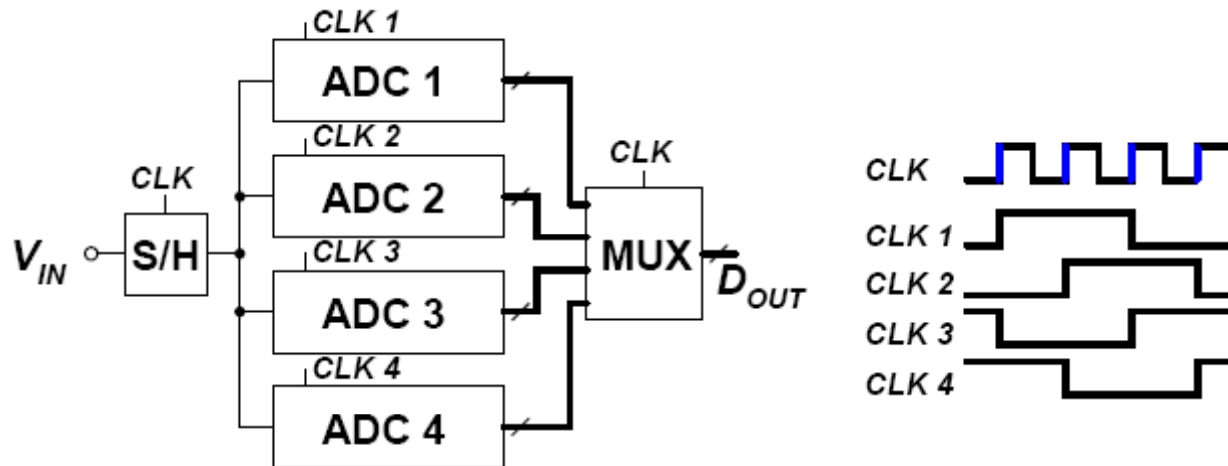
由DLL中不同时钟相合成ADC工作所需的控制时钟:

- 1) 双相不交叠
- 2) 提前下降沿
- 3) 时钟相细分
- 4) 多路交织控制时钟

DLL-Based Clock Time-interleaved ADC



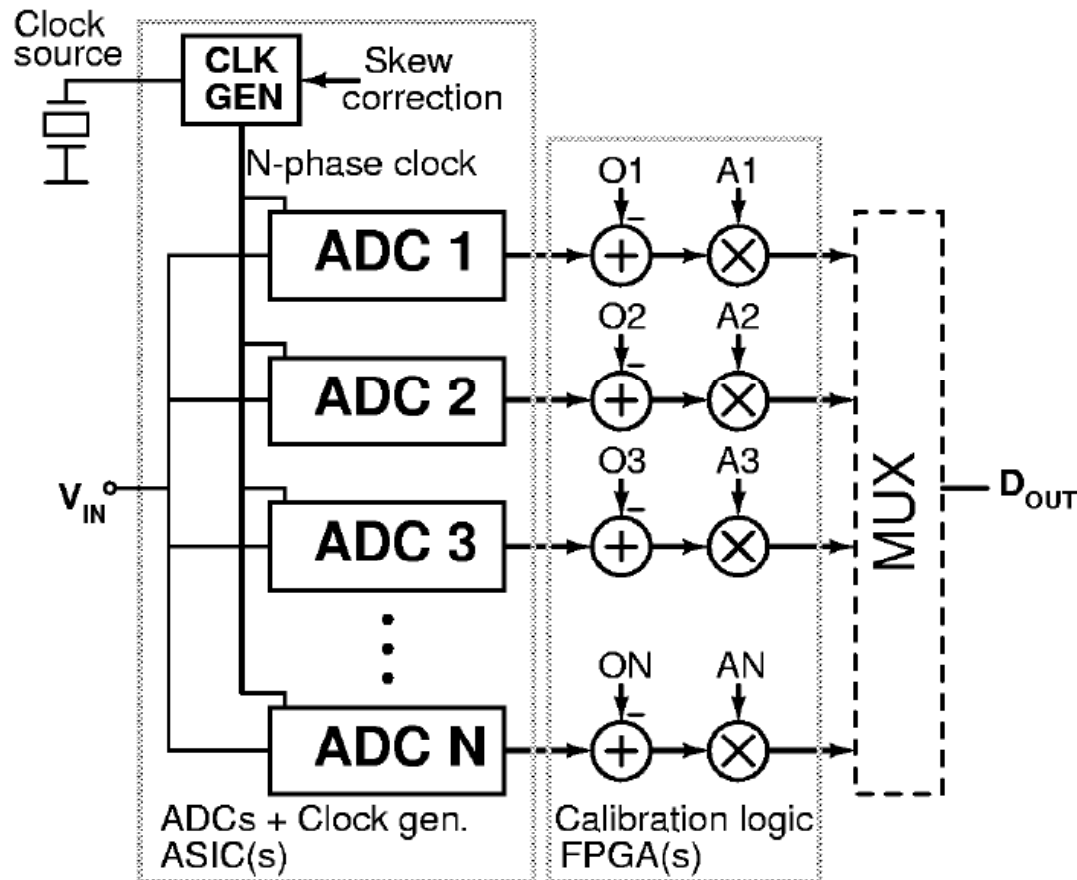
Four-channel time-interleaved ADC and its clock signals.



Time-interleaved ADC with front-end S/H circuit.

采用**DLL**可避免高频时钟或者时钟倍频器

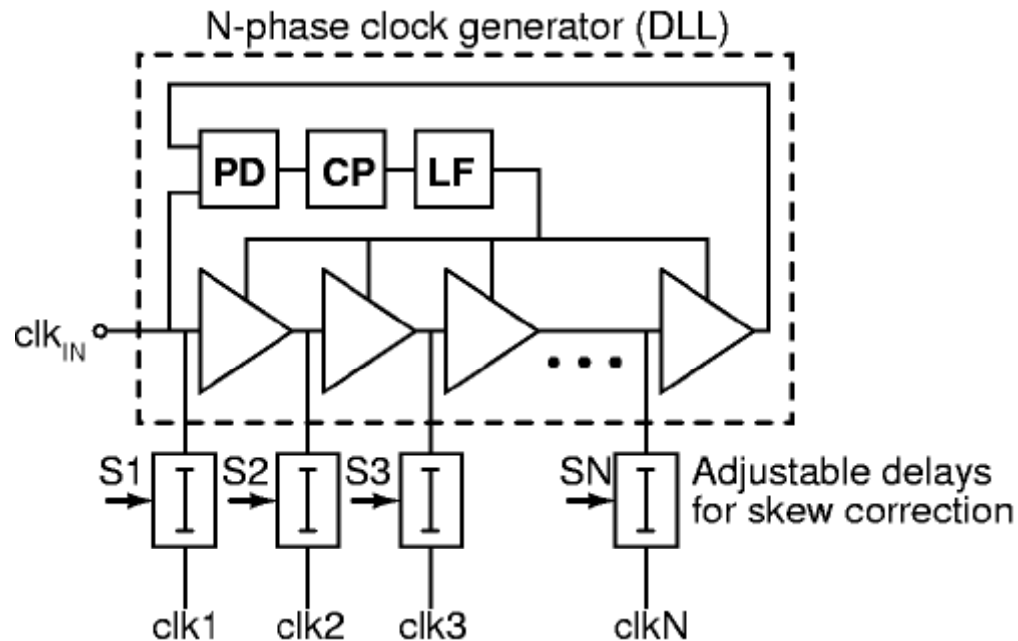
Clock for Time-interleaved ADC



- Time-interleaved structure
- Low-jitter clock with skew correction
- Digital calibration of channel gain and offset mismatch

Ref: http://microelectronics.esa.int/mpd2004/ADC_pre_0204_v3.pdf

Clock for Time-interleaved ADC

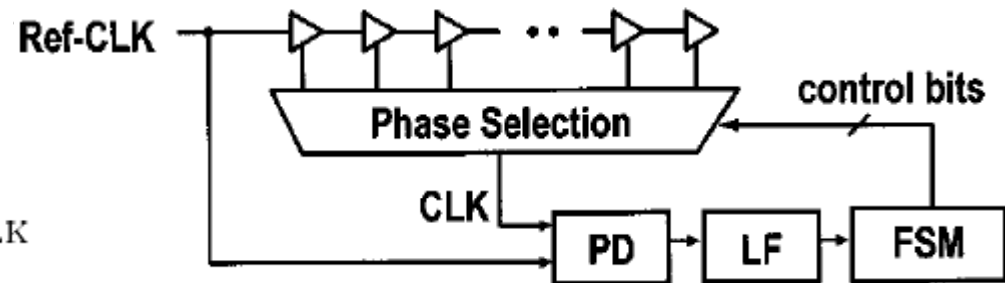
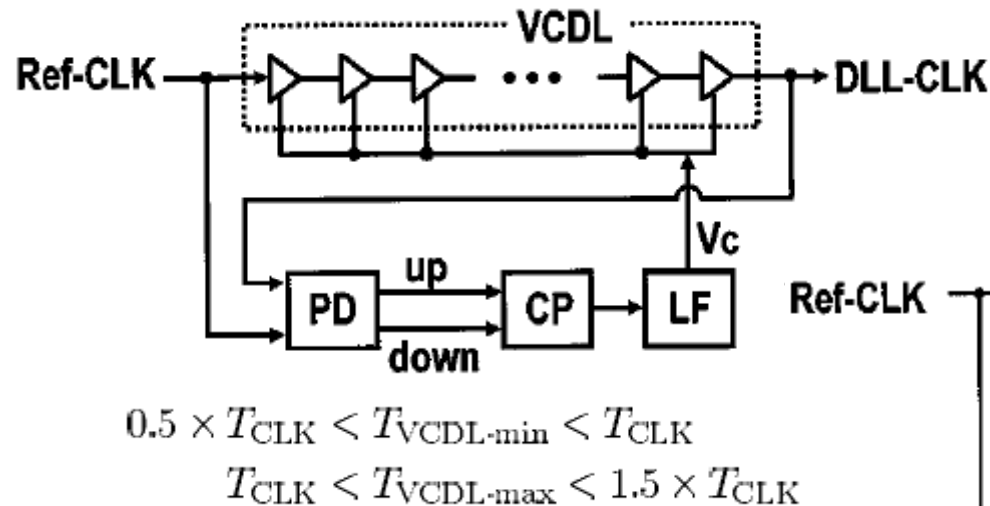


Thermal jitter $\sim N^2$ (N=length of DLL)

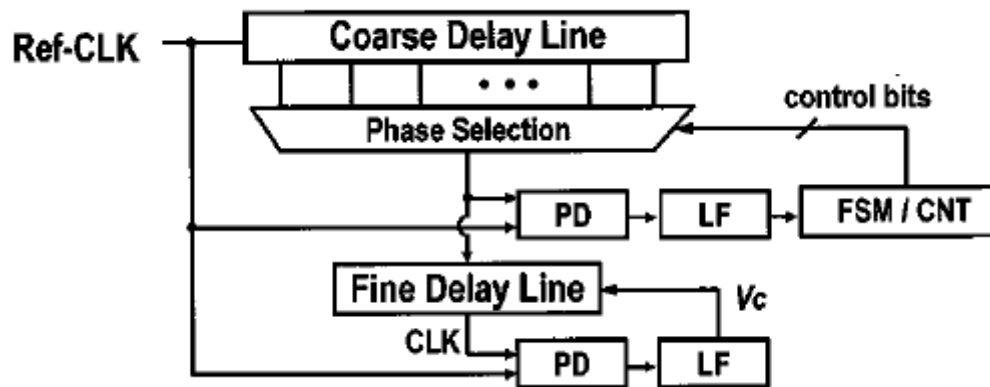
Additional jitter from on-chip components minimized

Ref: http://microelectronics.esa.int/mpd2004/ADC_pre_0204_v3.pdf

Wide-range DLL?



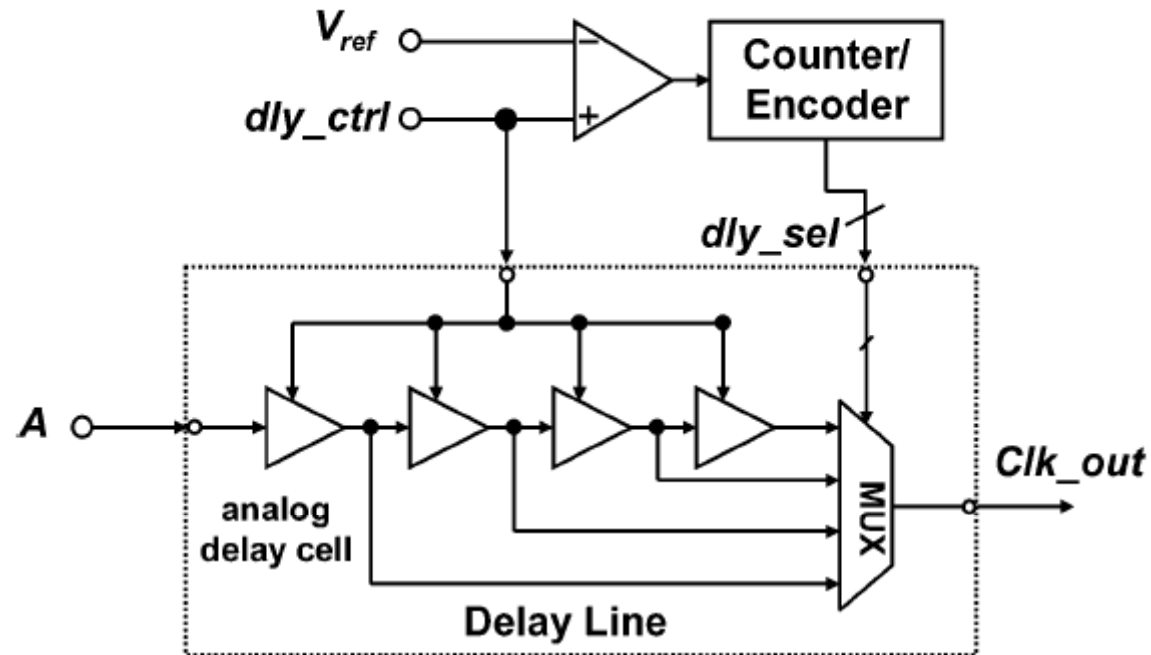
Digital DLL



Dual-loop DLL

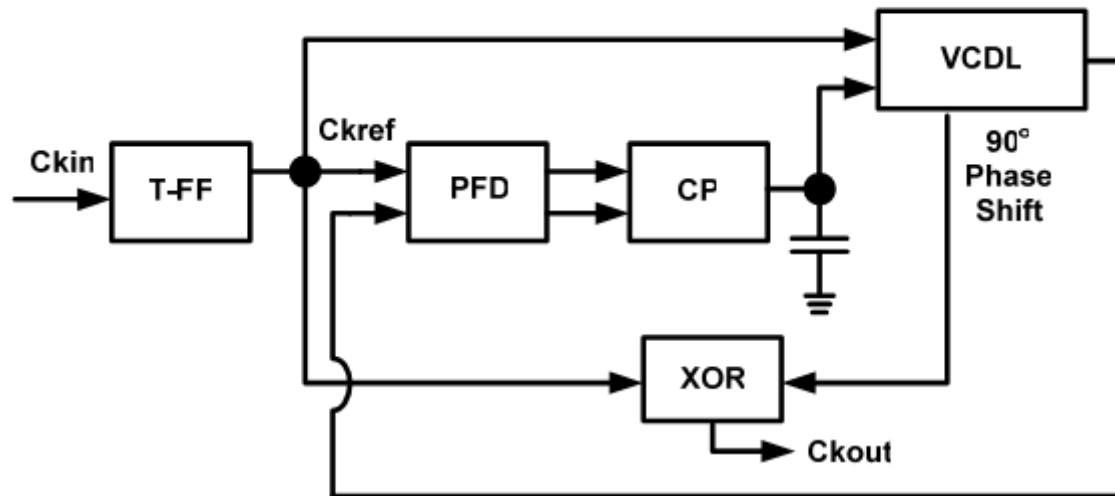
Ref: Yongsam Moon, etc. An ALL-Analog Multiphase Delay-Locked loop Using a Replica Delay Line for Wide-range operation and low-jitter performance

Tuning Range Expanding

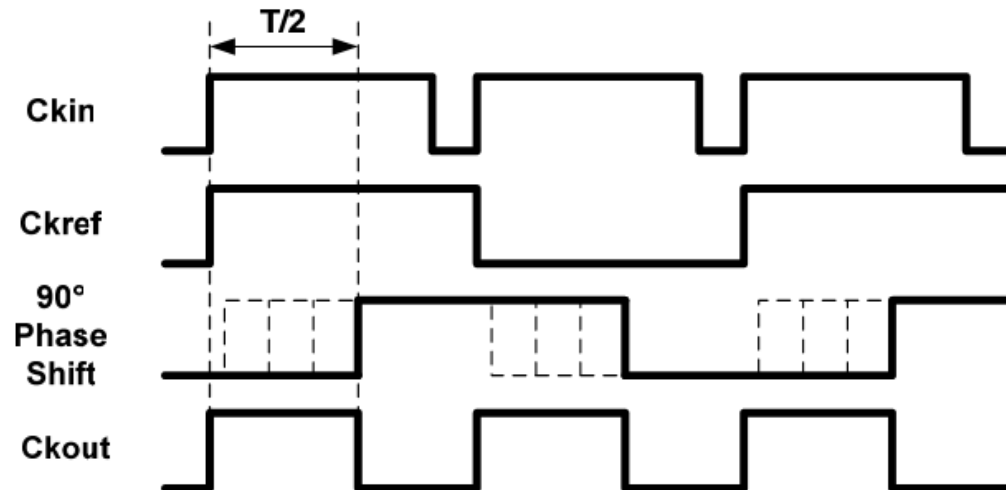


Expanding the tuning range

Duty Cycle Stabilizer

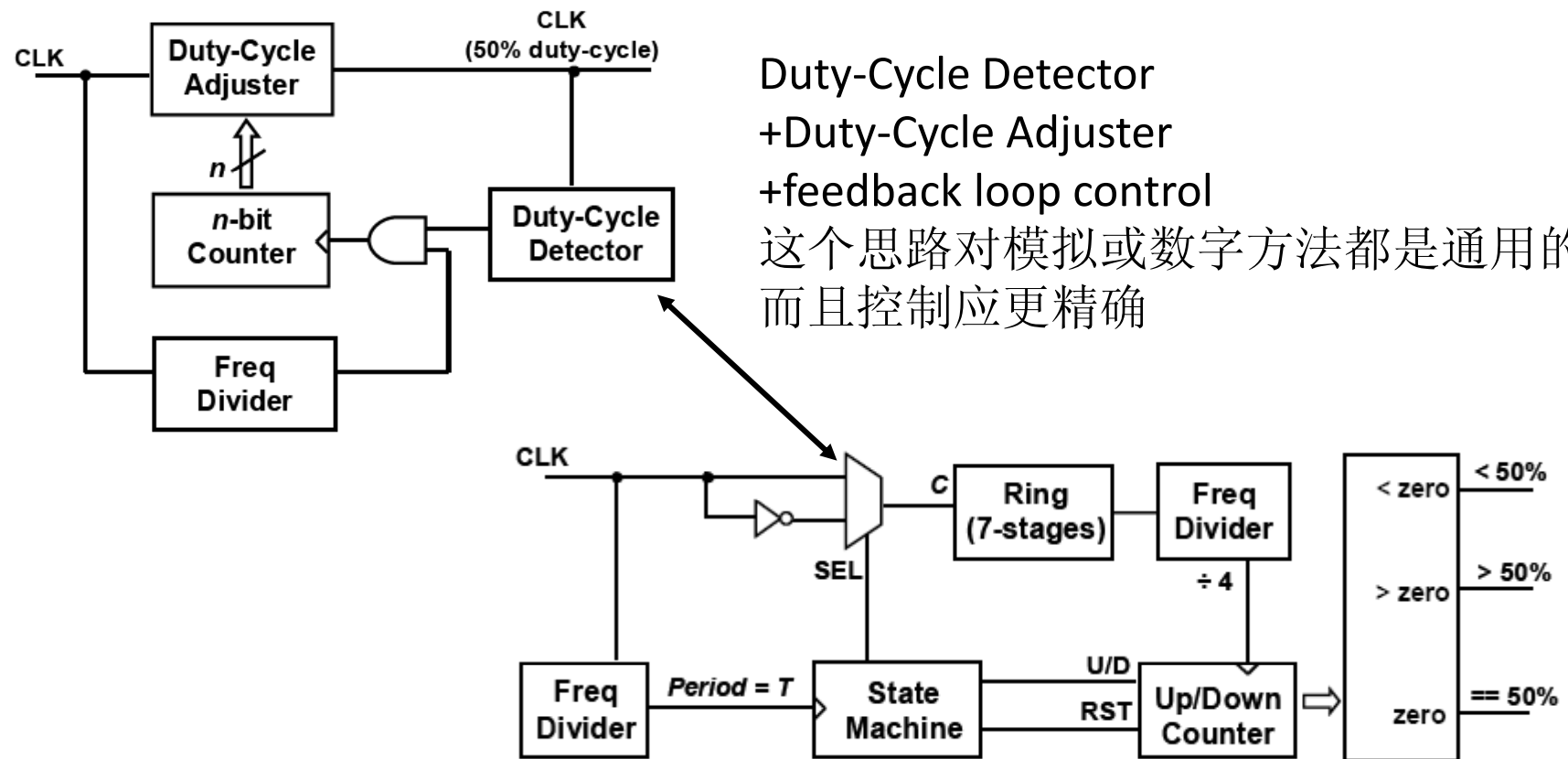


延迟控制
精度取决于各delay
Cell之间的延迟匹配



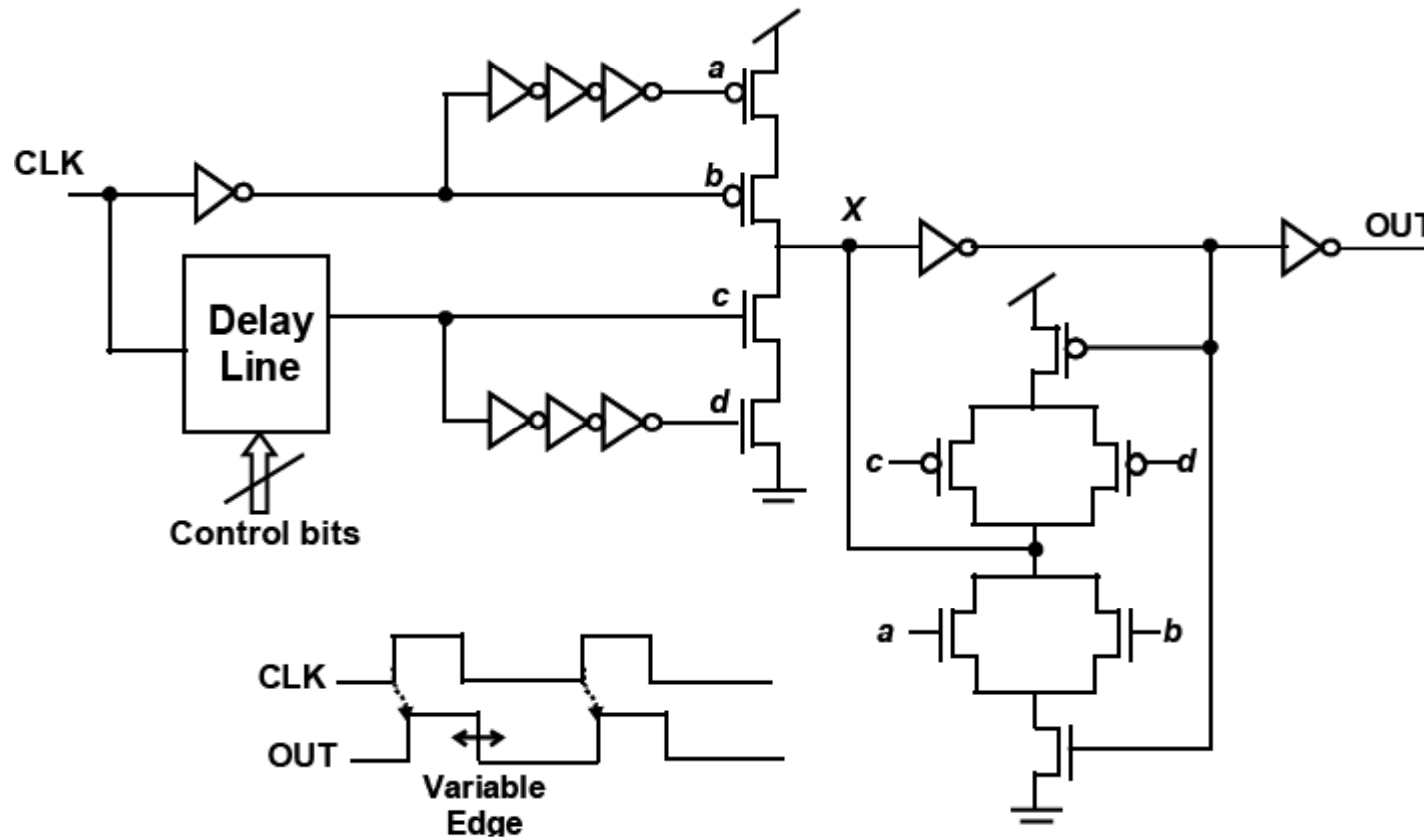
Jaeyong Lee, et al.
“12bits 40MHz
Pipelined ADC with
Duty-Correction
Circuit. “

Duty Cycle Stabilizer



Ref: Kanak Agarwal, etc. "A Duty-Cycle Correction Circuit for High-Frequency Clocks"

Duty Cycle Adjuster



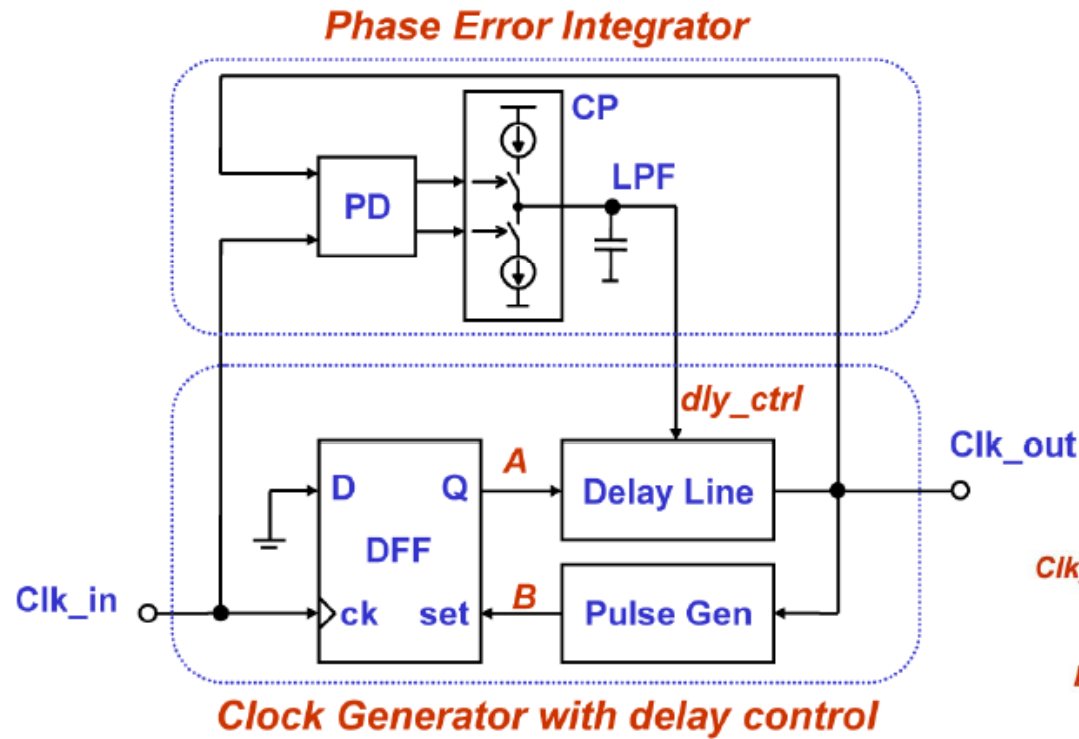
The duty-cycle adjuster circuit

Ref: Kanak Agarwal, etc. "A Duty-Cycle Correction Circuit for High-Frequency Clocks"

李福乐, 清华大学微电子所, 北京, 中国

lifule@mail.tsinghua.edu.cn

Duty Cycle Stabilizer

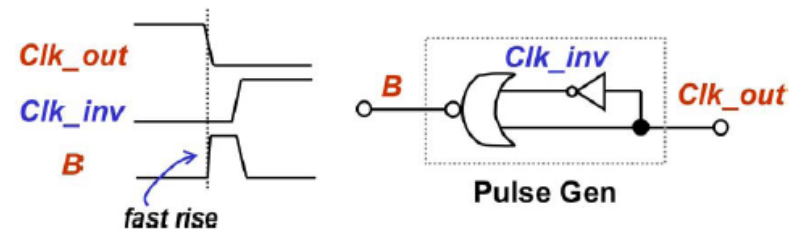
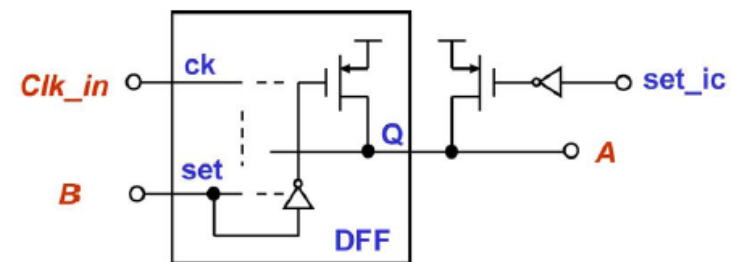
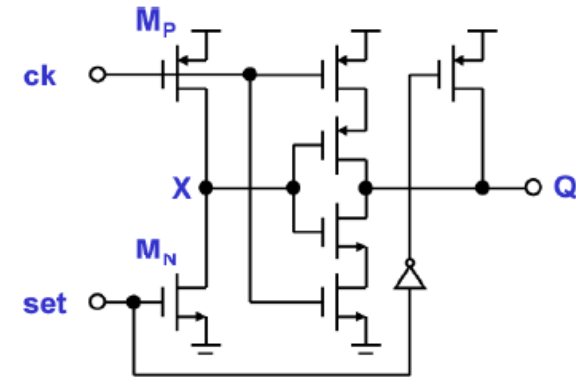


$$T_{POS} = T_{DFF,ck} + T_D$$

$$T_{NEG} = T_{PG} + T_{DFF,set} + T_D$$

Ref: Tsung-Hsien Lin, etc. "A Synchronous 50% Duty-Cycle Clock Generator in 0.35-um CMOS

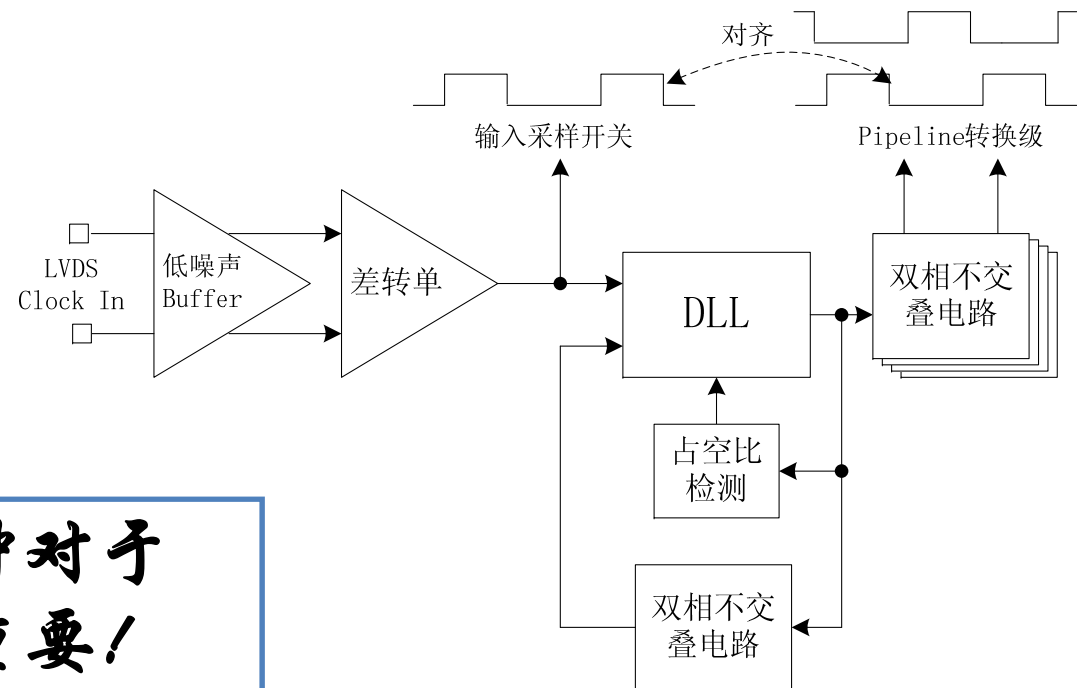
李福乐, 清华大学微电子所, 北京, 中国



一个时钟电路的例子

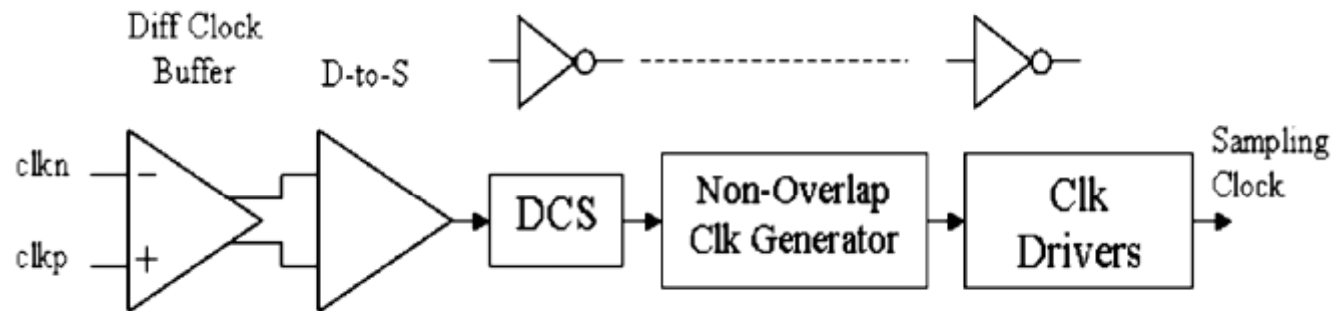
- 采用LVDS和低噪声放大整形
- 占空比稳定
- 时钟去偏移

$$R_{\text{SN}} = -20 \lg (2\pi f_{\text{analog}} t_{\text{jitter}})$$



◆低抖动的采样时钟对于
宽带输入性能至关重要!

Example



Simplified schematic of the low-jitter clock path.

30mA @3V !

Low noise BiCMOS input buffer

About 50fs jitter

Carefully substrate isolation and clock shielding

Heavily supply decouple

Ref: Ahmed M. A. Ali, etc, "A 14-bit 125 MS/s IF/RF Sampling Pipelined ADC With 100 dB SFDR and 50 fs Jitter",
JSSC, 41.8

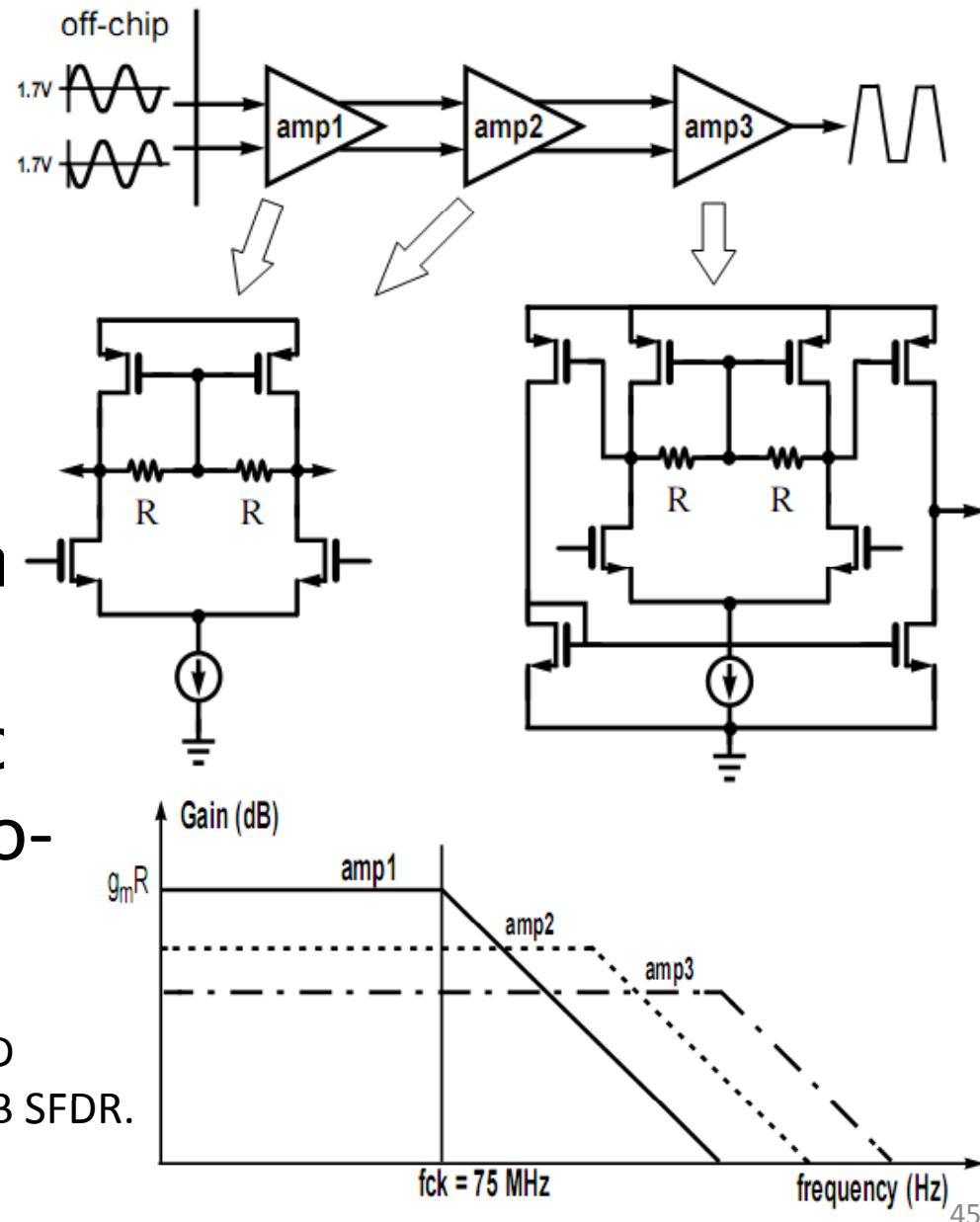
李福乐, 清华大学微电子所, 北京, 中国

lifule@mail.tsinghua.edu.cn

Clock input buffer

- $V_{GS} - V_T = 0.44V$
- $BW1 \approx \text{Clk Freq.}$
 - Dummy cap to adjust BW
- Amp2&3 with lower gain and wider BW to sharp clock edges at zero-crossings

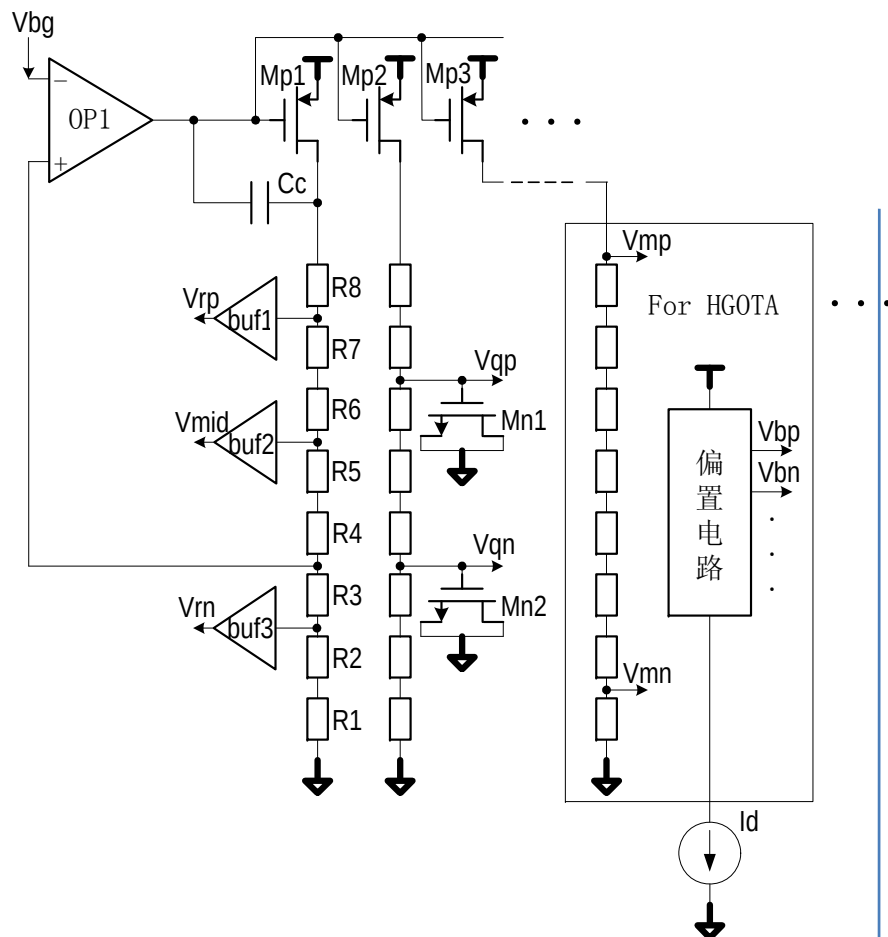
Ref: Hui Pan, A 3.3V 12-b 50-MS/s A/D Converter in 0.6- μm CMOS with 80-dB SFDR.
Phd thesis, 1999



Reference Buffer

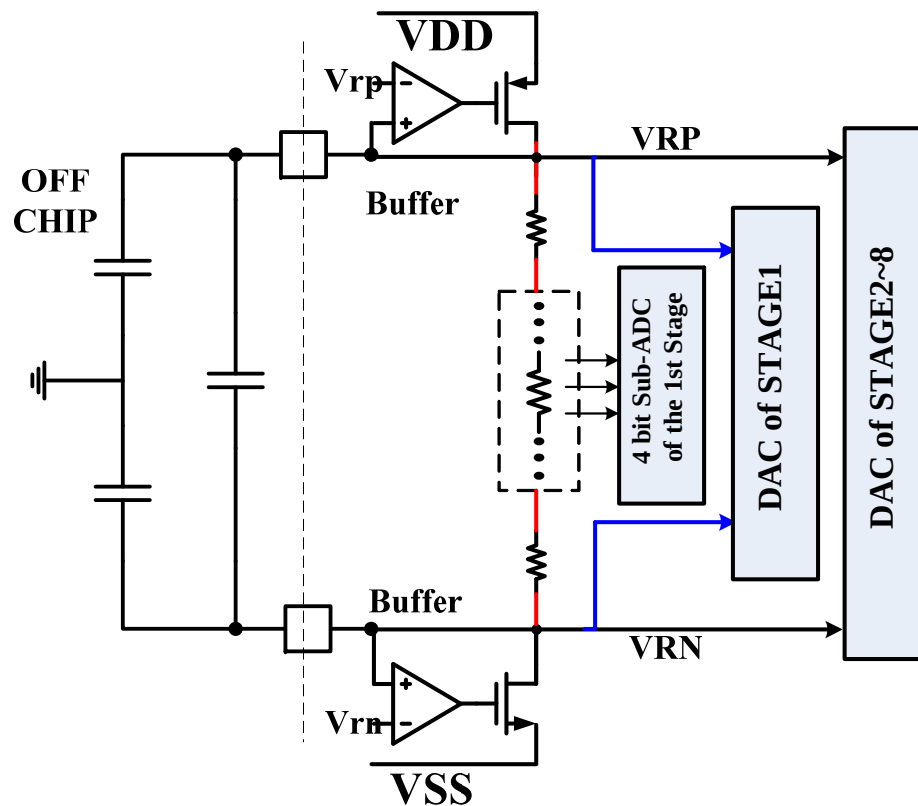
- 驱动参考，参与开关电容放大器的建立过程，其速度应快于对信号的建立要求
- 带片外解耦电容的 buffer
 - 利用片外大电容做AC驱动
 - 片内电压buffer只需补充DC电流
 - 对电压buffer的速度要求低，电路功耗低
- 无片外解耦电容的buffer
 - 依靠电压buffer的带宽实现快速驱动建立
 - 功耗高，一般可占总功耗的20%以上
 - 或者通过无源驱动办法，缺点是功耗高
- 高性能的转换器一般要求片上高速buffer

带片外解耦电容的buffer

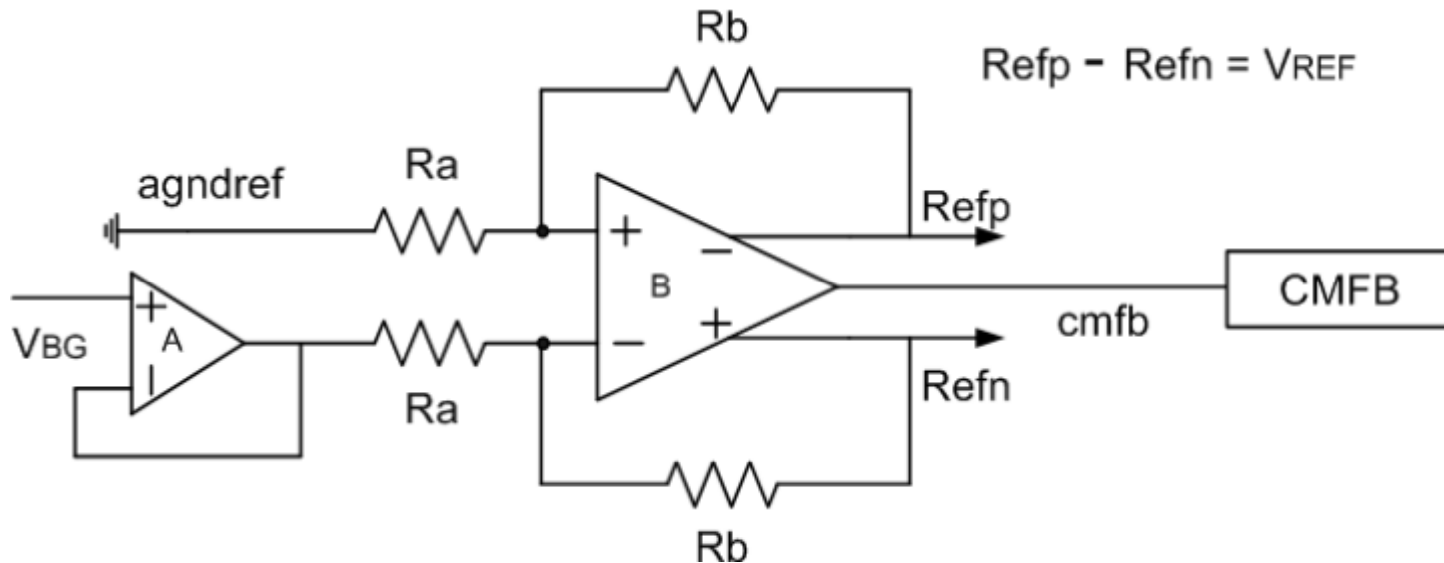


$$C_{off-chip} > 2^{N+1} \cdot C_L$$

由于解耦电容在片外，在高速转换下容易受到bonding wire inductor的限制



Differential reference buffer



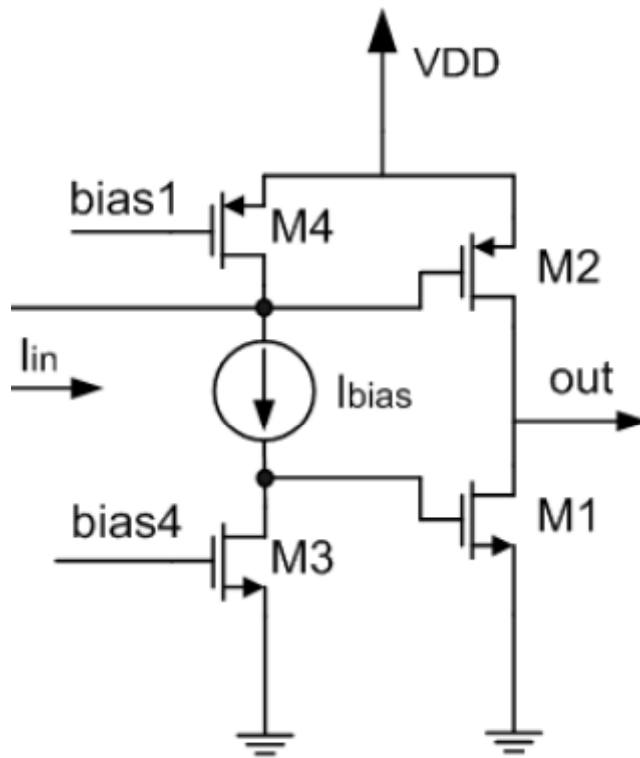
$$V_{REF} = \frac{R_b}{R_a} V_{BG} \quad \text{For ideal opamps}$$

Class-AB opamp is preferred for amplifier B

Driving requirement for opamp A?

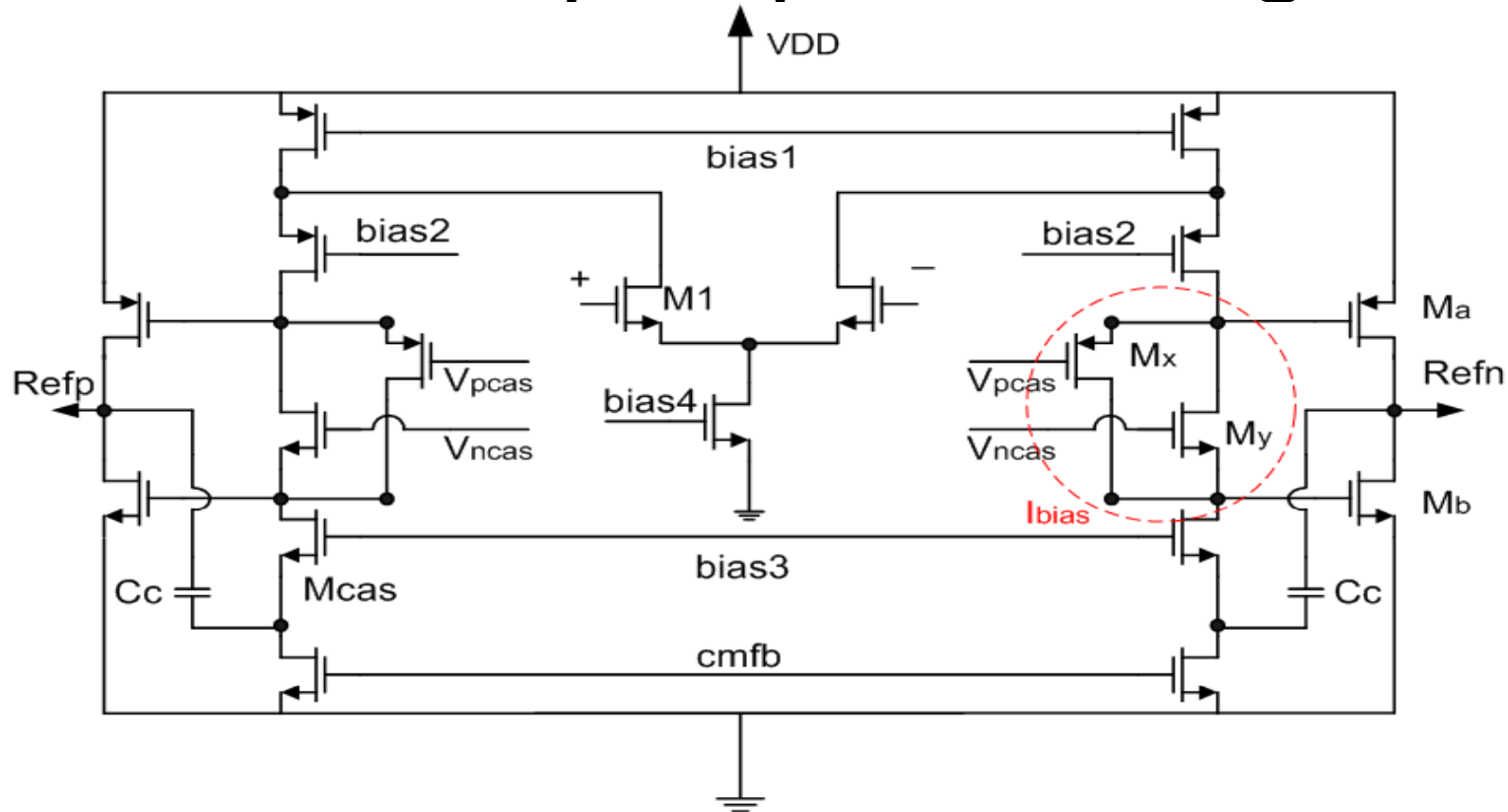
DC gain and GBW requirements for opamp B?

Class-AB output structure



Push-pull for expanding driving current
Rail-to-rail output swing

Class-AB opamp main stage



2-stage miller capacitor compensation structure

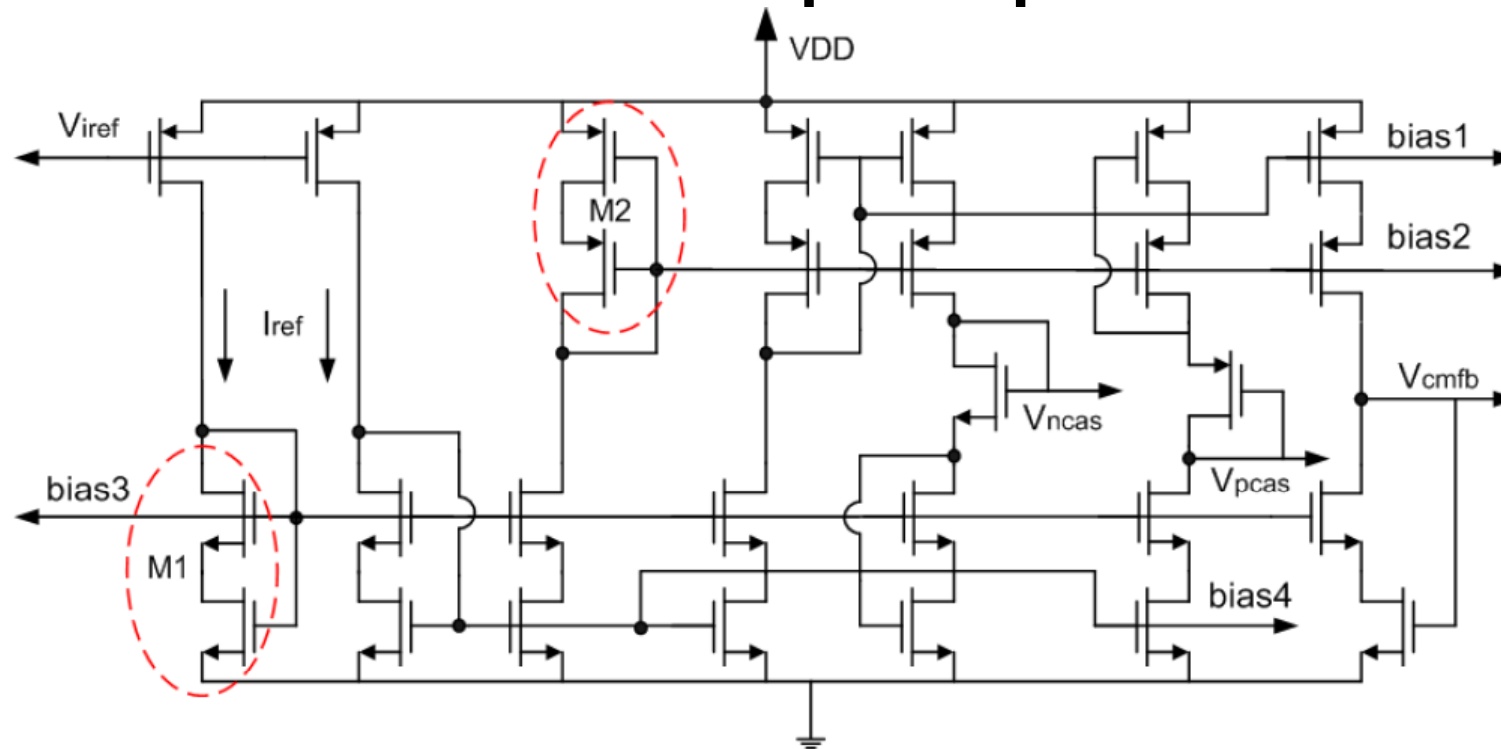
The first stage: Folded-cascode for larger input range & voltage gain

The second stage: inverter structure for expanding driving current

The static current of the 2nd stage is defined by trans-linear loops

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Class-AB opamp bias



Trans-linear loops define the static biases of the class-AB stage

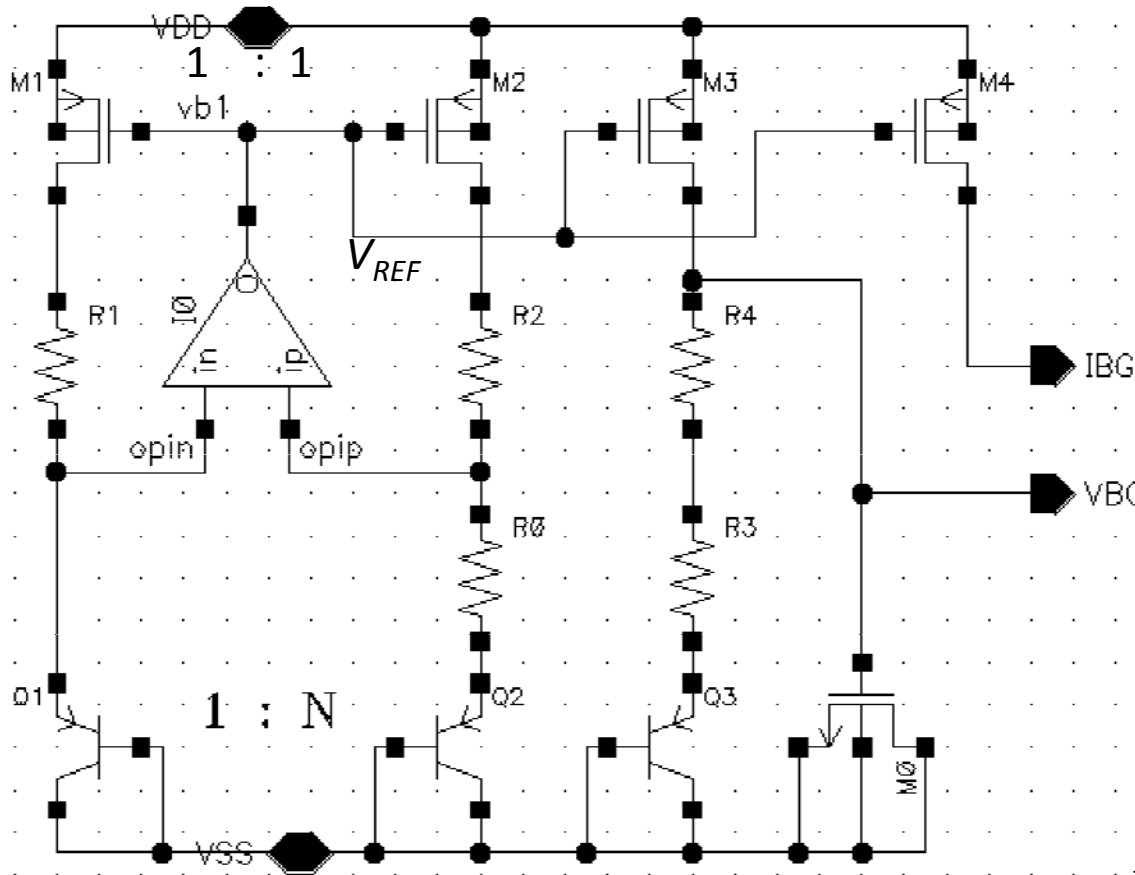
M1 & M2 split into two in order to suppress the channel-length modulation

The lowest Supply voltage?

Bandgap设计目标

- 提供与温度系数尽可能小的电压基准
- 尽可能大的电源电压范围
- 尽可能小的静态工作电流
- 尽可能大的PSRR
- 尽可能小的输出分布范围和Noise
- 具有可靠的启动电路
- 尽可能小的面积

Bandgap 1



基本的bandgap电路结构,提供对温度不敏感的基准电压, 以及PTAT基准电流

工作原理:

$$\Delta V_{BE} = V_T \ln n$$

$$\frac{\partial \Delta V_{BE}}{\partial T} = \frac{k}{q} \ln n > 0$$

$$V_{REF} = V_{BE1} + \frac{\Delta V_{BE}}{R_0} \cdot R_2$$

$$\frac{\partial V_{REF}}{\partial T} = \frac{\partial V_{BE1}}{\partial T} + \frac{R_2}{R_0} \cdot \frac{k}{q} \cdot \ln N = 0$$

$$R3=R0, R4=R2, M3=M2, Q3=Q2, \text{ So } V_{BG}=V_{REF}$$

考虑版图, N可取8

运放的失调必须控制好

具体设计请参考sansen教材第16章

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李福乐, 清华大学微电子所, 北京, 中国

lifule@mail.tsinghua.edu.cn

基准电压与电流

$$V_{REF} = V_{BE1} + \frac{\Delta V_{BE}}{R_0} \cdot R_2$$

$$V_{REF} = V_{BE1} + \frac{kT}{q} \cdot \ln N \cdot \frac{R_2}{R_0}$$

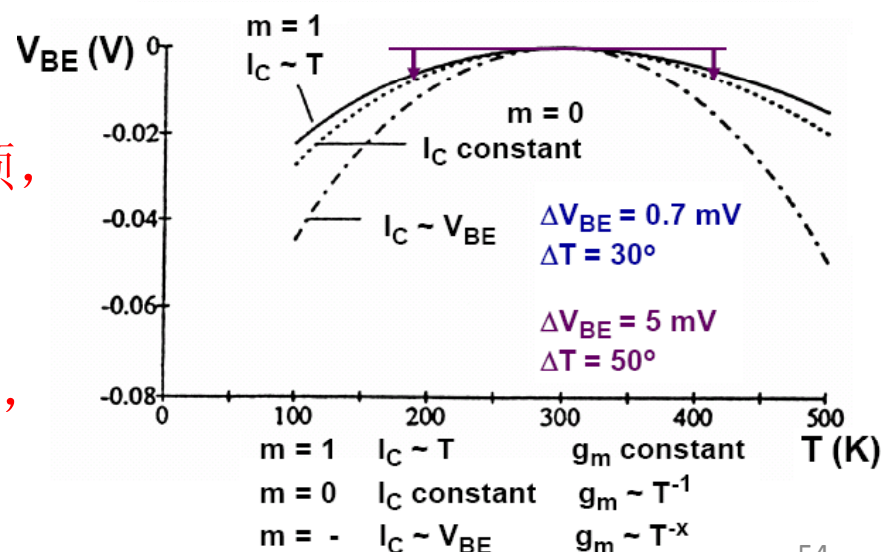
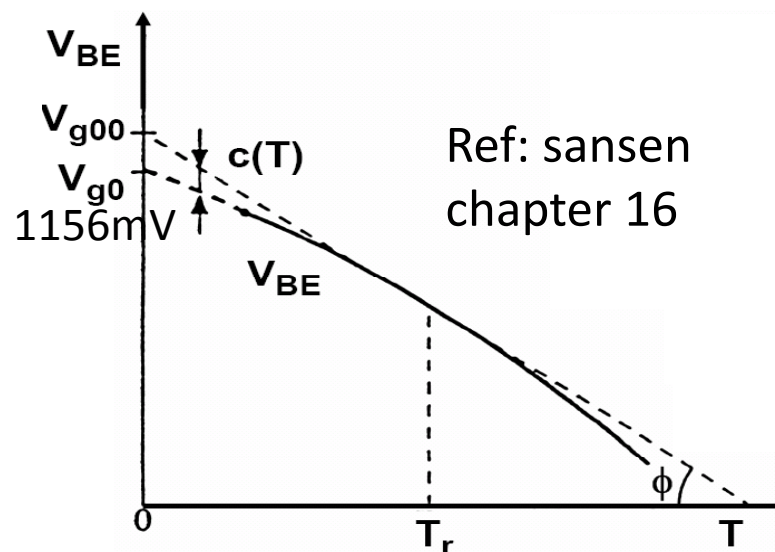
$$V_{BE} = V_{g00} - \lambda T + C(T) \quad \text{For } I_{DS} \sim T^m$$

$$I_{BG} = \frac{kT \cdot \ln N}{qR_0} \quad \text{Is PTAT}$$

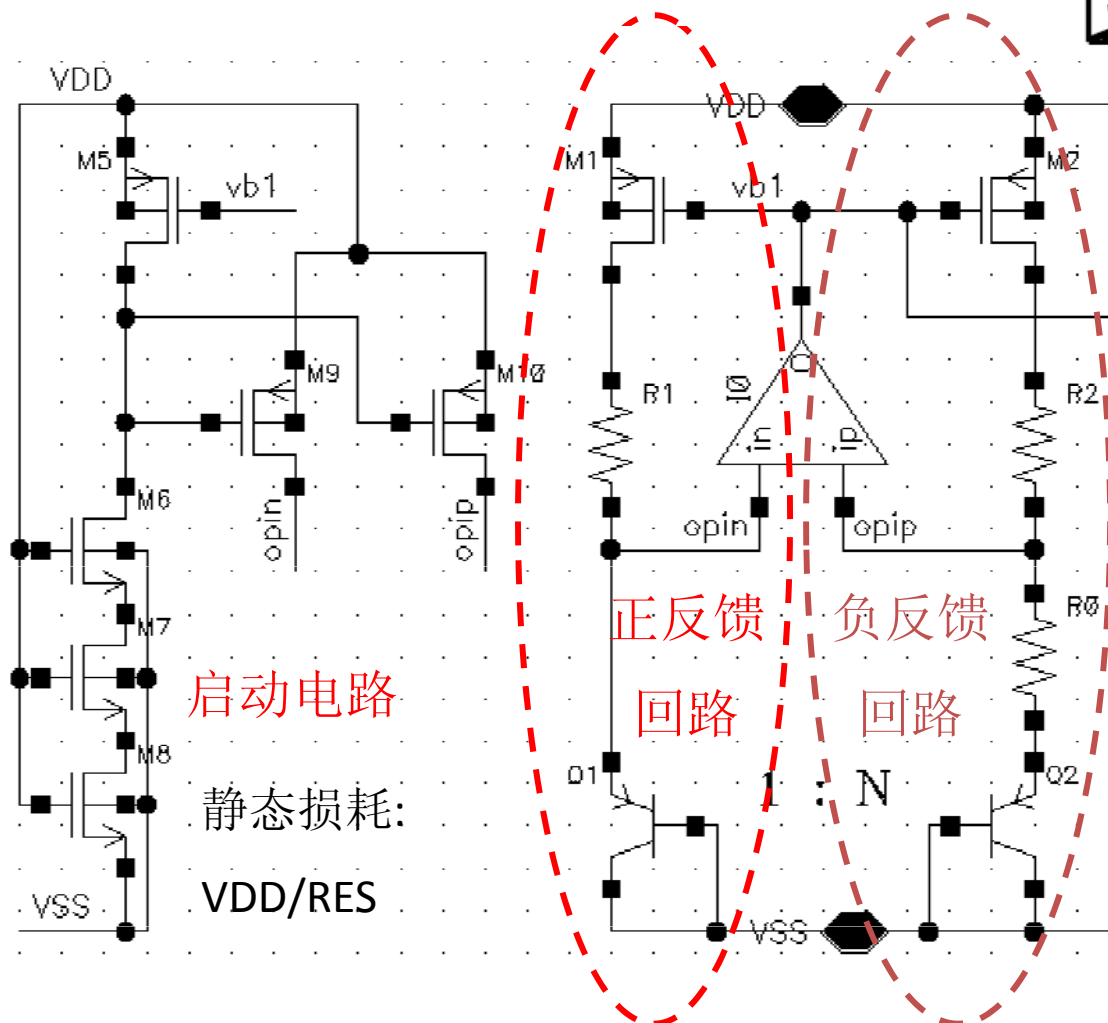
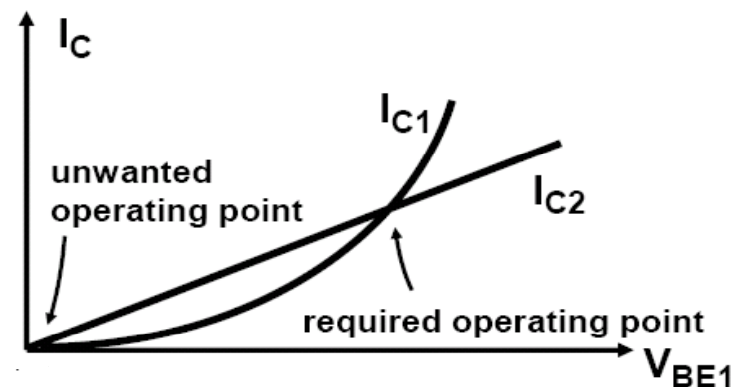
VBG通常在1.2左右

VBG的温度特性包含抛物线型非线性项，设计中令顶点位于常温附近，或采用曲率补偿电路

IBG~T，在高温与低温下其值变换较大，若用它直接去偏置其他电路，这一点须特别注意



启动与稳定性



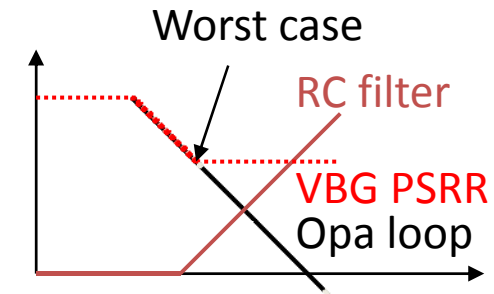
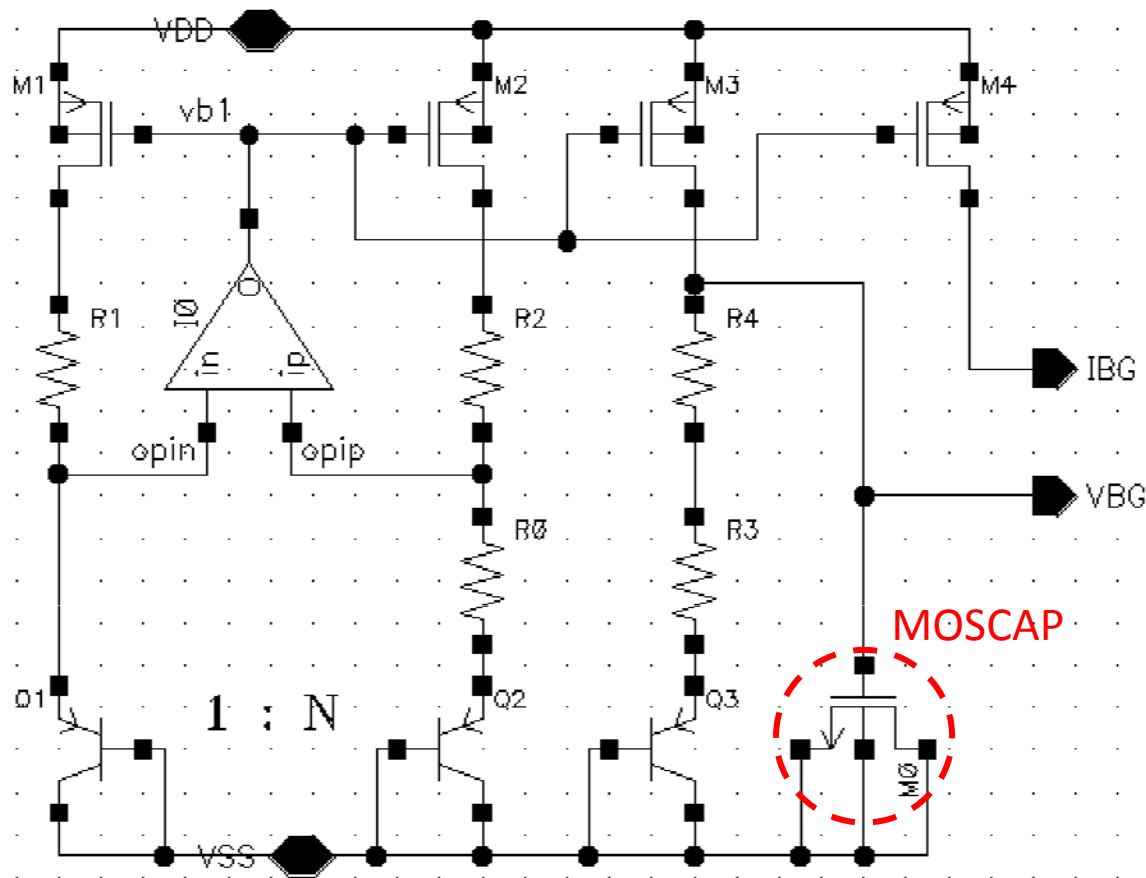
启动：这个电路有两个稳定工作点，其中一个是两个支路电流都为0，设计中需加启动电路来避免这个工作点；

稳定：正、负反馈环路共存，运放输入端需正确连接，保证负反馈强于正反馈

启动电路中的M9要大于M10，使进入正反馈回路的刺激大一些

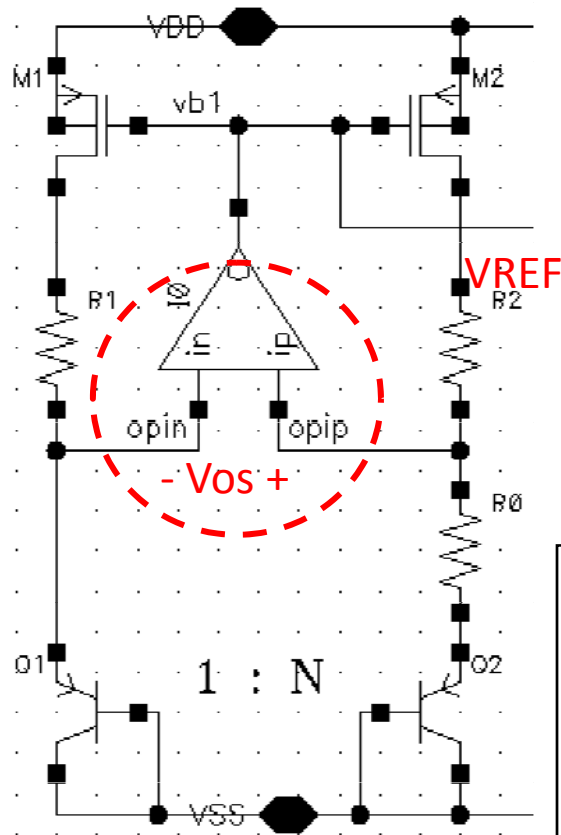
M6~M8作为MOS RES，其值应保证正常工作后M9, M10关闭

PSRR



低频：OPA环路增益保证
 高频：输出RC滤波保证
 最差点在中间，提高需要：
 提高环路带宽，与功耗
 tradeoff
 降低RC带宽，与面积
 tradeoff

电压离散性与噪声



抑制 $1/f$ 噪声还可采用chopping技术

$$\Delta V_{REF} \approx V_{os} \cdot \frac{R_2 + R_0}{R_0}$$

$$\sigma(V_{REF}) \approx \sigma(V_{os}) \cdot \left(1 + \frac{R_2}{R_0}\right)$$

例：由 $\frac{kT}{q} \cdot \ln N \cdot \frac{R_2}{R_0} \sim 2mV/^{\circ}C$

推得： $\sigma(V_{REF}) \approx \sigma(V_{os}) \cdot 10$

$$\sigma_{V_T} \approx \frac{A_{VT}}{\sqrt{WL}}$$

输出离散性：运放失调的控制很重要！

噪声：

*. 热噪声可由输出滤波电容来控制；

*. $1/f$ 噪声：与 V_{os} 一样会被放大, 且很难被滤波, 因此要特别注意电路中的NMOS; 运放输入管应用PMOS, 与失调要求一样, 要增加 W, L

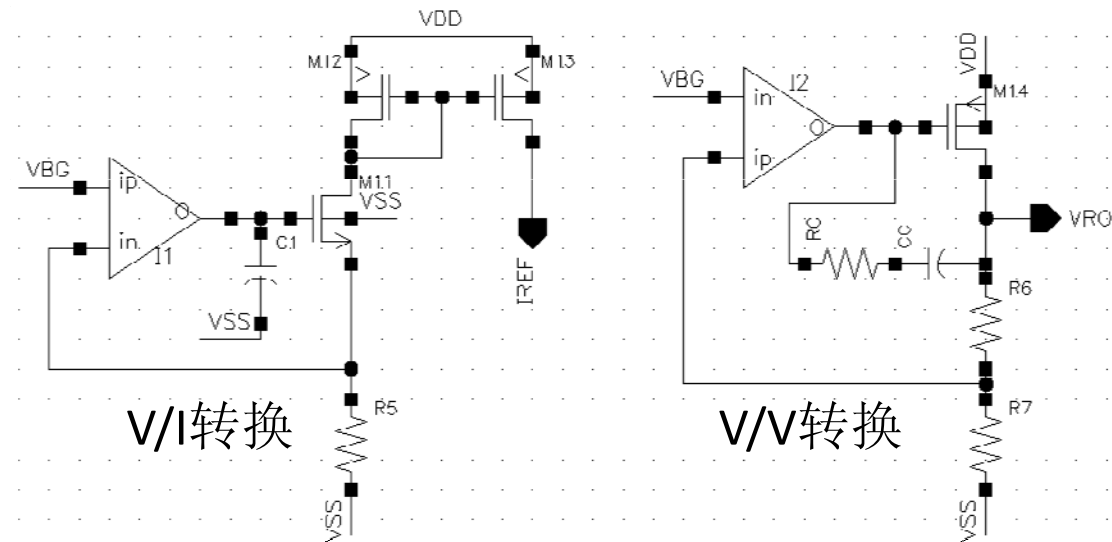
局限性与解决办法

- 输出基准电压固定为1.2V左右, 不够灵活
- 输出基准电流为PTAT, 在全温度范围内变化很大
- 电源电压的限制
 - 最低工作电压 $>1.2+V_{dsat}$

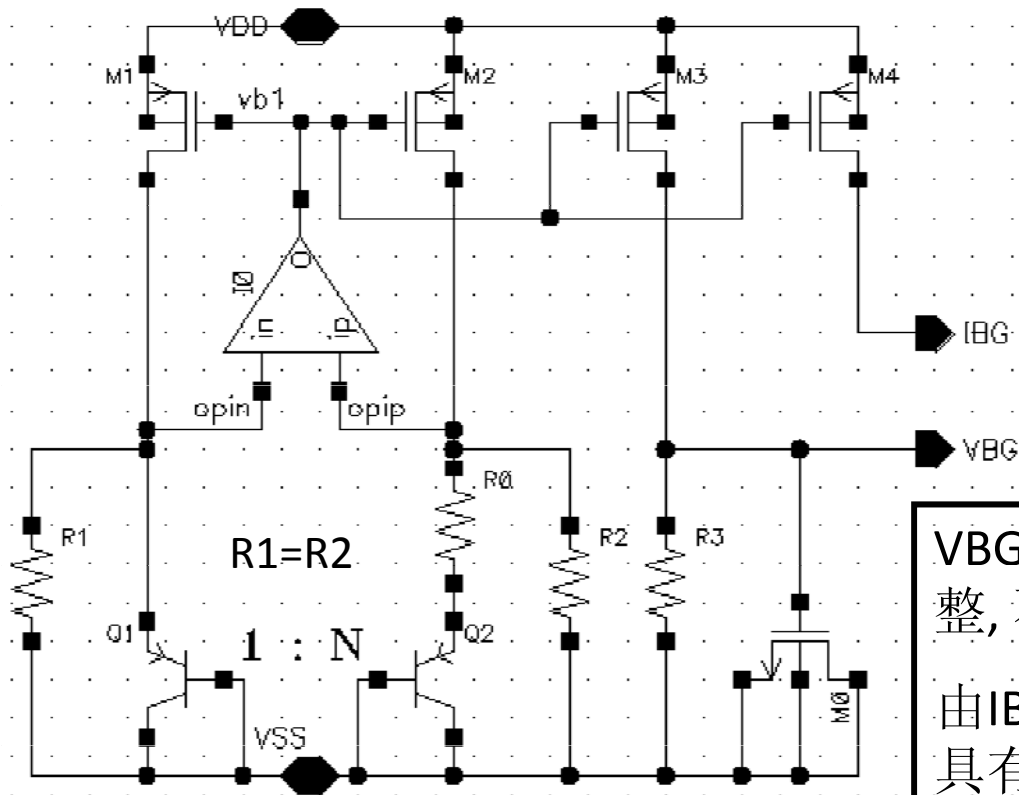
Solution:

*. 通过V/V, V/I转换电路来解决前两个问题, 代价是电路复杂性和功耗

*. 或采用下面的bandgap2, 解决上述各问题



Bandgap 2



$$V_{BG} = \frac{R_3}{R_2} \left(V_{BE} + \frac{R_2}{R_0} \cdot \frac{kT}{q} \cdot \ln N \right)$$

$$\frac{\partial V_{BG}}{\partial T} = 0$$

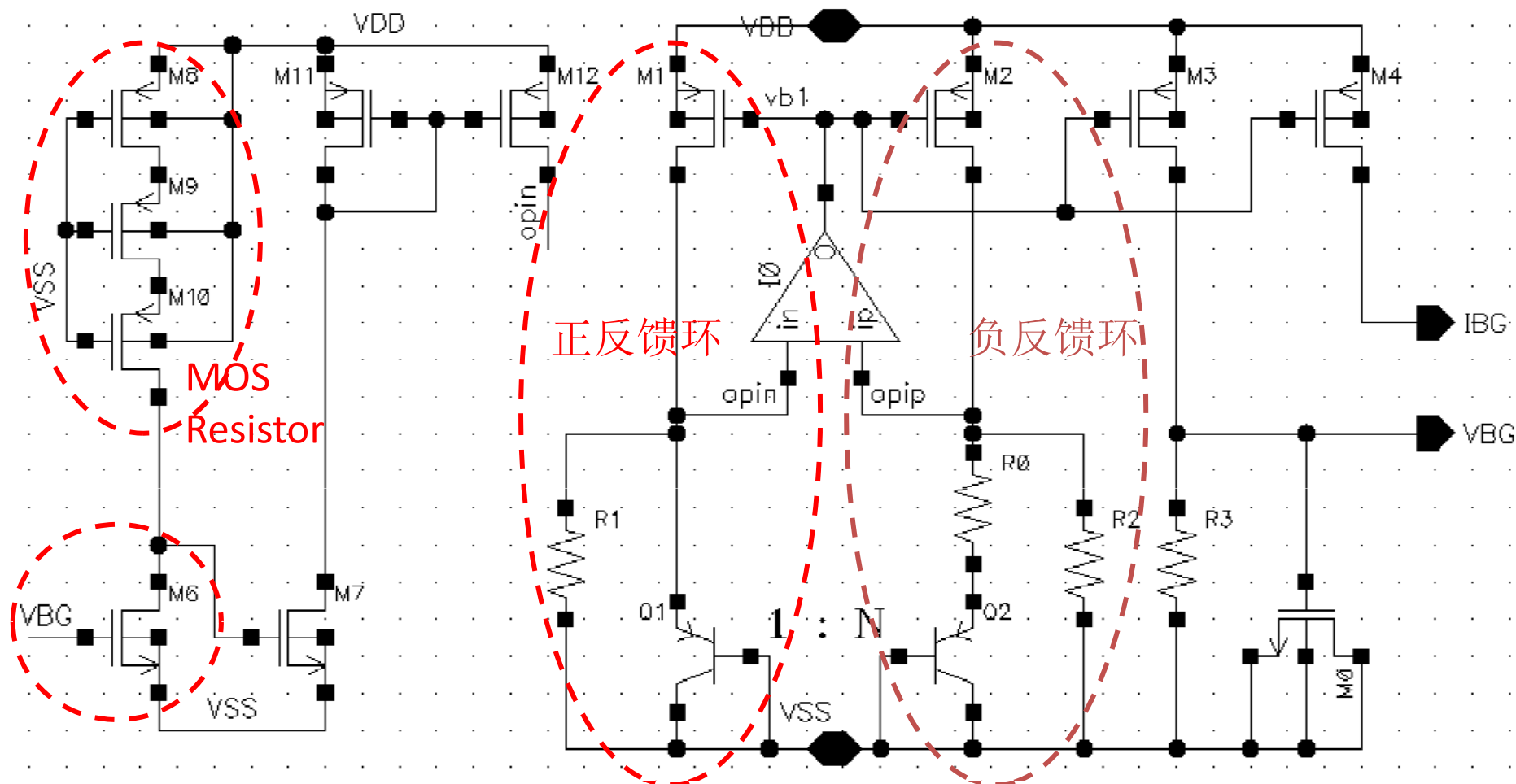
VBG的大小可通过R3/R2来灵活调整, 不影响温度系数

由 $I_{BG}=V_{BG}/R_3$, 可知输出电流参考具有与电压参考一样的稳定性

最低电源电压可 $<1V$

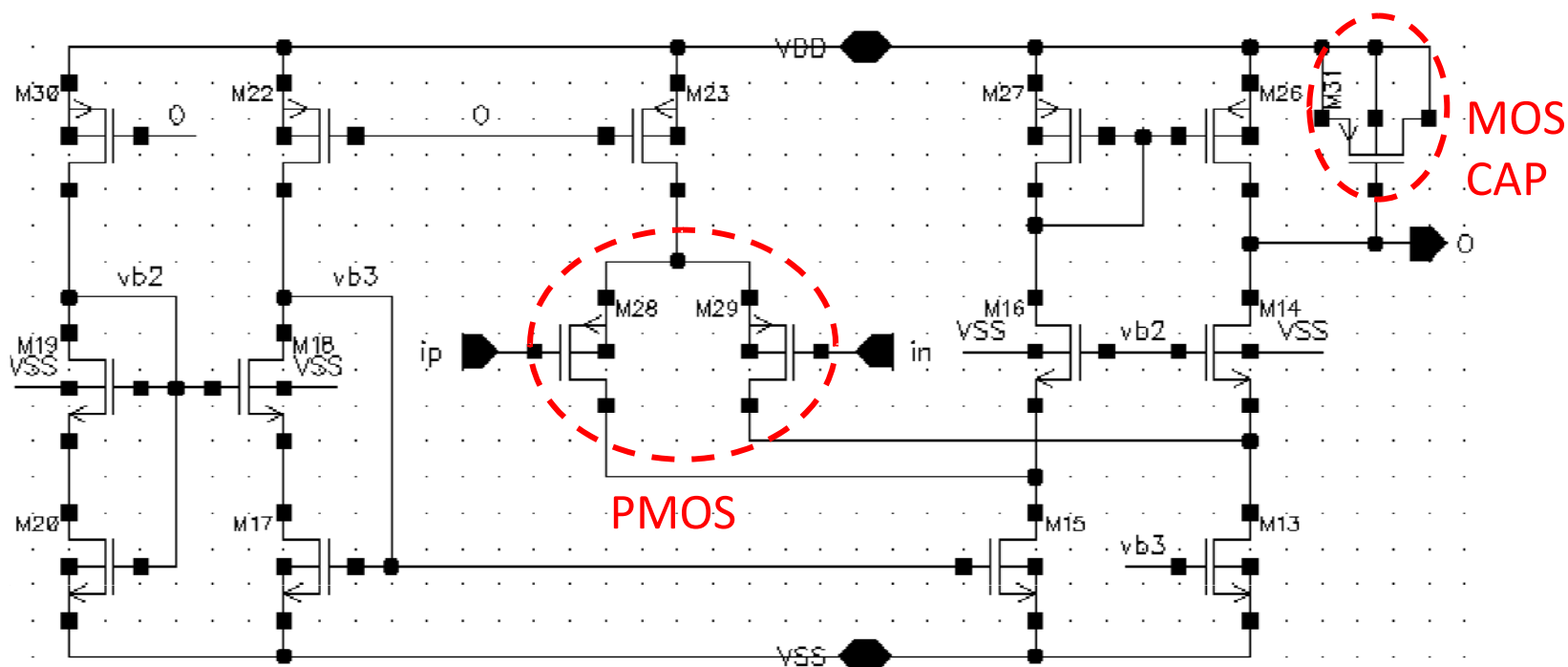
Ref.:
Malcovati, JSSC
July 01, 1076-1081

启动电路



正常输出下 $VBG > V_{Tn}$, 否则应采用其他形式的启动电路

自偏置运放电路

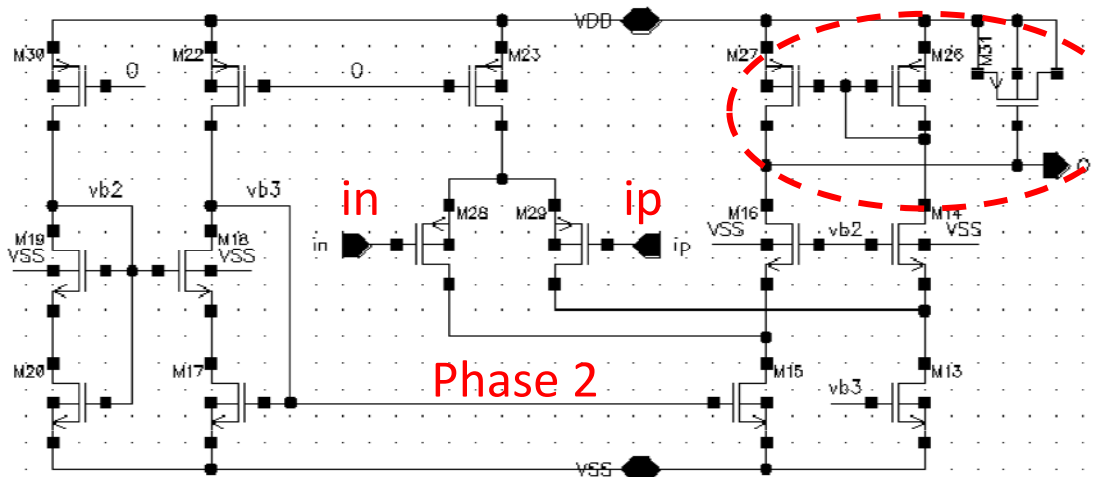
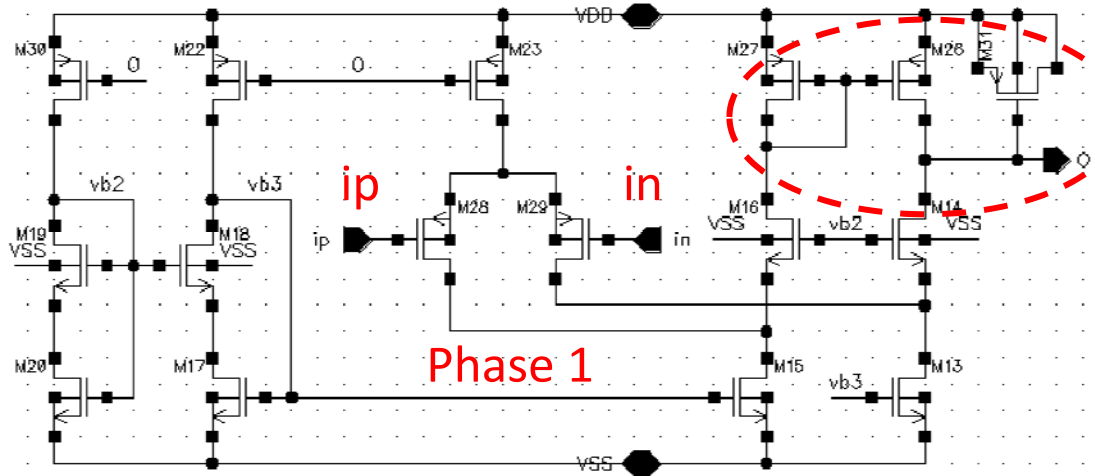


采用folded结构

采用PMOS输入差分对, 减少 $1/f$ 噪声, 并适合于低输入共模的情况

输入差分对 $V_{GS}-V_T$ 小, 其他电流镜 $V_{GS}-V_T$ 大, 晶体管的L要大一些, 减少失调, $1/f$ 噪声, 以及沟道调制效应

Chopping & filtering



加chopping控制后:

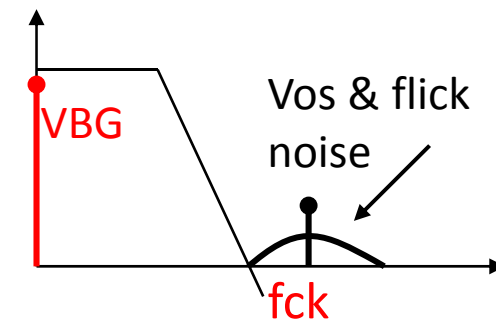
Phase 1: $VBG' = VBG + aVos$

Phase 2: $VBG' = VBG - aVos$

低通滤波后:

$avg(VBG') = VBG$

此方法对1/f噪声也有效



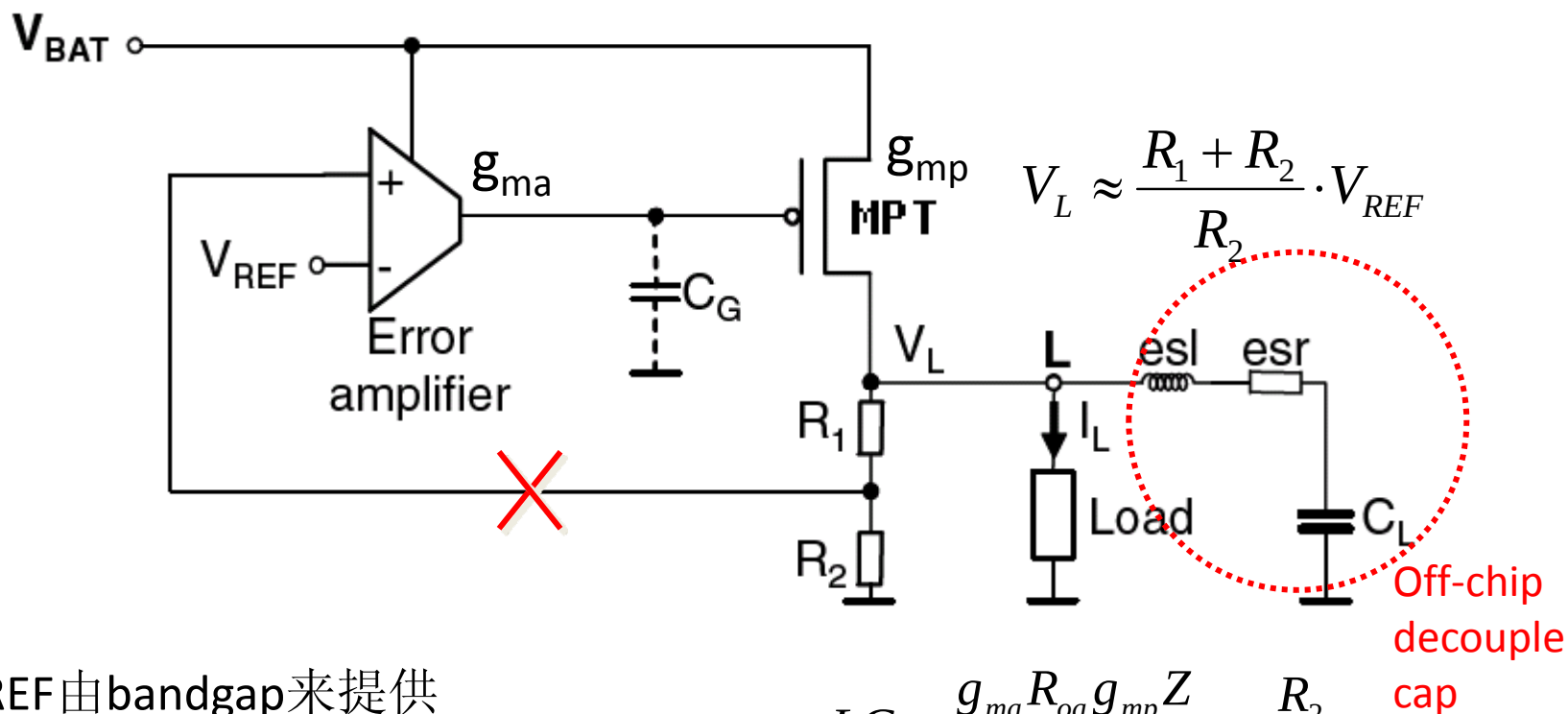
版图设计

- PNP之间, 电阻之间, 电流镜MOST要严格按照匹配要求来设计版图
- 运放的失调要特别注意控制, 这要求晶体管的WL要比较大, 版图绘制时注意匹配
- 给其他模块提供电压基准时要注意与这些模块的共地
- 在SOC中, 要注意与noisy block的隔离
- 面积优化

LDO设计目标

- 提供稳定的电源电压
 - 无论环境变化，输入变化，负载变化
 - TC, PSRR, Load Regulation, Transient Response
- 尽可能小的静态工作电流
- 尽可能小的输入输出Dropout
- 尽可能大的环路增益和带宽
- 在各种corner和负载条件下都必须稳定
- 具有过流保护能力
- 减小对片外元件的要求，最好是不用片外元件

基本LDO电路



V_{REF} 由bandgap来提供

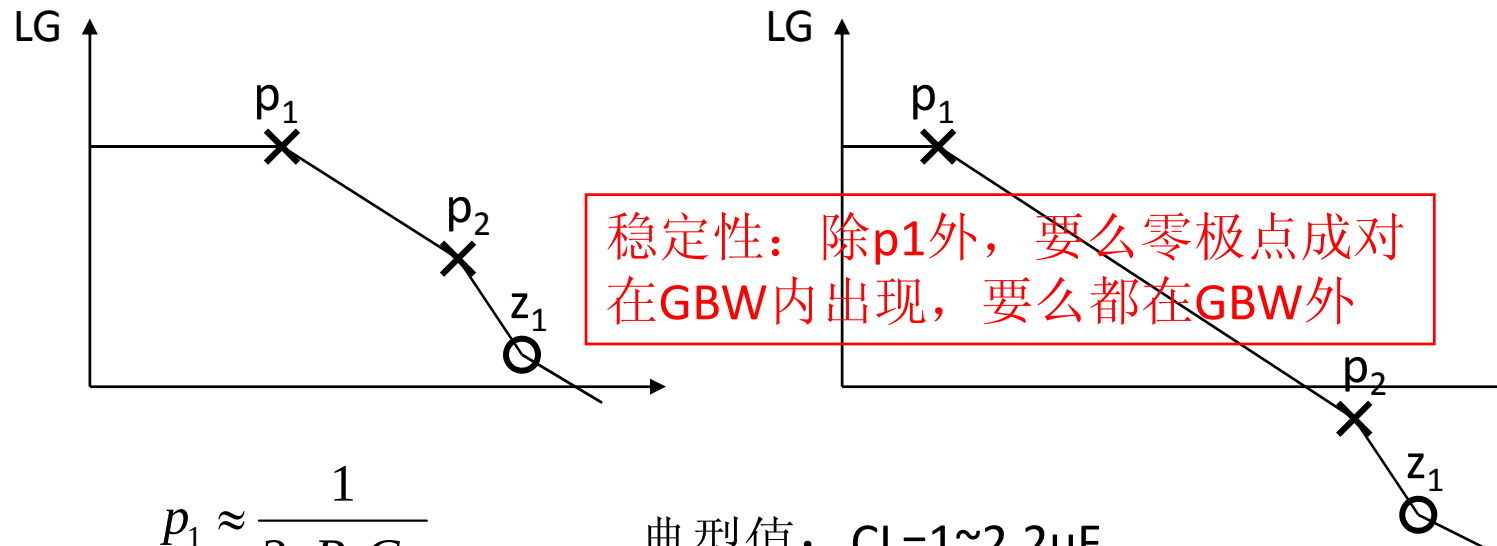
开环为无miller反馈电容的两级运放

环路增益LG为：

$$LG = \frac{g_{ma} R_{oa} g_{mp} Z}{1 + s R_{oa} C_G} \cdot \frac{R_2}{R_1 + R_2}$$

$$Z \approx \frac{1 + s R_{esr} C_L}{1 + s R_L C_L}$$

AC特性



$$p_1 \approx \frac{1}{2\pi R_L C_L}$$

$$p_2 \approx \frac{1}{2\pi R_{oa} C_G}$$

$$z_1 \approx \frac{1}{2\pi R_{esr} C_L}$$

典型值：CL=1~2.2uF

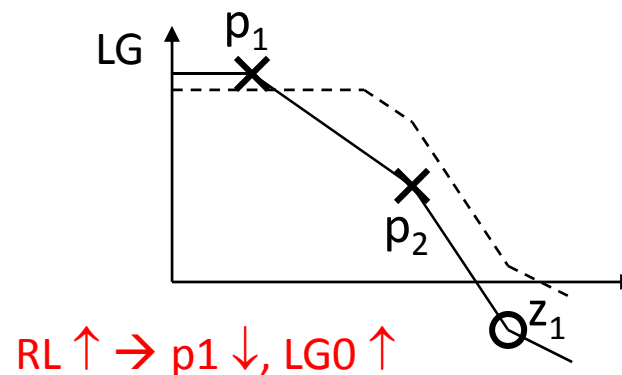
$R_{esr}=20\sim 200m\Omega$

$C_G \sim C_{ox} * (WL)_{MPT}$

增大p2，意味着减小Roa，而第一级增益为 $g_{ma} * R_{oa}$ ，为维持增益将使得error amplifier的静态功耗增大

基本LDO的局限性

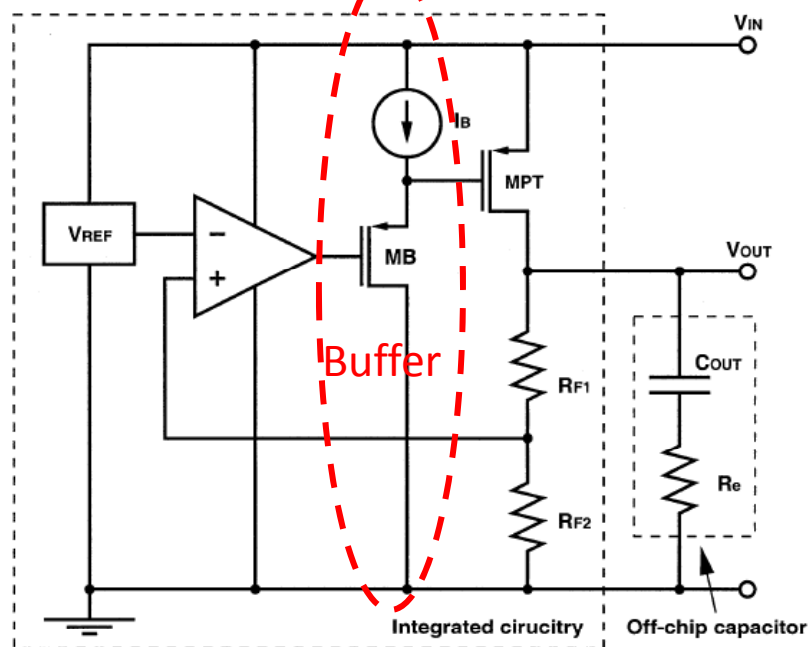
- 由于Error amplifier直接驱动MPT，其负载较大，对应的静态功耗也较大
- Error amplifier的增益较小， p_2 与 z_1 的间距较大，这两者都导致GBW变小，瞬态响应速度变慢；
- 在负载 RL 变化时， p_1 相应变化，可能出现 p_2 在GBW内， z_1 在GBW外的情况，导致不稳定
- 解决办法：加buffer来降低error amplifier的负载



加中间级buffer的LDO

静态功耗

稳定性设计!



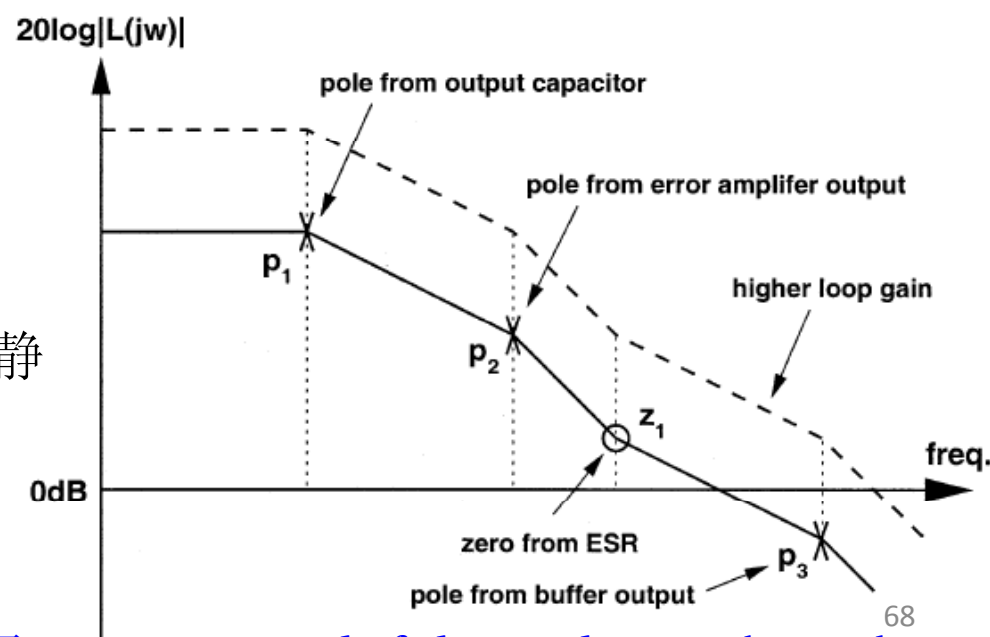
$$\text{Line Regulation} = \frac{\Delta V_{\text{OUT}}}{\Delta V_{\text{IN}}} \approx \frac{g_{mp} r_{op}}{L_o} + \frac{1}{\beta} \cdot \left(\frac{\Delta V_{\text{REF}}}{\Delta V_{\text{IN}}} \right)$$

$$\text{Load Regulation} = \frac{\Delta V_{\text{OUT}}}{\Delta I_{\text{OUT}}} = -\frac{r_{op}}{1 + L_o}$$

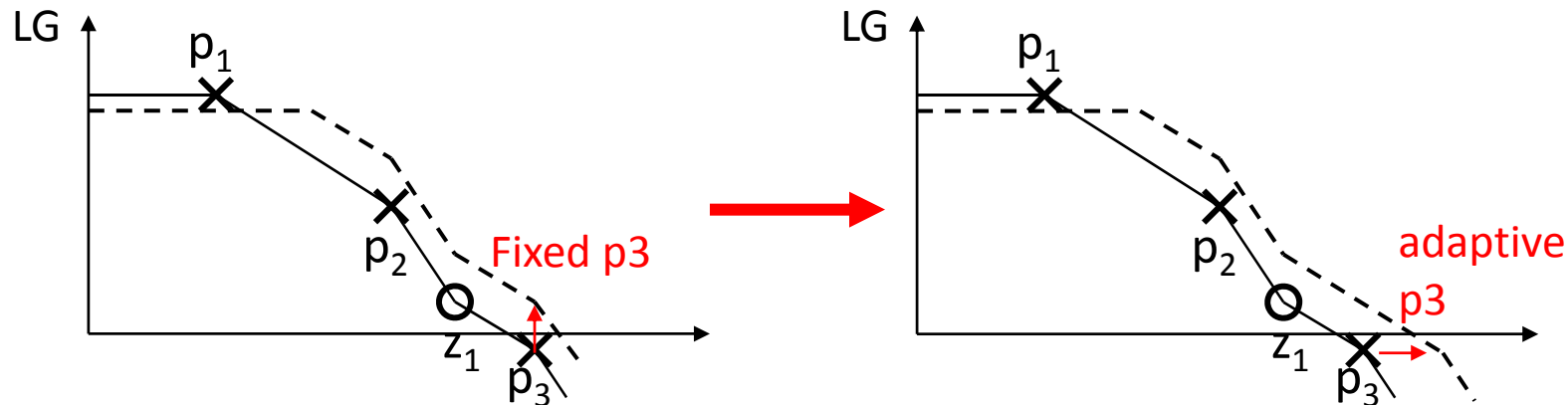
加buffer的好处:

1 减小err amp的负载电容, 降低其静态功耗, 更易使p2与z1接近

2 与MPT栅电容并联的是1/gm, not 1/gds, 更容易实现高极点



加buffer LDO局限性



Problem: 引入了新的极点 p_3 , 且要求 $p_3 > \text{GBW}$

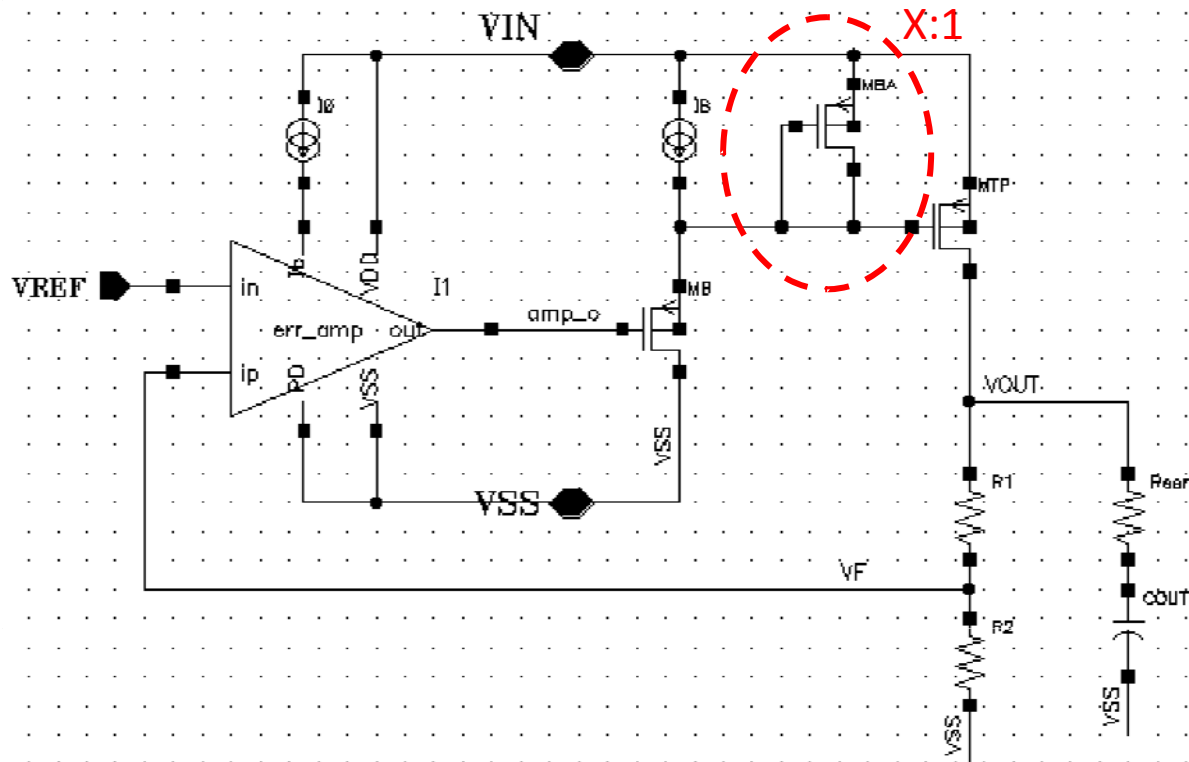
当 $RL \downarrow \rightarrow p_1 \uparrow, \text{GBW} \uparrow \rightarrow \text{unstable}$. 因此 p_3 应足够大

即buffer级的静态电流要足够大, 保证各种负载下的稳定性

这导致在小负载(RL 大)情况下, 效率较低;

Solution: adaptive bias for buffer

Adaptive bias for buffer



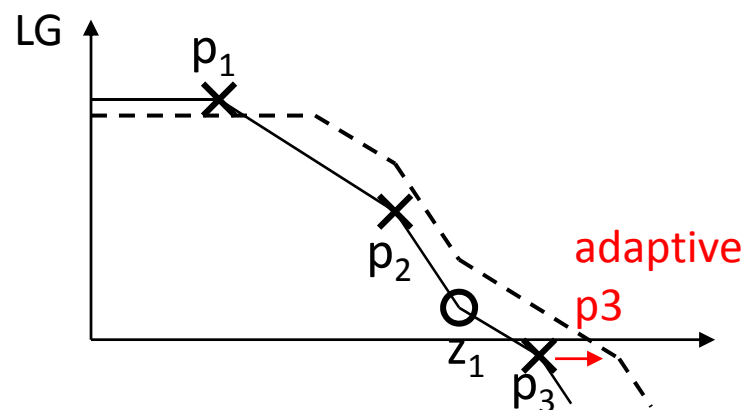
Buffer bias: $IMB = IB + IMBA = IB + X \cdot IMTP$ $IB = ?$ $X = ?$

$RL \downarrow \rightarrow IMTP \uparrow \rightarrow IMB \uparrow \rightarrow p3 \uparrow \rightarrow \text{stable}$

$RL \uparrow \rightarrow IMTP \downarrow \rightarrow IMB \downarrow \rightarrow \text{keep high efficiency at low load}$

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片外解藕电容



$$z_1 \approx \frac{1}{2\pi R_{esr} C_L}$$

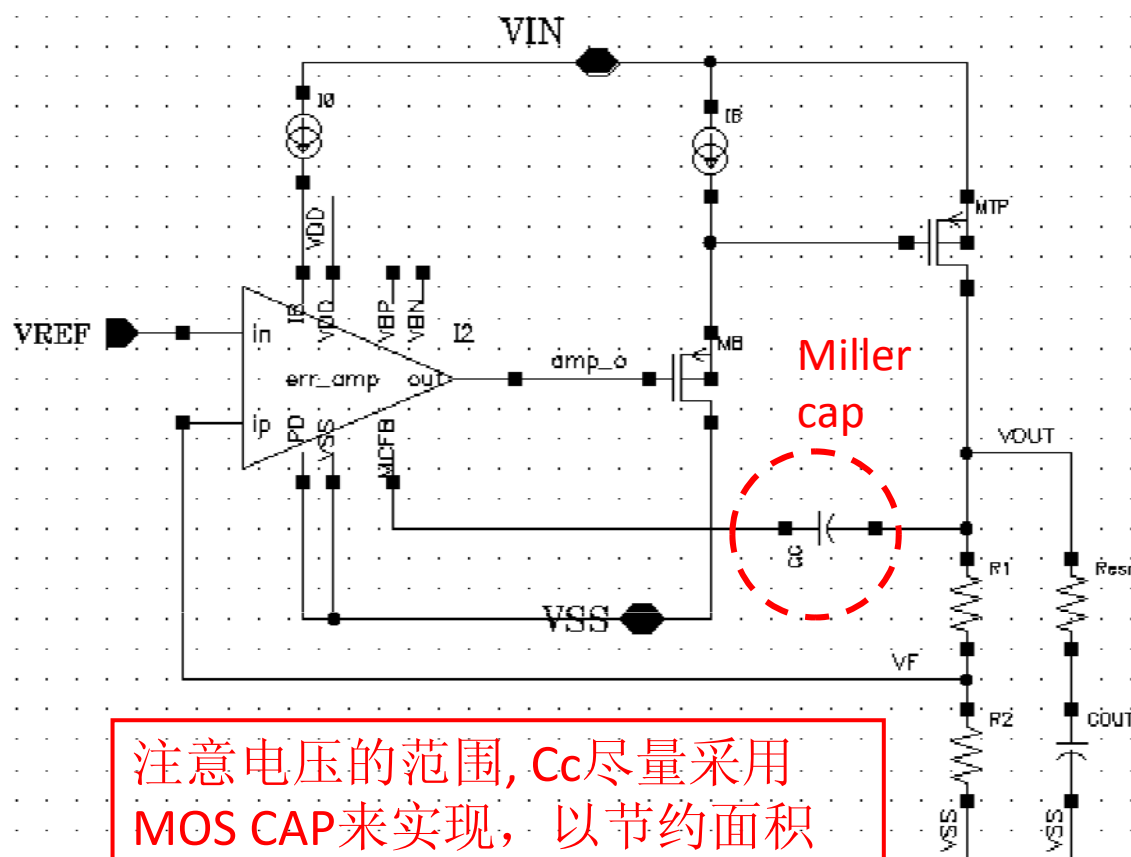
局限性：对片外电容的ESR有要求，若电容类型选择不当，可能导致不稳定！

Solution:

选择合适的片外解藕电容

或者，改进设计，如后面所示

带miller补偿的LDO



注意电压的范围, C_c 尽量采用 MOS CAP 来实现, 以节约面积或兼容逻辑工艺;

MTP 须工作在饱和区, 以保证 g_m , 保证稳定性

结构: 带有miller电容补偿的两级放大器

Err_amp采用folded cascode结构, 通过其cascode管减弱 C_c 的前馈

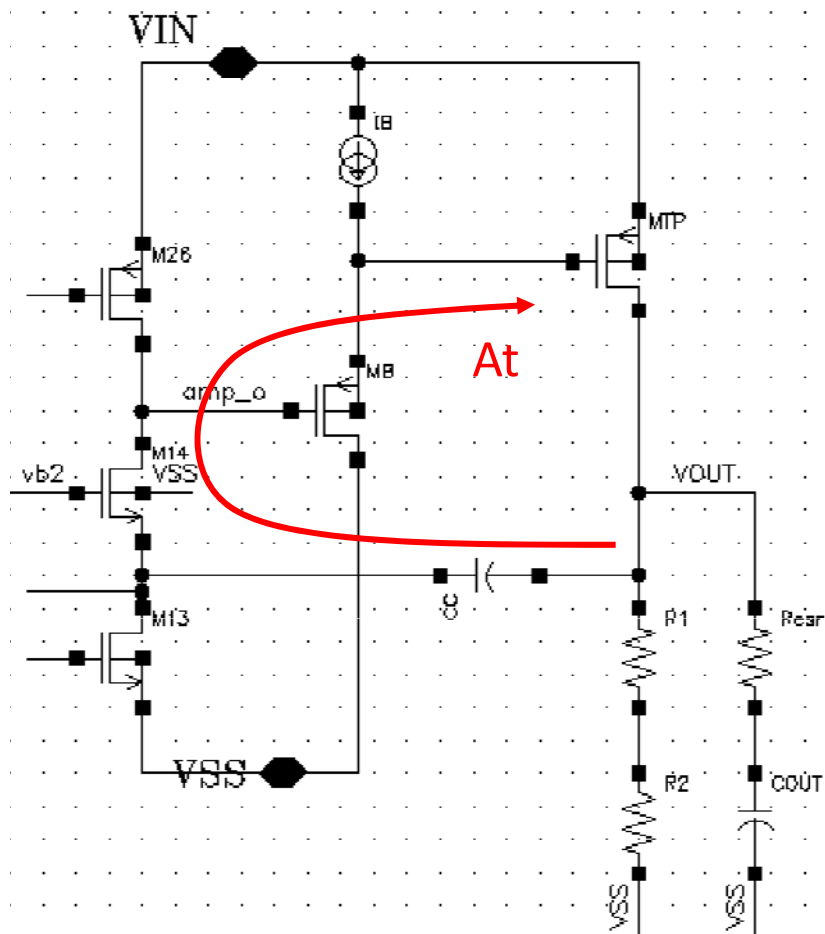
$$GBW = g_{m1}/C_c$$

$$P_2 = g_{m2}/C_{OUT}?$$

Problem:

C_{OUT} 很大, PM?

稳定性



假设从VOUT到晶体管MTP的G的增益为At，则：

$$P2 \sim At * gm2 / COUT$$

$At \sim Cc / Cn$, Cn为amp_o点寄生电容

注意到p2被推高了At倍，这就是这种结构的一个优点，对于低速电路很有用；

稳定性：

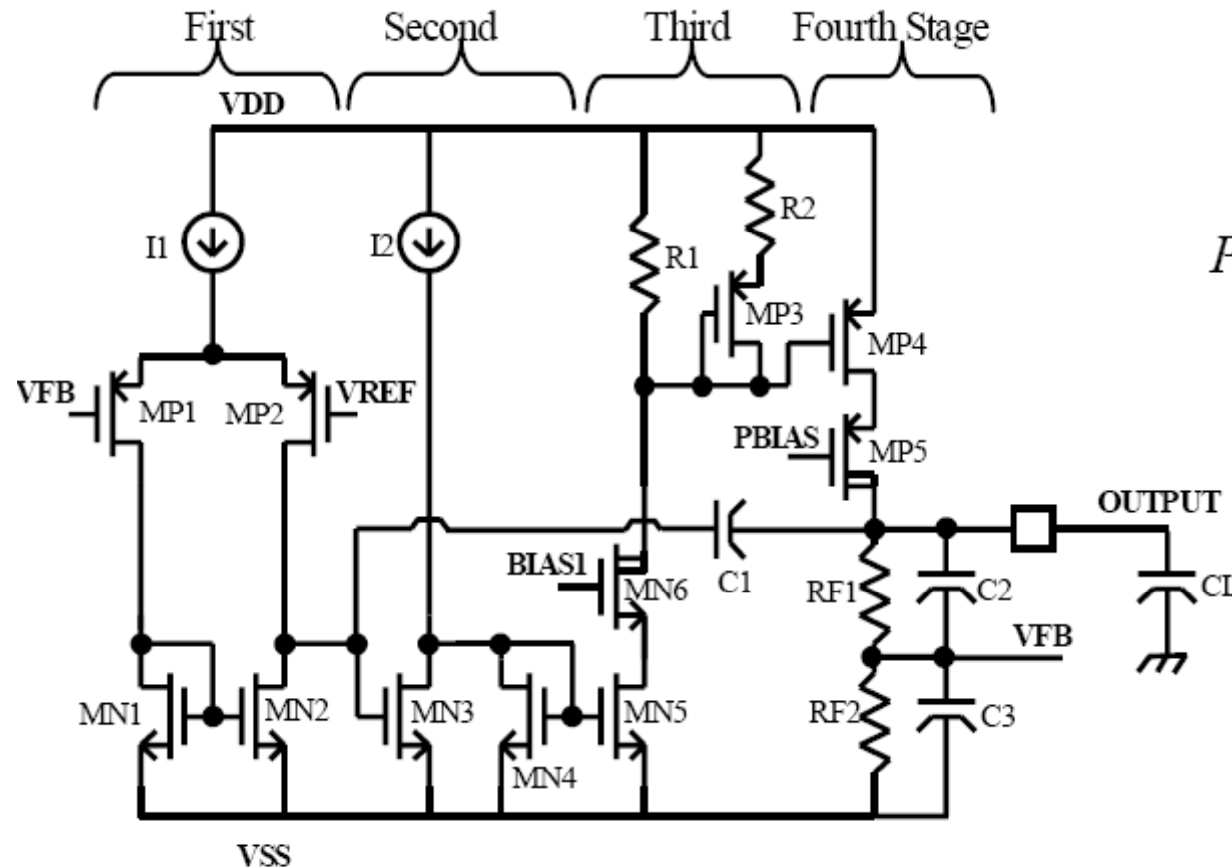
在IL足够小时，p1在VOUT处

在IL足够大时，miller补偿起到作用，p1在amp_o处，p2=?

当IL在两者之间时，PM=worst case?

稳定性不依靠片外电容ESR； 注意控制BW；

P2 boosting example



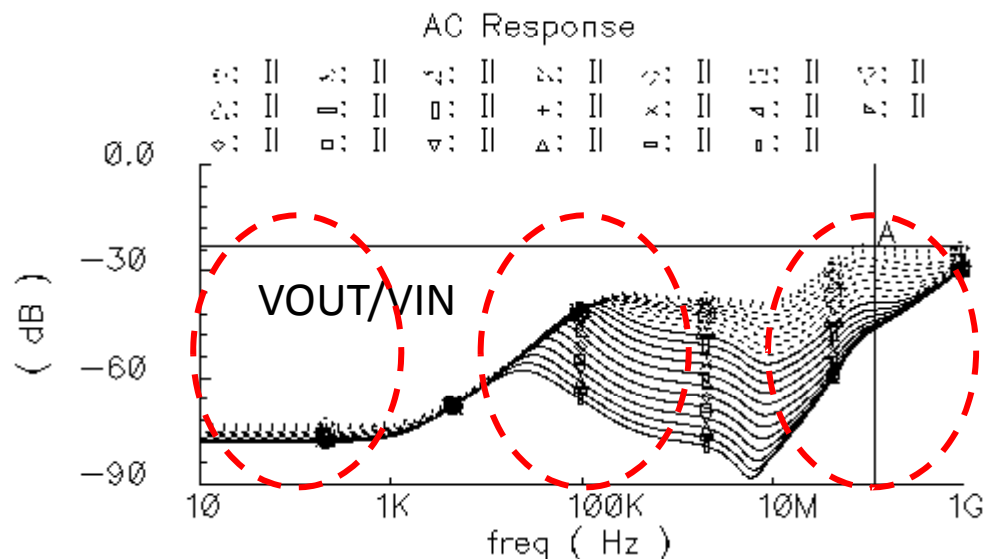
$$P_2 \approx \frac{gm_2''}{CL} \approx \frac{A_2 \cdot A_3 \cdot gm_4}{CL}$$

通过miller补偿和
miller boosting效应,
来推高p2, 保证LDO的
稳定性

Fig. 2. Simplified architecture of the LDO.

Ref: Kae Wong: A 150mA Low Noise, High PSRR Low-Dropout Linear Regulator in 0.13um Technology for RF SoC Applications

PSRR?



片外大解藕电容+片内小解藕电容
使得高PSRR在中高频段得以保持

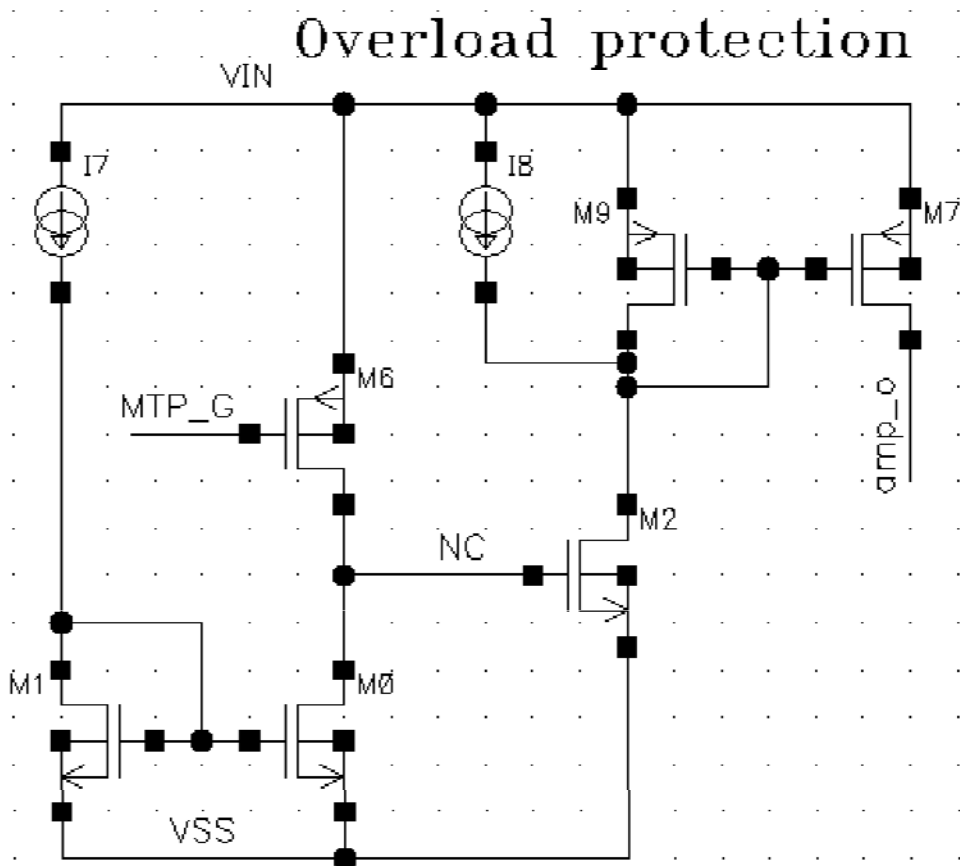
在低频处, 环路增益比较大,
对VIN上的干扰能有效抑制;

频率升高, 环路增益下降,
PSRR有下降的趋势;

由于片外解藕电容的作用,
在中间频率处, 仍可保持比较
高的PSRR;

在高频处, 环路反馈和片外电
容都不起作用, 此时依靠片内
解藕电容, 可保持高的PSRR;

过载保护



一个过载保护电路的例子

功能：当负载很大时，限制输出电流，以保护电路；

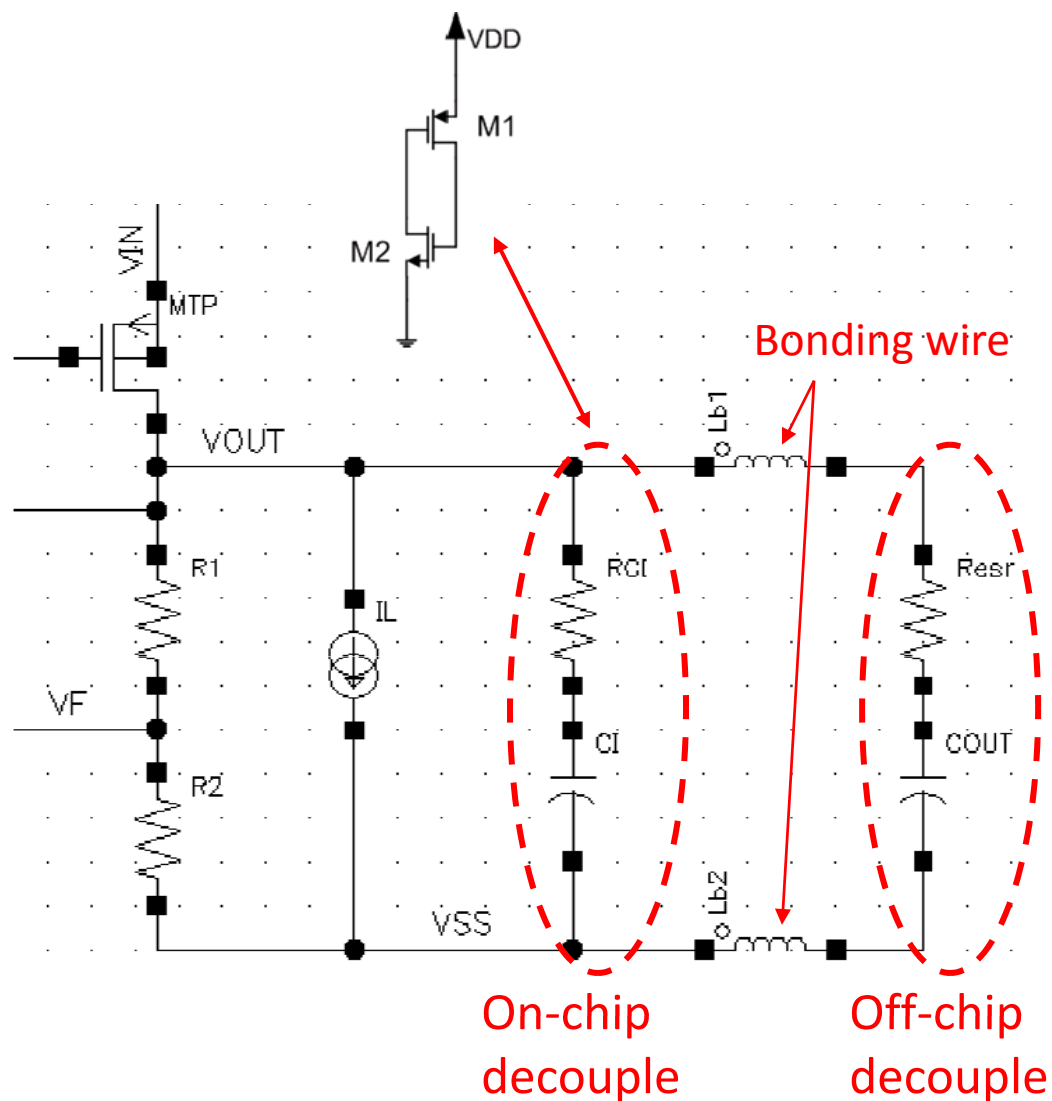
过载阈值应大于最大工作电流;

原理，以右图为例：

M6与MTP构成电流镜, 以检测输出电流

IDS6与设定的IDS0进行比较, 结果决定NC点的电压, 正常情况下, $IDS6 < IDS0$, NC点低, M7 G端高, M7 cutoff; 在过载情况下, $IDS6 > IDS0$, NC高, M7_G低, M7导通将amp_o拉高, 进而限制通过MTP的电流, 即用比较结果实现反馈控制

片内解藕电容



Bonding Wire使得片内高频扰动难以被片外电容滤除

一般要在片内加解藕电容

为防止出现LC震荡, 片内电容上要串联damping电阻

片内解藕RC支路是否会改变LDO稳定性?

$R1, R2 \gg R_{CI} > R_{esr}$

$C_{OUT} \gg C_I$

如: $R1, R2: x0k, R_{CI}: 10, R_{esr}: 0.1$

$C_I: 1nF, C_{OUT}: 1\mu F$

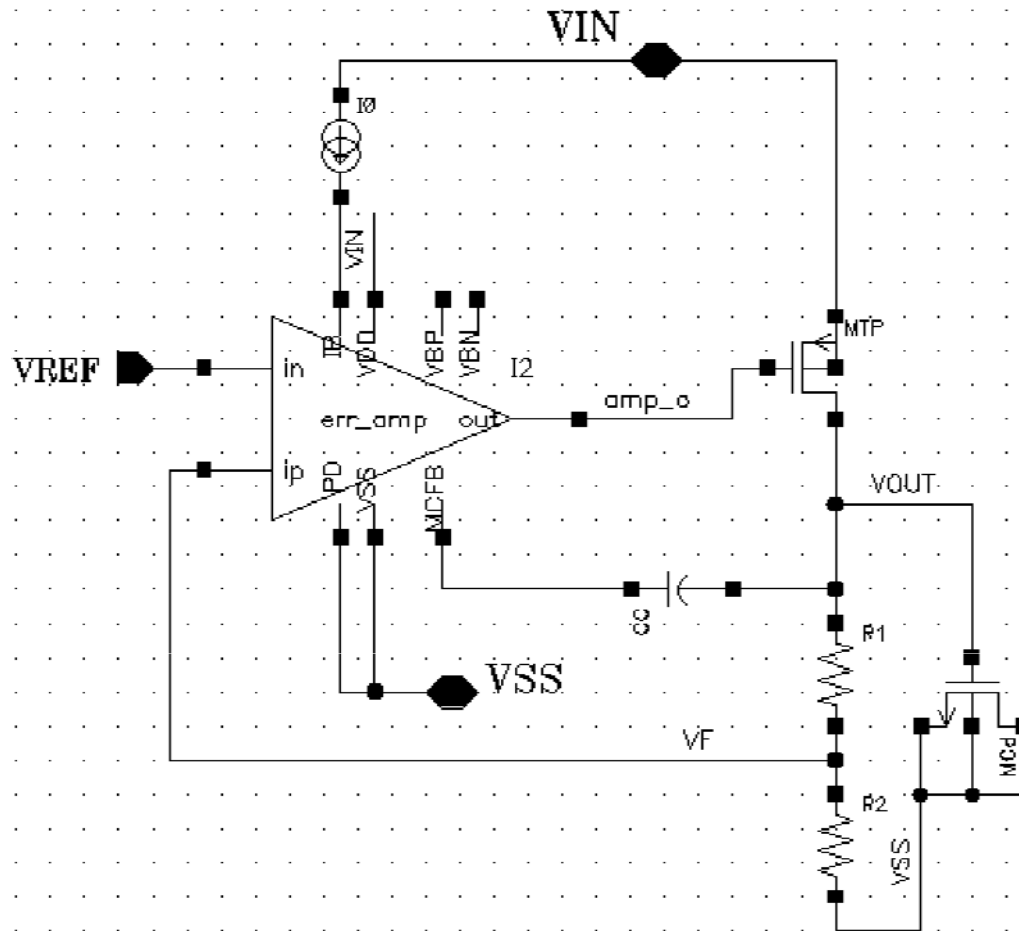
$$Z_O \approx \left(R_{CI} + \frac{1}{sC_I} \right) // \left(R_{esr} + \frac{1}{sC_{OUT}} \right)$$

$$Z_O \approx \left(R_{esr} + \frac{1}{sC_{OUT}} \right)$$

全集成LDO?

- 片外解藕的问题
 - 增加了片外元件, 增加了PCB面积
 - 增加了芯片面积, 封装管脚数
 - 一些应用, 比如卡, 不允许加片外电容
- 无片外电容LDO的实现?
 - 片内解藕电容?
- 下面的例子仅供参考, 有兴趣的话可自己去调试, 也可参考多级放大器 (见sansen教材chapter 11) 的设计方法来设计全集成LDO

Miller补偿结构



采用miller补偿实现极点分离:

$$BW = (gm1/Cc) * R2 / (R1 + R2)$$

$$P2 \sim (gm2/CL)*At$$

At $\sim C_c/C_G$, C_G 为MTP的栅电容

注意到片内解藕电容远小于 C_{OUT} ，故有可能以较小的静态功耗实现稳定；

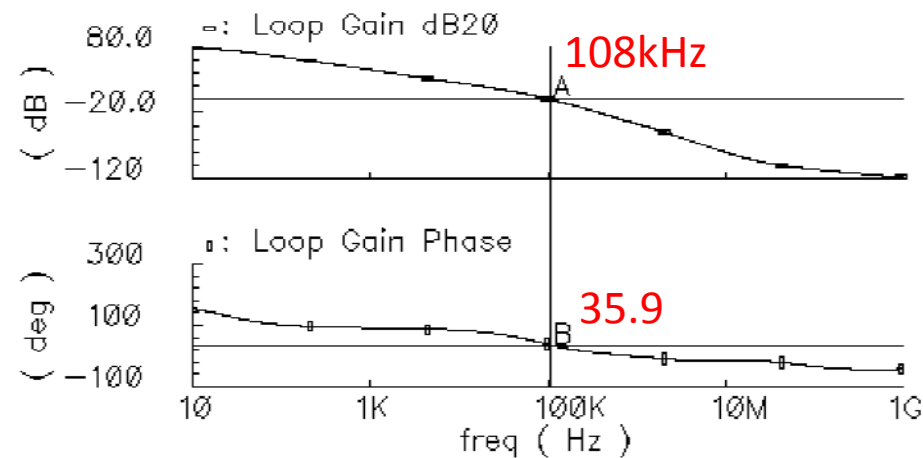
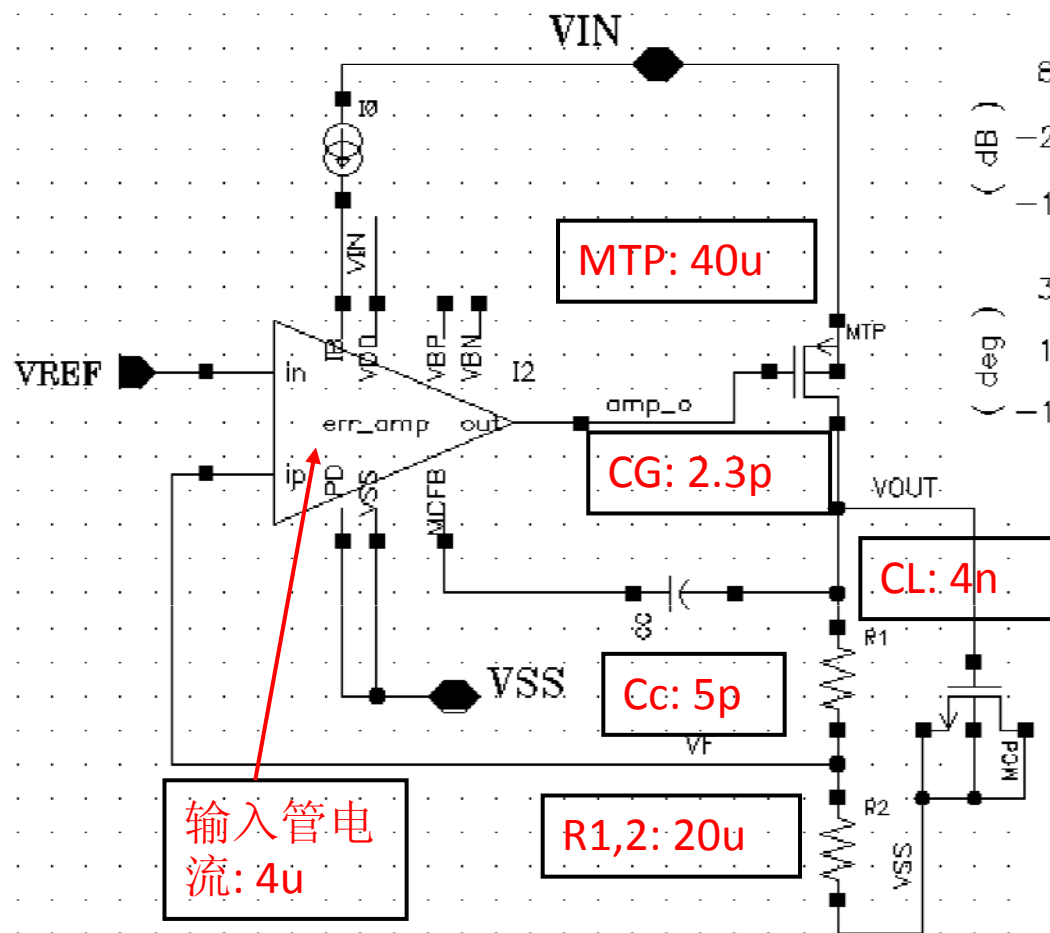
在负载较小时稳定性差;

Tradeoff:

带宽：减小Cc

稳定：增大Cc

特性与不足



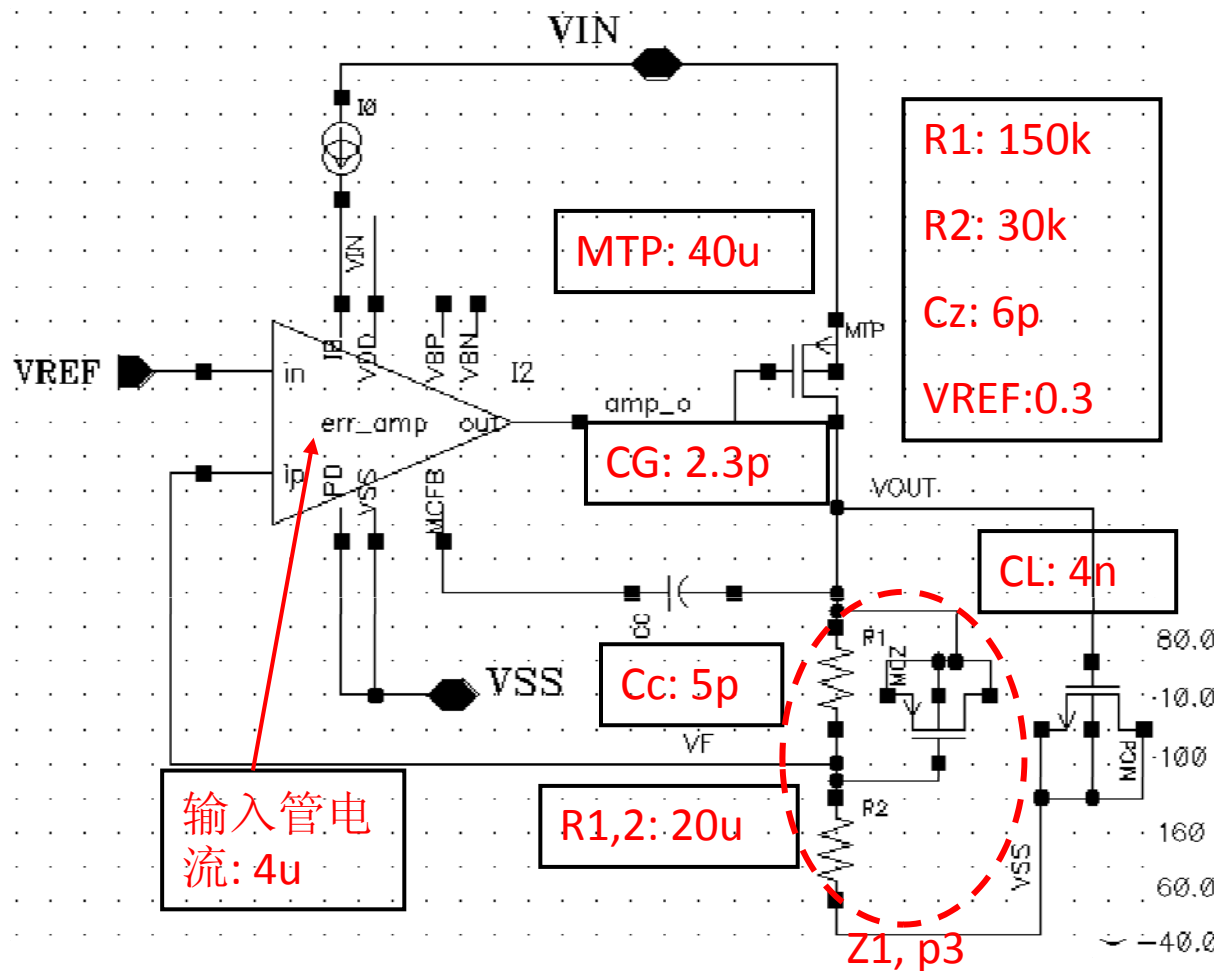
Problem:

IL小时, g_{m2} 小, PM太低!

Solution:

零点补偿!

零点补偿



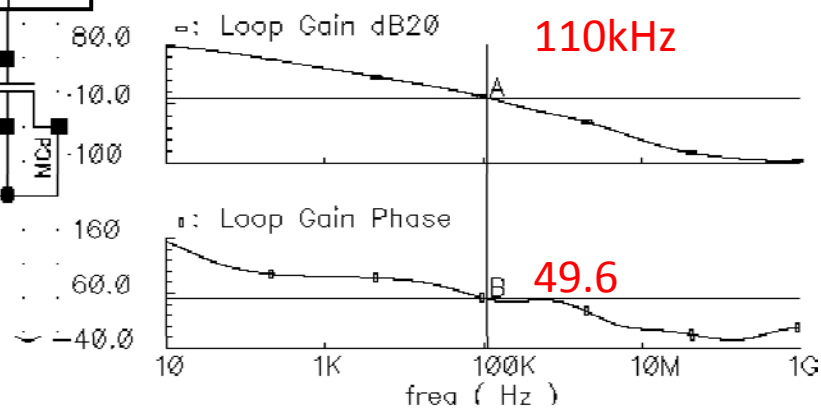
增加与R1并联的电容Cz

反馈网络引入 z_1, p_3

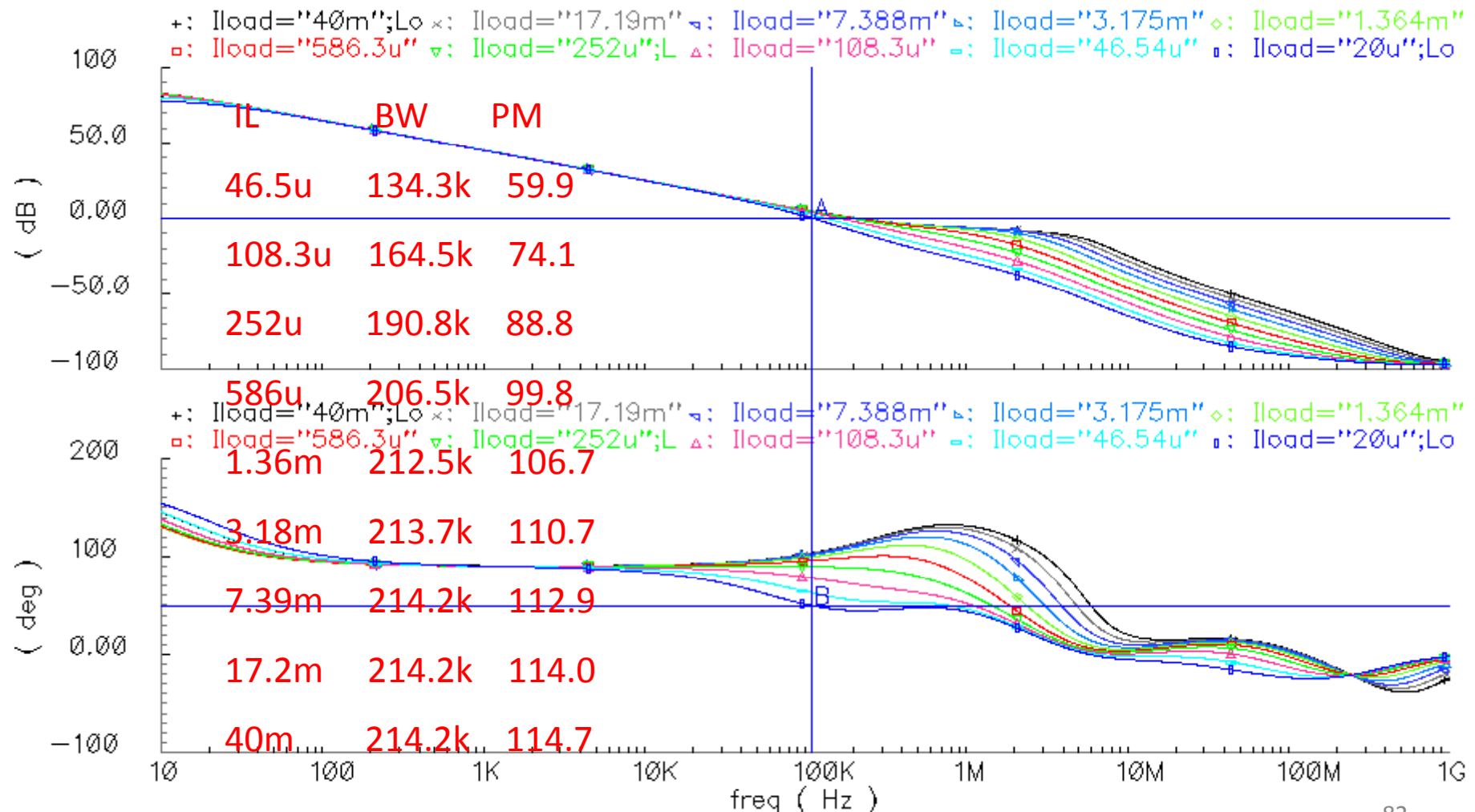
$$Z1=1/(R1Cz)$$

$$P3=1/(R2Cz)$$

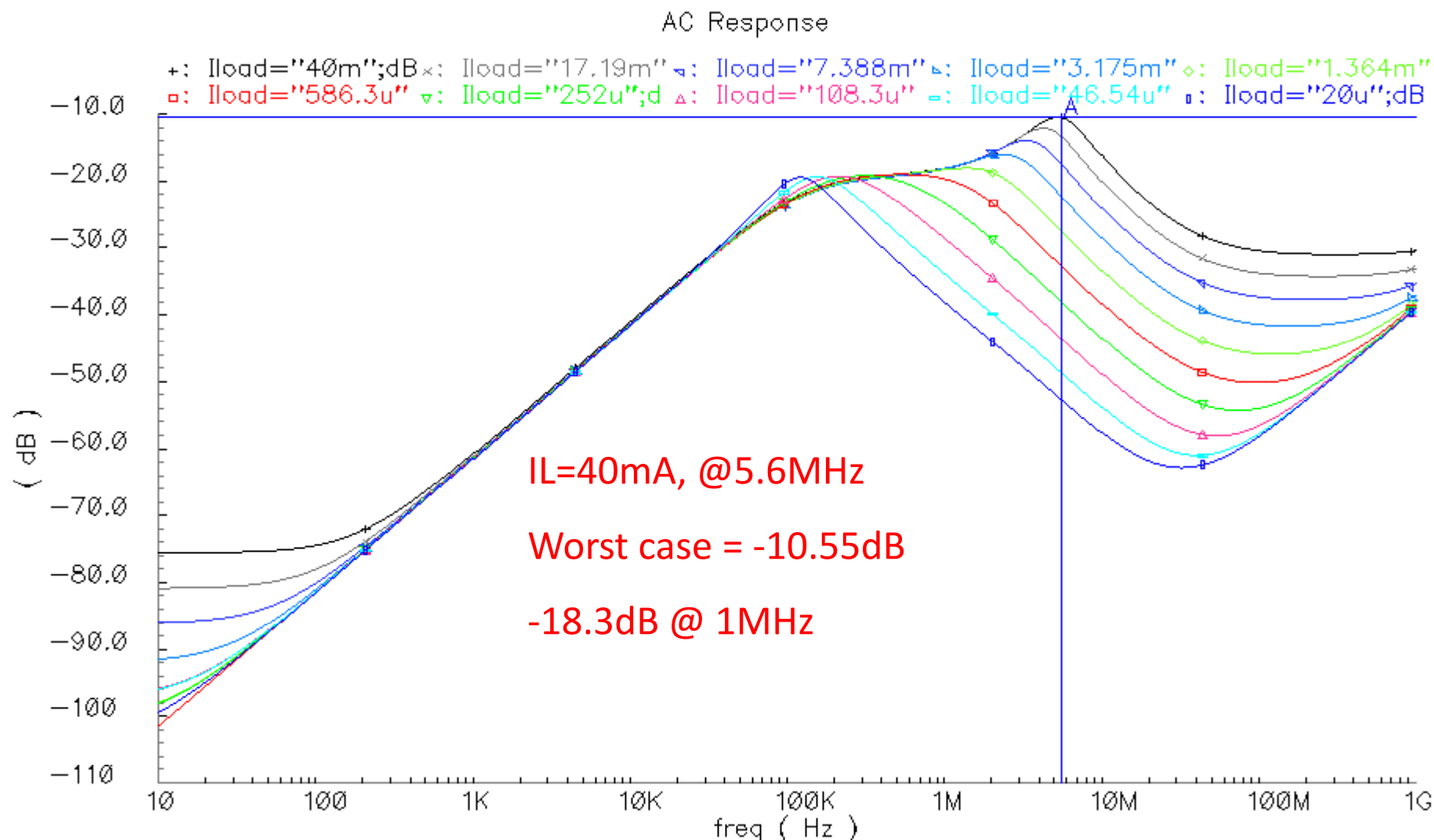
为使零点对PM起作用，令 $z_1 \ll p_3$, 故减小VREF, 增加R1/R2



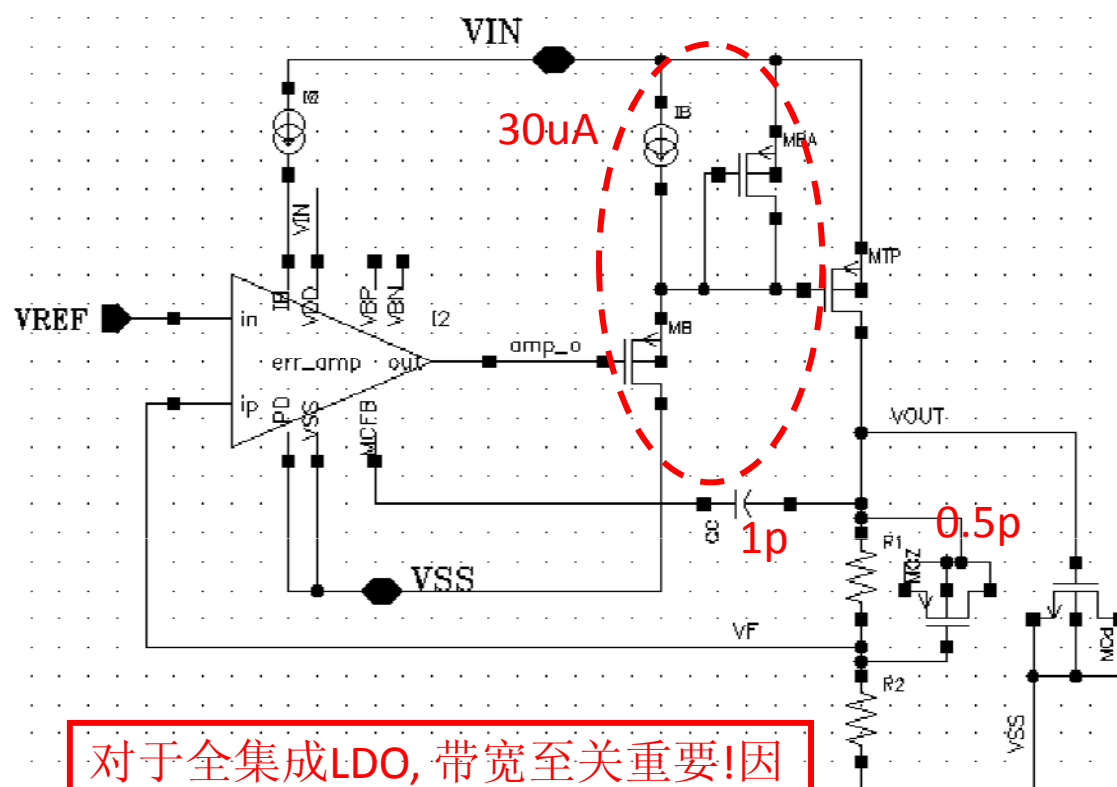
全负载稳定性



PSRR



带buffer结构



对于全集成LDO, 带宽至关重要! 因为此时无片外大解藕电容缓冲, 负载的变化需要环路自己迅速作出响应, 避免电压出现大的波形

P2 boosting:

$$P2 \sim (gm2/CL) * At$$

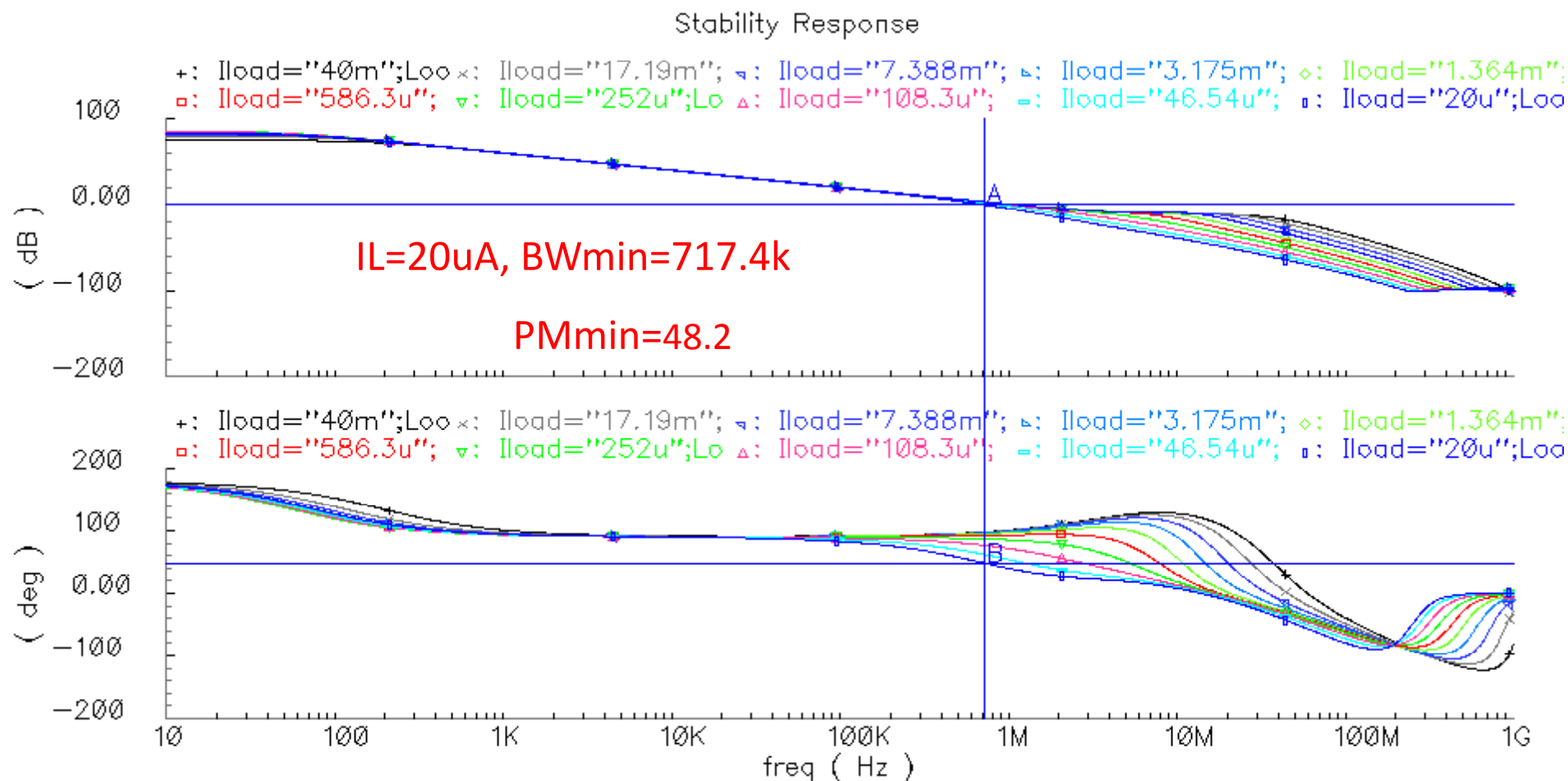
$At \sim Cc/CG$, CG为MTP的栅电容
造成Cc不能太小, 带宽较小!

若在MTP前加上buffer, 则:

$At \sim Cc/Cn$, Cn为amp_o点上的寄生电容, 其值远比CG小, 故At更大, P2更高, 因此可减小Cc, 增加带宽, 而保持稳定

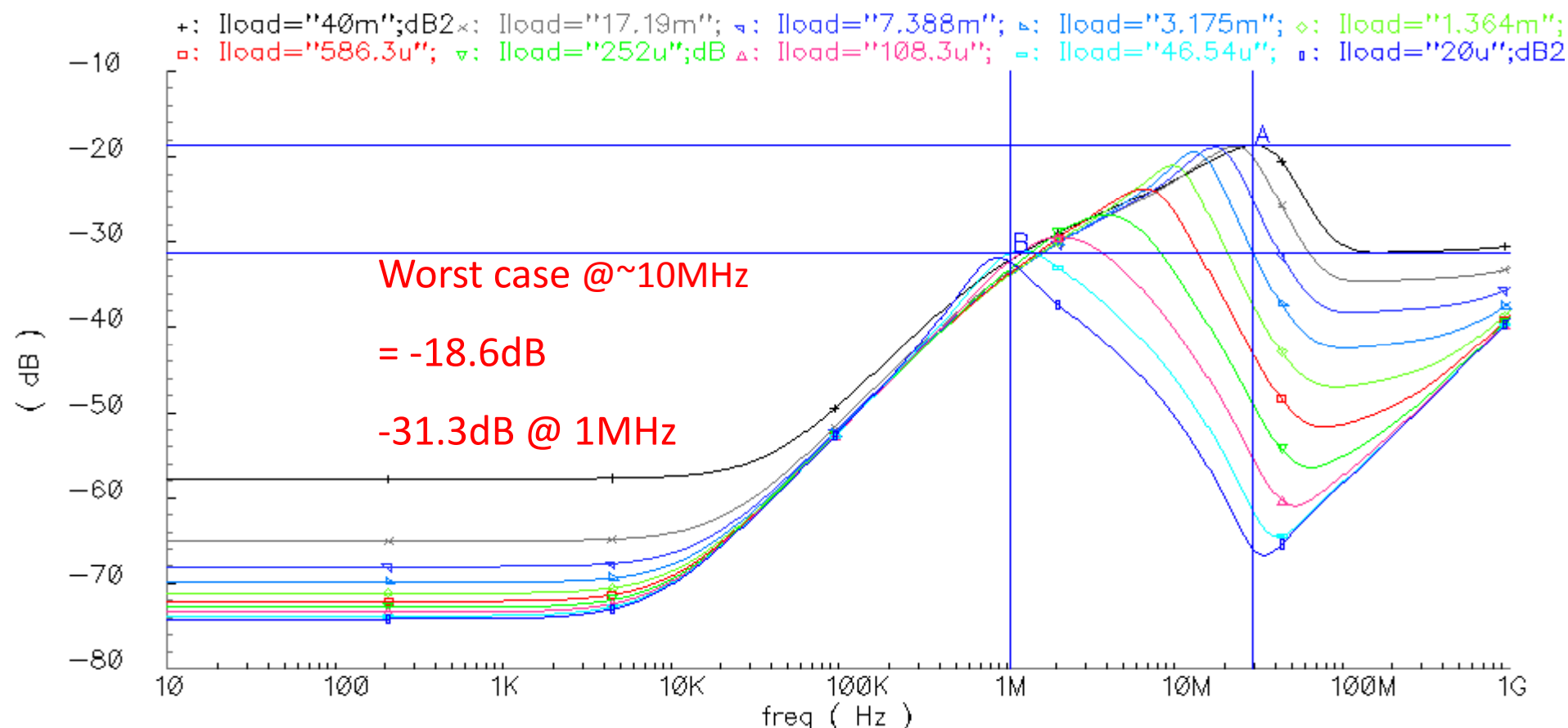
带宽增加, 则Cz相应地减小

全负载稳定性



PSRR

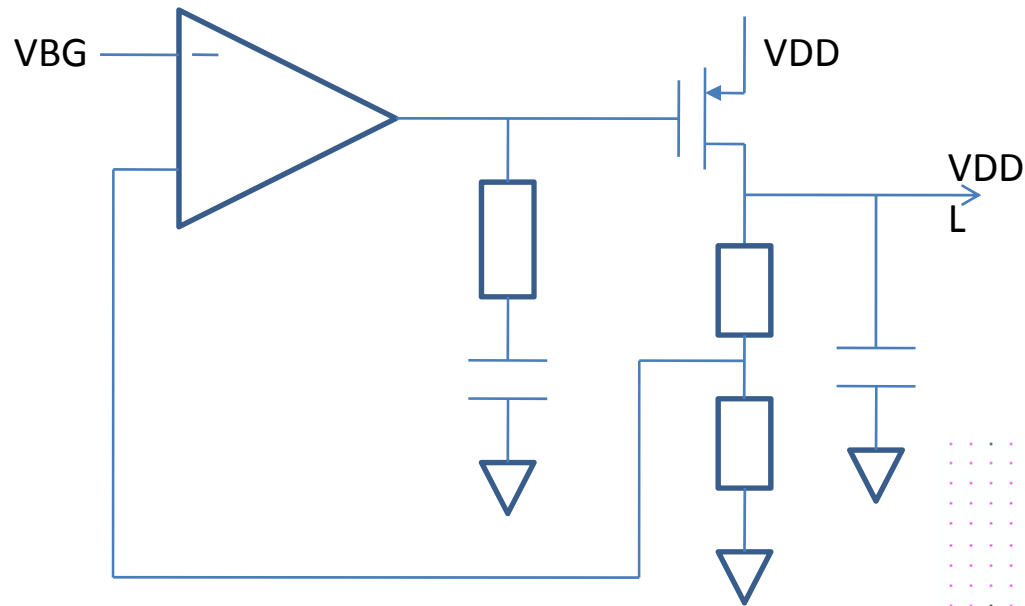
AC Response



全集成LDO要点

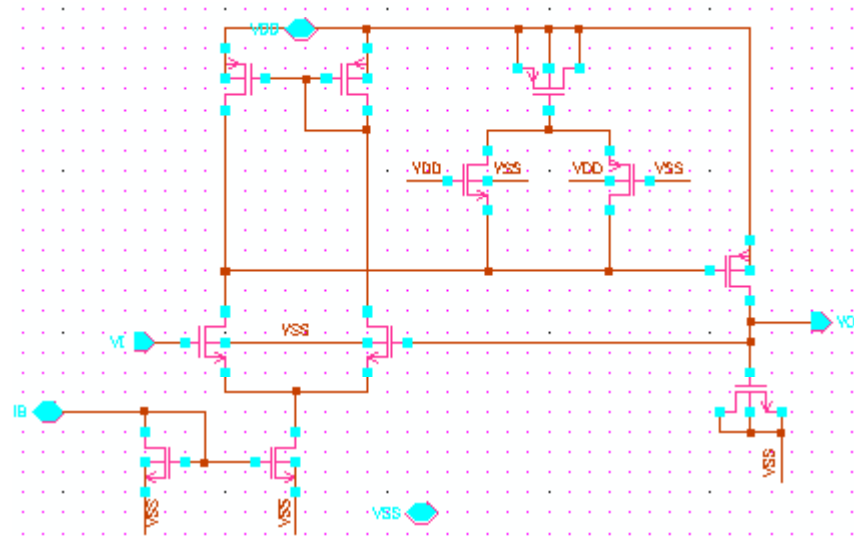
- **Err_amp+MTP**, 最简单结构, 最可靠
- **加中间buf, 可增大带宽**, 但要注意buf的能力, buf太弱, 可能带来不稳定;
- 反馈网络加并联电容, 可改善小负载下的**PM**, 但小负载需要大电容, 大负载需要小电容, 可权衡取值, 或加变容管
- 总解藕电容比较小, 当负载快速变化时, 要特别核对过流保护电路是否会误触发!

另一种全集成LDO

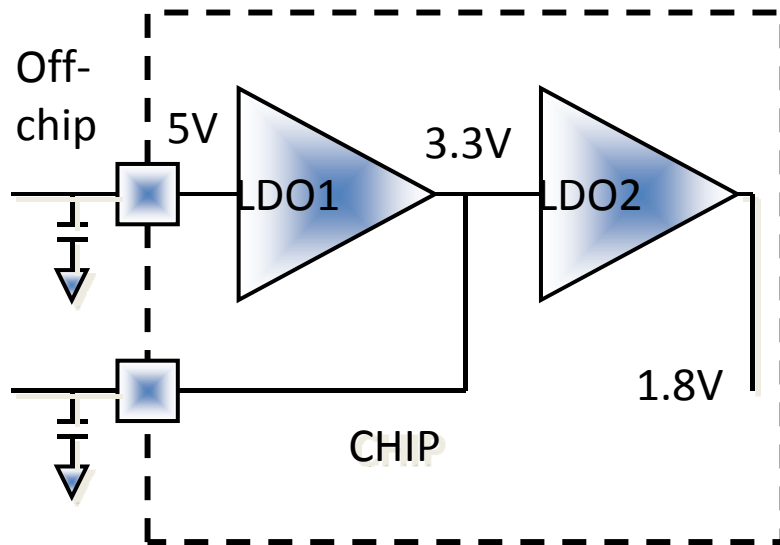


第一级输出加零点补偿

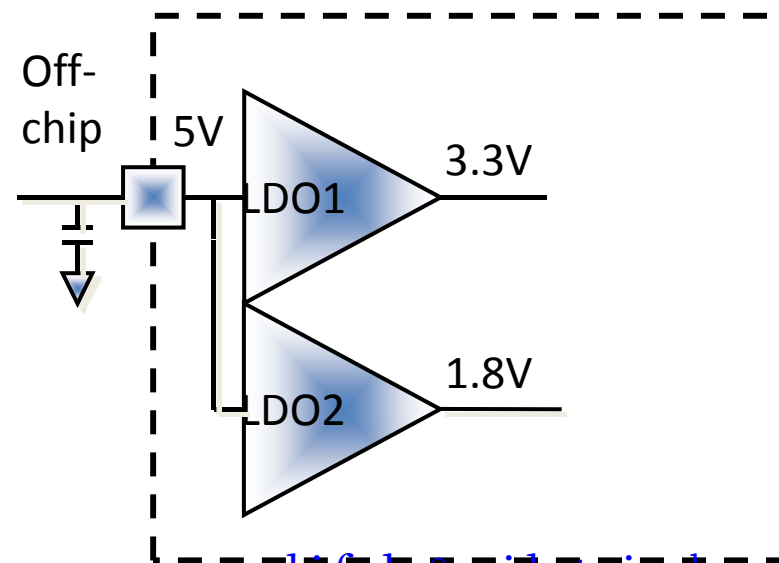
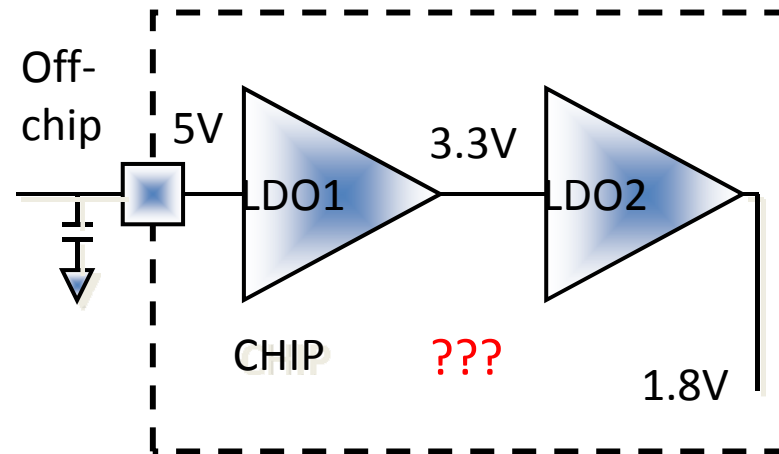
电阻和电容均可用MOS管实现



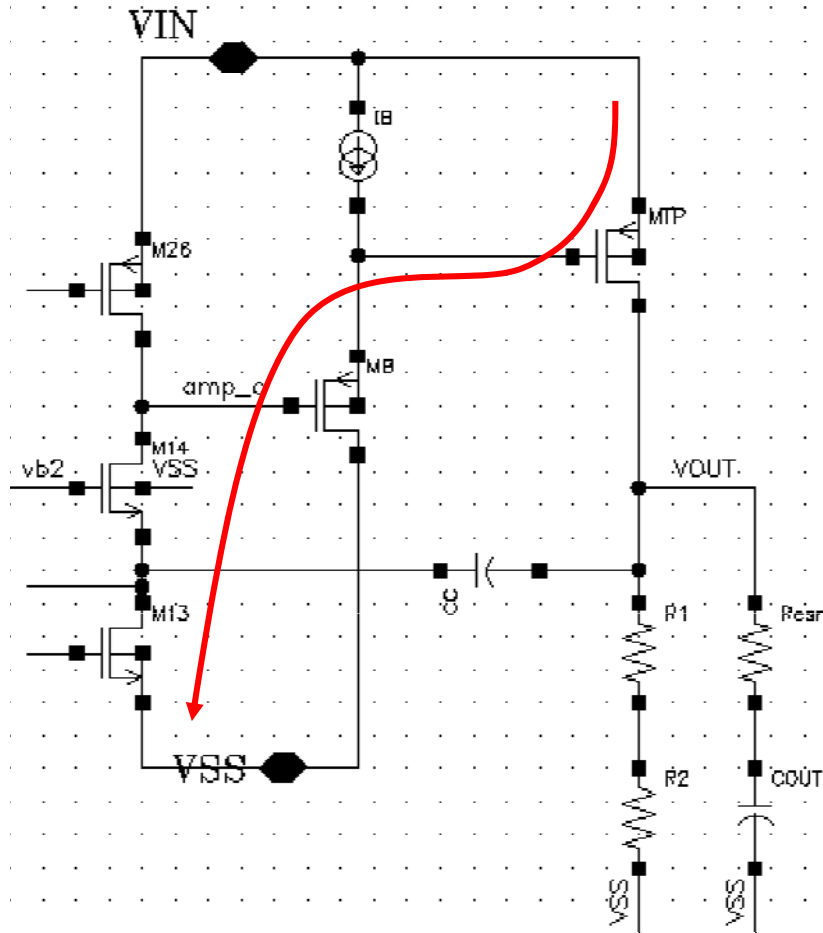
LDO in SOC



在SOC设计中，可能存在多种电源电压和多级LDO, 此时要特别注意LDO, 片外解藕电容的接法, 注意输入电源的频率成分和LDO本身的PSRR特性, 保证输出电源的稳定干净



电源电压范围?



Problem:

$$V_{IN} > 2V_{GS} + 2V_{dsat} = 2V_T + 4V_{dsat}$$

在大电流输出情况下，VIN的下限更高

Solution:

降低中间Buffer级所带来的电平移位

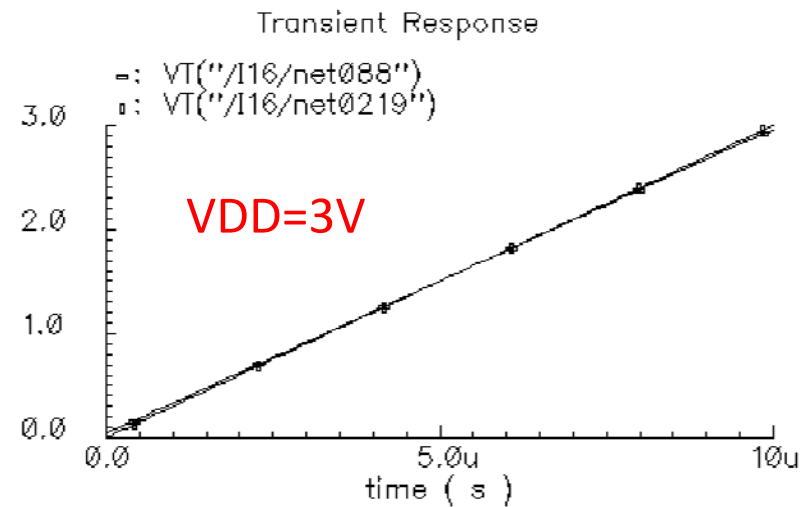
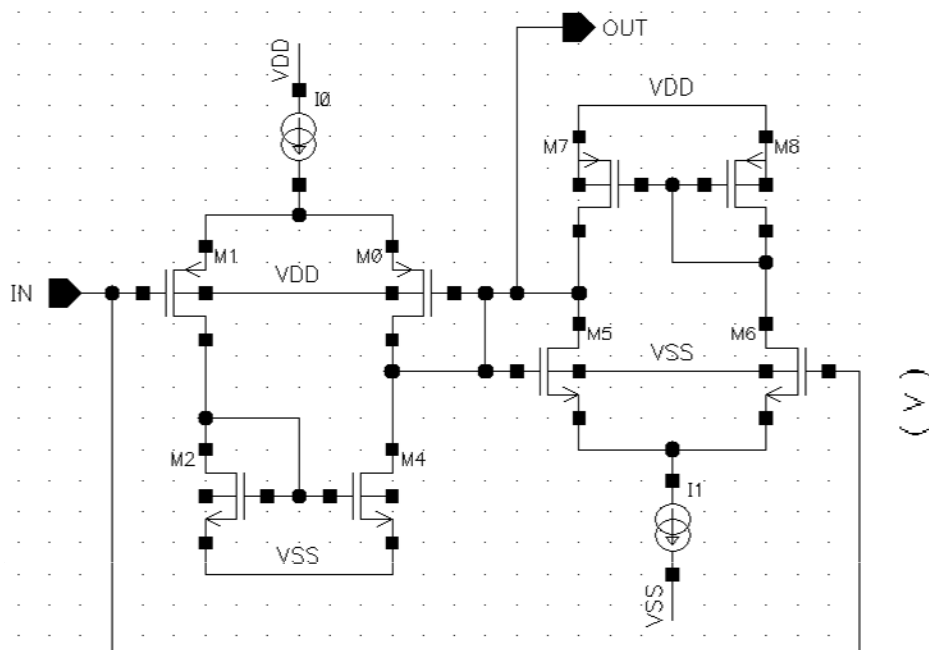
如:

取消中间buffer

采用depletion MOST(工艺兼容?)

采用基于互补差分对的rail-to-rail buffer, 如下页所示

Rail-to-rail buffer

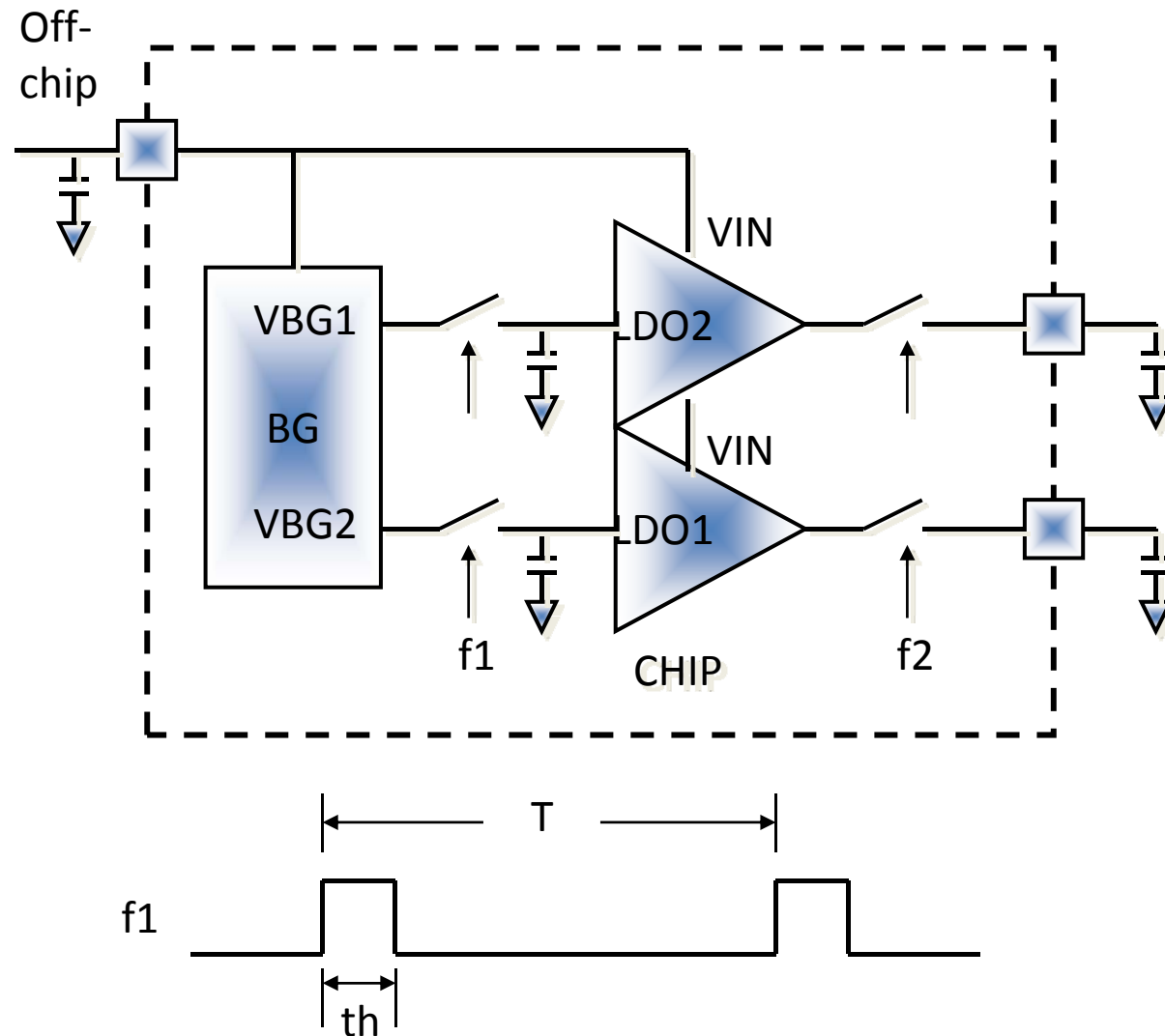


输入偏高时, 右边的放大器工作, 偏低, 则左边的工作, 在中间, 两个都工作

Gm没有做均衡处理, 对此应用不重要

最低电源: $V_{GS} + 2V_{dsat}$

Low power?



可采用间歇工作方法来降低功耗

如对于bandgap, 可通过开关电容来采样基准电压, 电容保持电压, bandgap只是间歇打开刷新电容上的电压

这样处理后的功耗:

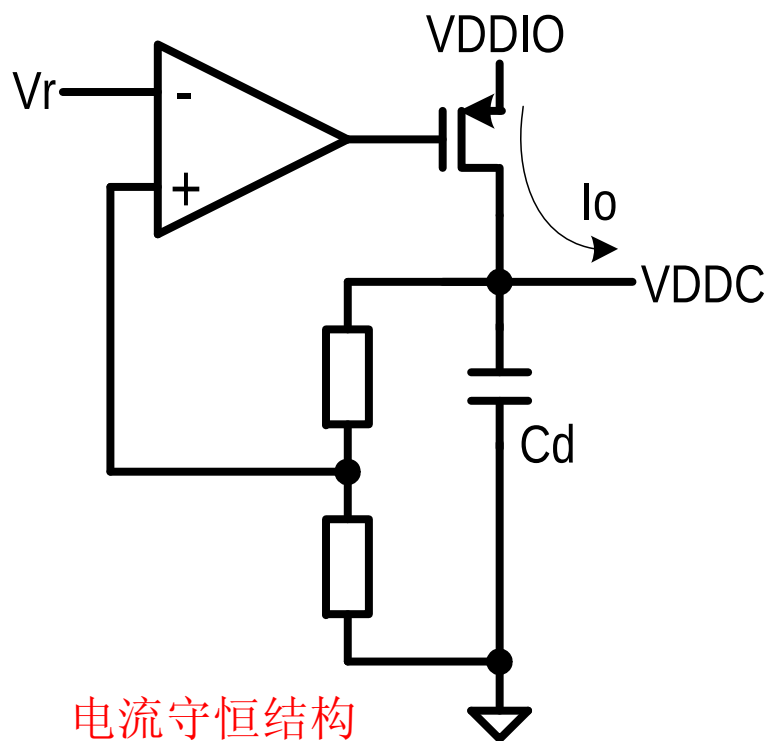
$$P_{bg}' = P_{bg} * (t_h / T)$$

LDO或许也可以这样, 但要注意其建立特性

版图设计

- 注意匹配设计
 - MTP, MB, M6之间的匹配
 - 电阻R1, R2之间的匹配
 - 运放电路中的差分对和电流镜most的匹配
- 特别注意MTP是大电流工作
 - 对于有片外解藕的情况，其D,S端到PAD的连线要足够宽, 尽量短, 保证寄生电阻足够小, 并满足电流密度限制
 - 注意加宽衬底保护环

LDO的效率问题



$$Efficiency = \frac{V_{DDIO}}{V_{DDC}} \cdot \frac{(I_o + I_Q)}{I_o}$$

设计中:

降低LDO本身的静态功耗 I_Q

但是, 最大效率= V_{DDC}/V_{DDIO}

当 V_{DDIO} 与 V_{DDC} 相差较大时, 效率尤其低, 如5V USB -> 1.8V core

4V Battery -> 1.8V core

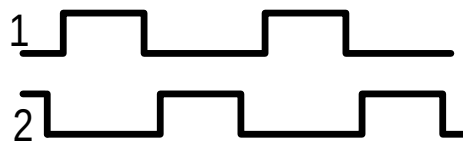
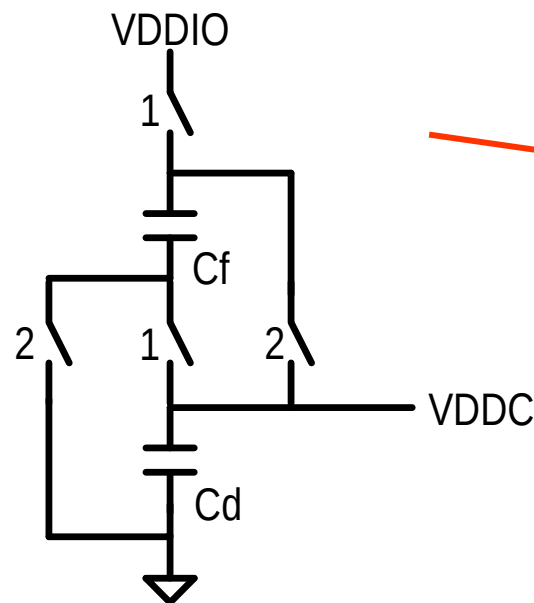
3V Battery -> 1.2V core

Etc.

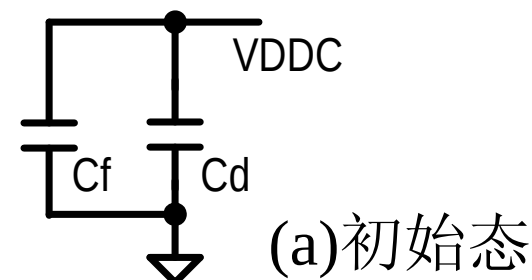
电荷泵高效降压原理

CHP BUCK

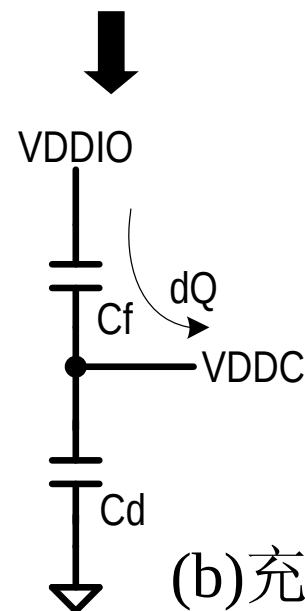
解决VDDIO, VDDC相差较大时效率过低的问题, 可采用Charge Pump BUCK降压, 它恰好适用于这种情况



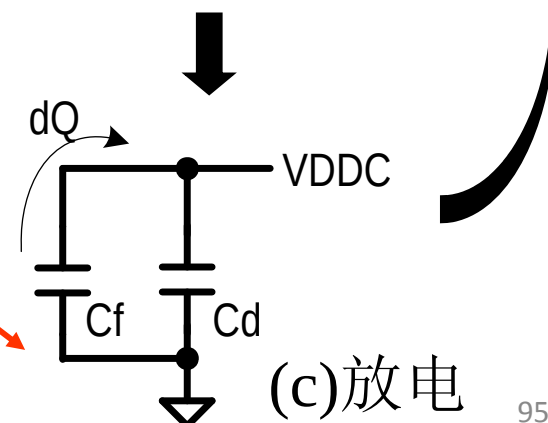
功率守恒结构



(a)初始态



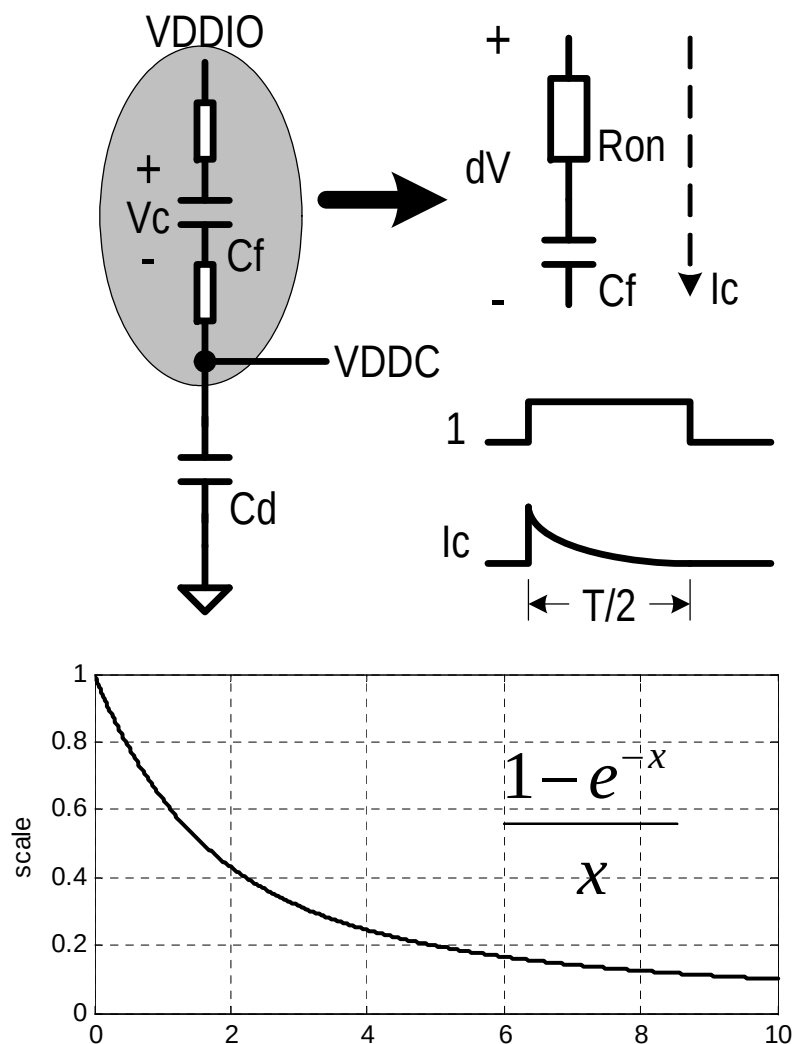
(b)充电



(c)放电

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充电电流



$$I_{avg} = \int_0^{T/2} I_c(t) dt / (T/2)$$

$$= \frac{dV}{R_{on}} \cdot \frac{1 - e^{-x}}{x}$$

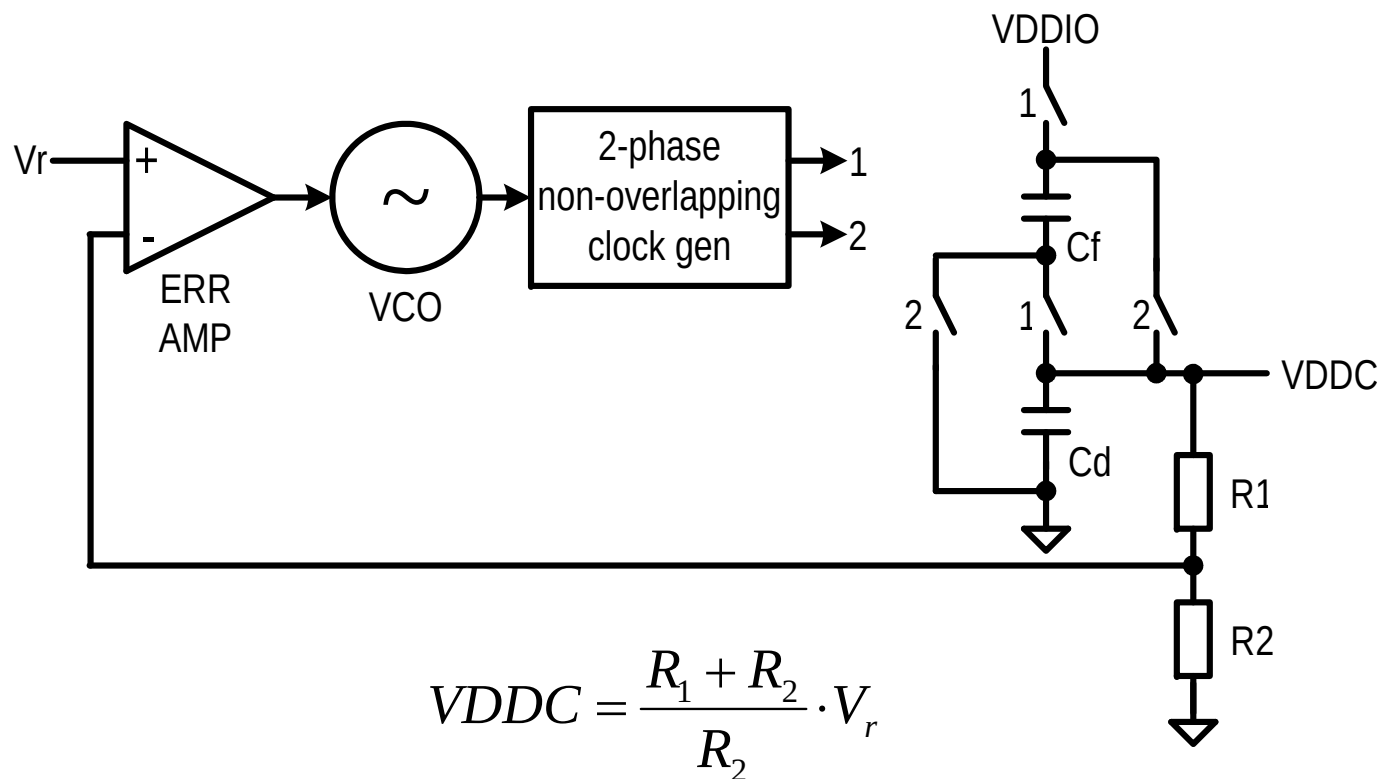
$$x = \frac{T}{2\tau} = \frac{T}{2R_{on}C_f}$$

$$dV < V_{DDIO} - 2V_{DDC}$$

要点:

- 1) 最大充电电流: $I_{max} = dV / R_{on}$
- 2) 调节充放电频率 $\rightarrow$ 调节平均充电电流

稳压回路



利用充放电频率与充电电流的关系，
建立负反馈控制环路，实现稳压目标

输出纹波

假设在充电相 C_f 上的电压被充到 V_{c1} ,则在放电相开始时,输出电压为:

$$V_{DDC_b} = \frac{C_f V_{c1} + C_d V_{DDC}}{C_f + C_d}$$

输出纹波电压:

$$V_{ew} = V_{DDC_b} - V_{DDC} = \frac{C_f (V_{c1} - V_{DDC})}{C_f + C_d}$$

$$\longrightarrow V_{ew} < \frac{C_f}{C_f + C_d} (V_{DDIO} - 2V_{DDC}) = \frac{C_f}{C_f + C_d} \cdot dV$$

这个结果可作为设计 C_f , C_d 值的重要参考

转换效率

一个充放电周期中从VDDC上损失的等效电荷为：

$$Q_l = C_{pb} \cdot V_{DDC} + C_{pt} \cdot (V_{DDIO} - V_{DDC})$$

If $C_{pb} = C_{pt}$

$$Q_l = C_{pb} \cdot V_{DDIO}$$

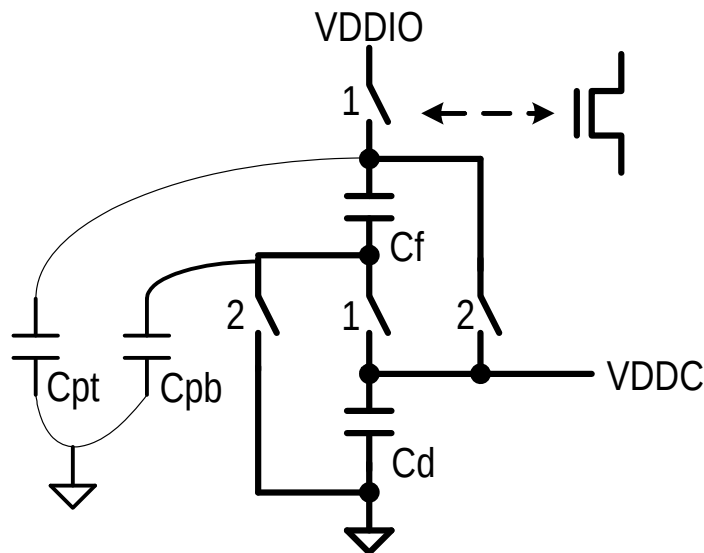
一个充放电周期中，VDDC上得到的有效电荷为：

$$Q_e = I_{avg} \cdot T = 2dV \cdot C_f \cdot (1 - e^{-x})$$

$$\eta_Q = \frac{Q_e}{Q_e + Q_l} \approx 1 - \frac{V_{DDIO}}{2dV} \cdot \frac{C_{pb}}{C_f} \cdot \frac{1}{1 - e^{-x}}$$

电荷搬移效率为：

总的转换效率为：

$$\eta = \eta_Q \eta_V \approx \left(1 - \frac{V_{DDIO}}{2dV} \cdot \frac{C_{pb}}{C_f} \cdot \frac{1}{1 - e^{-x}} \right) \cdot \left(1 - \frac{dV}{V_{DDIO}} \right)$$


极限指标

最高转换效率: $\eta_{\max} = \left(1 - \frac{dV}{V_{DDIO}}\right) / \left(1 + \frac{V_{DDIO}}{2dV} \cdot \frac{C_{pb}}{C_f}\right)$

最大充电电流: $I_{\max} = \frac{dV}{R_{on}}$

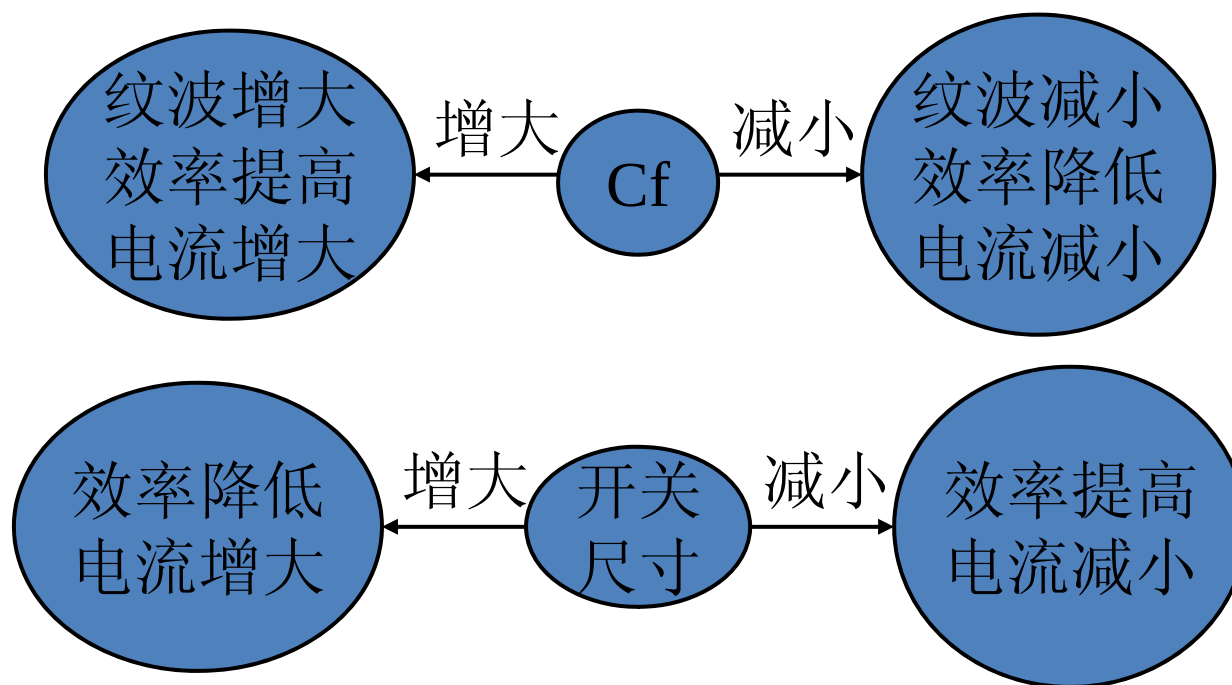
最大输出电压: $V_{\max} = \frac{V_{DDIO}}{2}$

最大输出纹波: $V_{ew_max} = \frac{C_f}{C_f + C_d} \cdot dV$

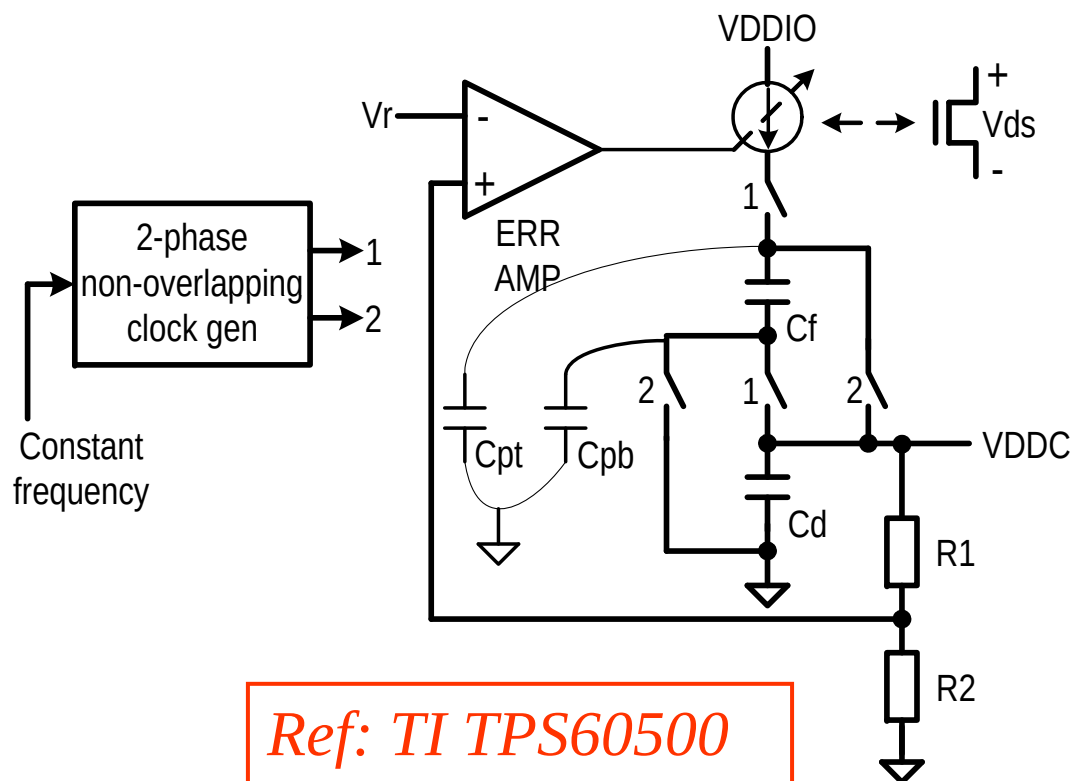
效率比LDO高的条件: $C_{pb} < \frac{2dV}{V_{DDIO}} \cdot C_f$

Flying Capacitor须采用片外电容!

Tradeoff



现有稳压方案



现有方案特点:

固定的充放电频率

通过压控电流源(vccs)来实现电流控制和稳压

问题:

寄生损耗不随负载变化而变化, 这样在小负载电流下, 效率降低

构成vccs的MOS管存在压降,降低了转换效率

寄生所致漏电流:

$$I_l = Q_l \cdot f_{ck} \approx C_{pb} \cdot V_{DDIO} \cdot f_{ck}$$

比较

	现有方案	所提方案
稳压控制	电流调节	频率调节
直接降压损耗	大 $V_{ds}+V_{sw}$	小 V_{sw}
寄生损耗	固定	负载大则大 负载小则小

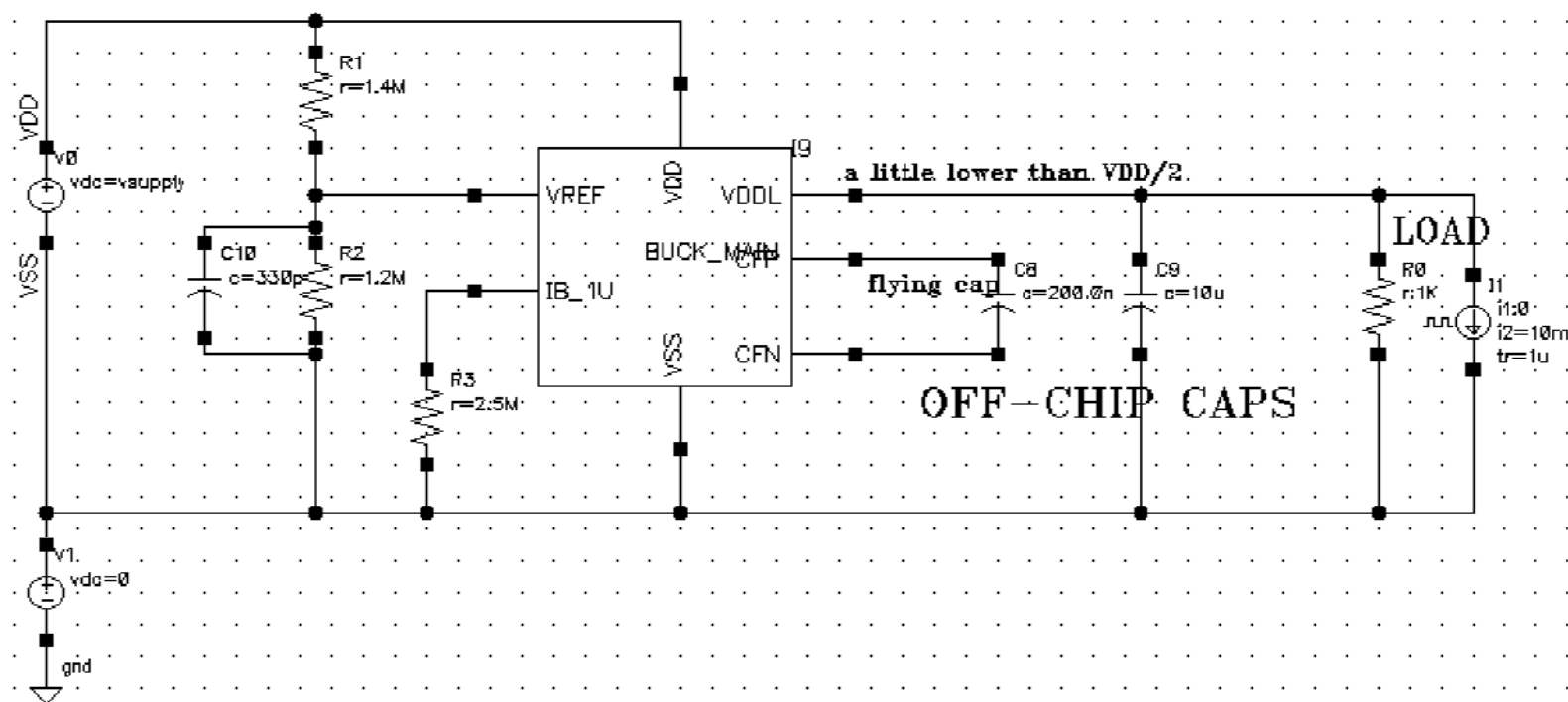
新方案优点：效率高，尤其是在小负载情况下

CHP BUCK设计实例

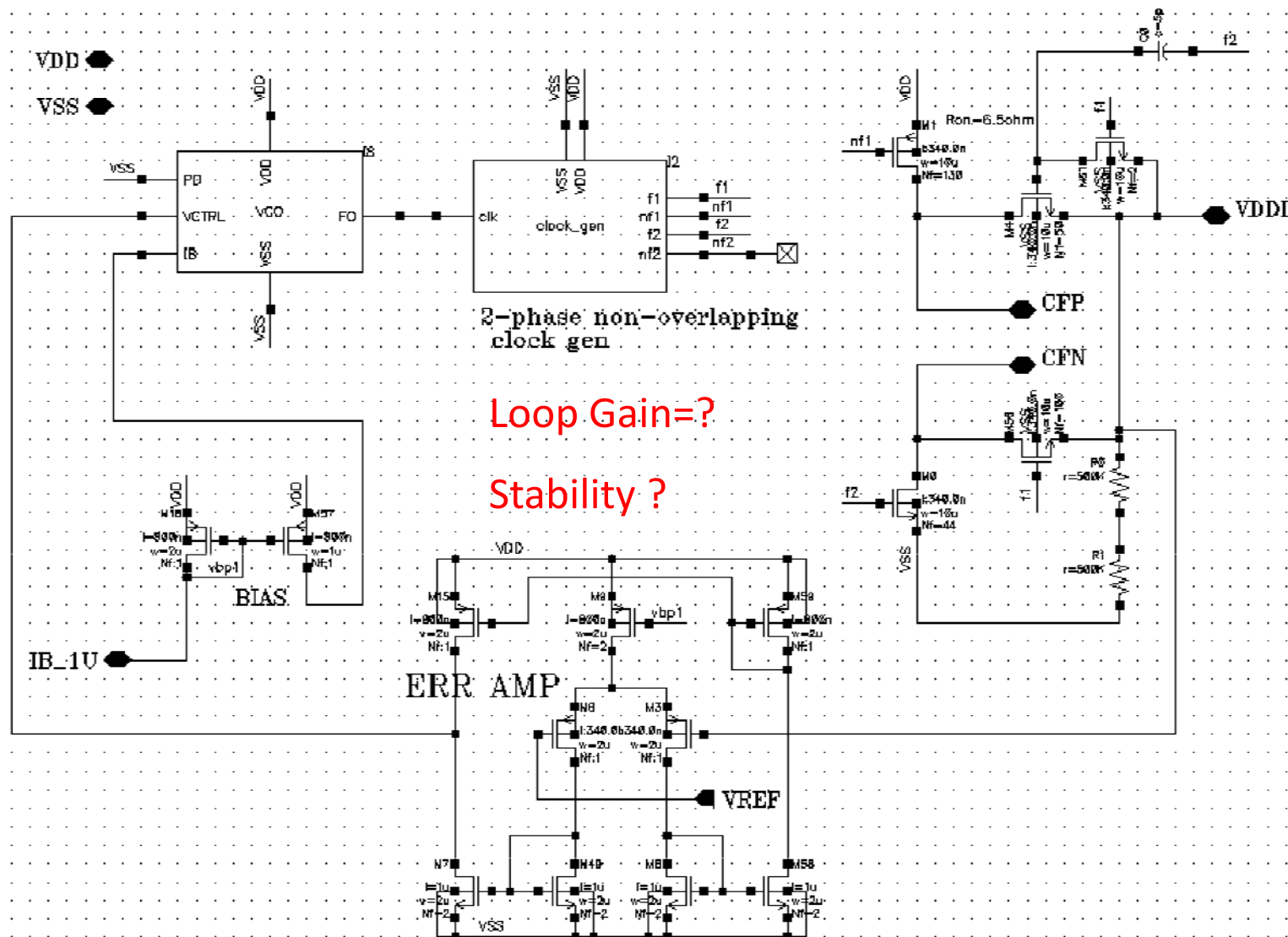
系统指标

- 输入电压: $>2.6V$
- 输出电压: $1.2V$
- 负载 $20mA$
- 空载电流: $<20\mu A$
- 效率: $>80\%$ ($I_L=1\sim 20mA, V_{IN}=2.6$)
- 输出纹波: $<5mV$
- $C_f=?$ $C_d=?$ Size of SW=? Clock freq=?

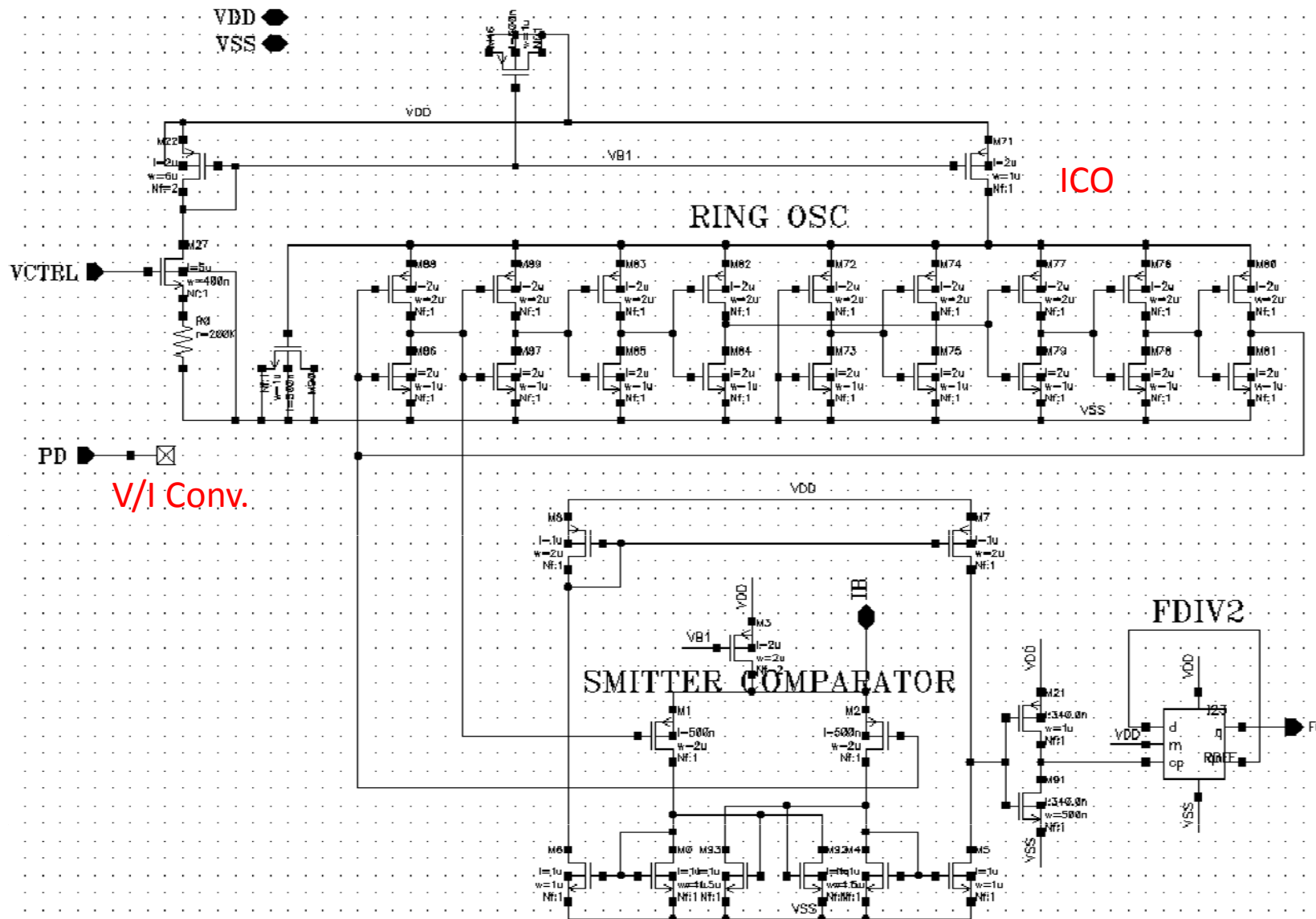
顶层仿真图



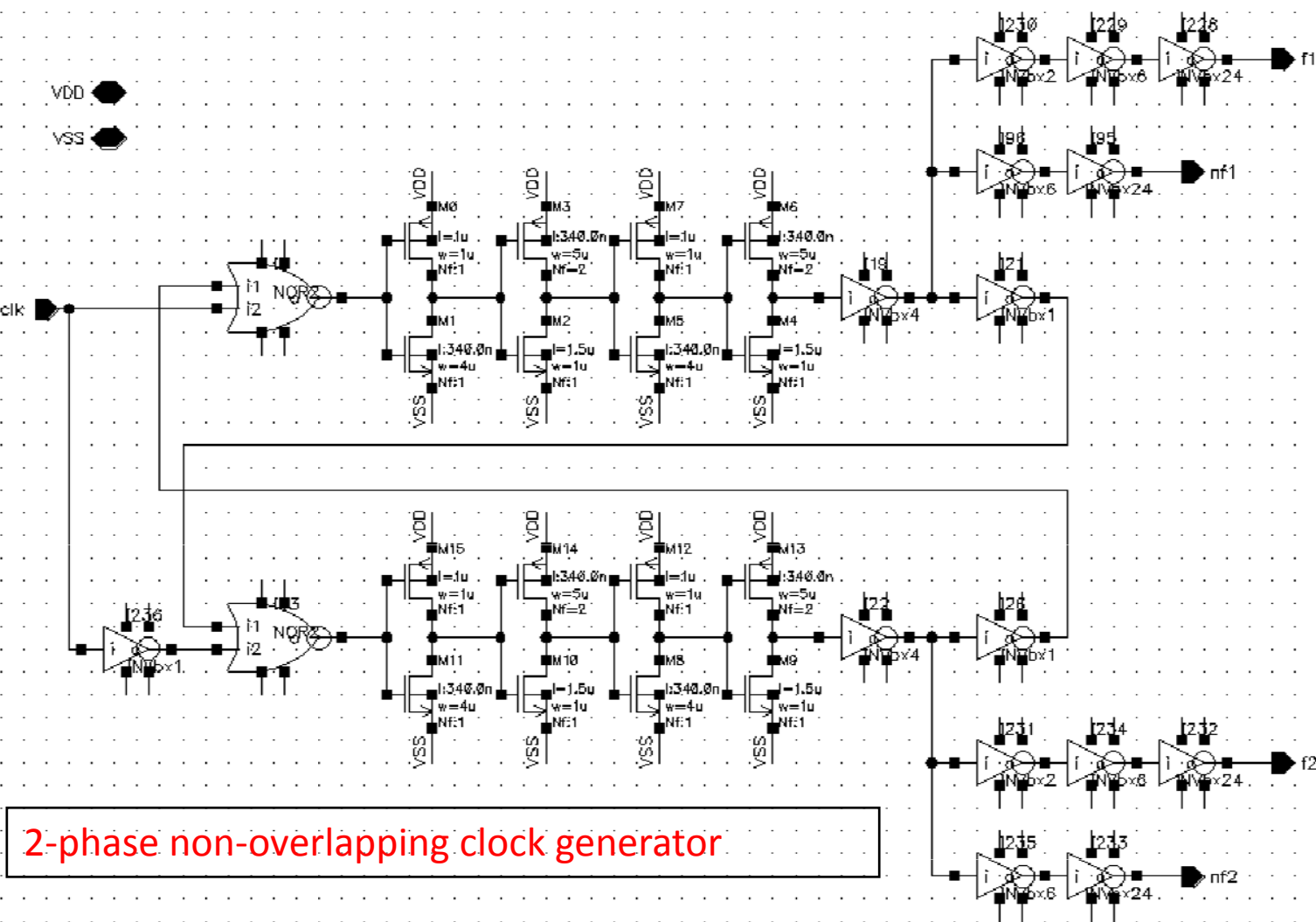
CHP BUCK主电路



VCO



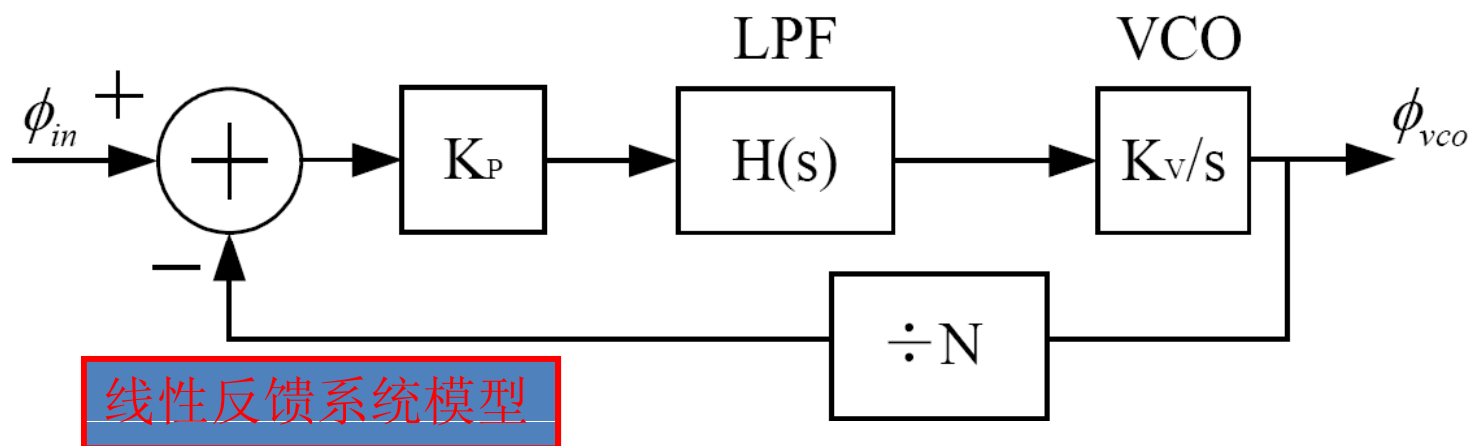
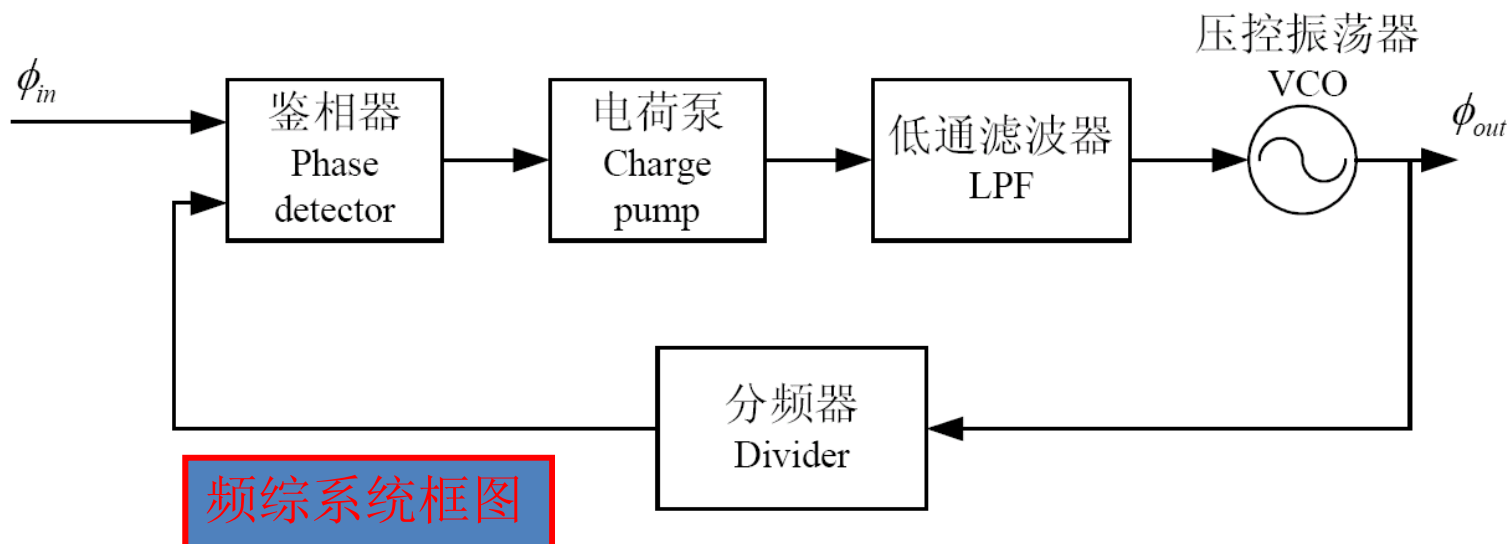
Clock generator



PLL频综设计

- 提供精确的, 相位相关的频率变换
- 尽可能小的phase noise, clock jitter
- 尽可能大的环路带宽
- 在各种corner下都必须稳定
- 尽可能小的面积与功耗

系统图与线性模型



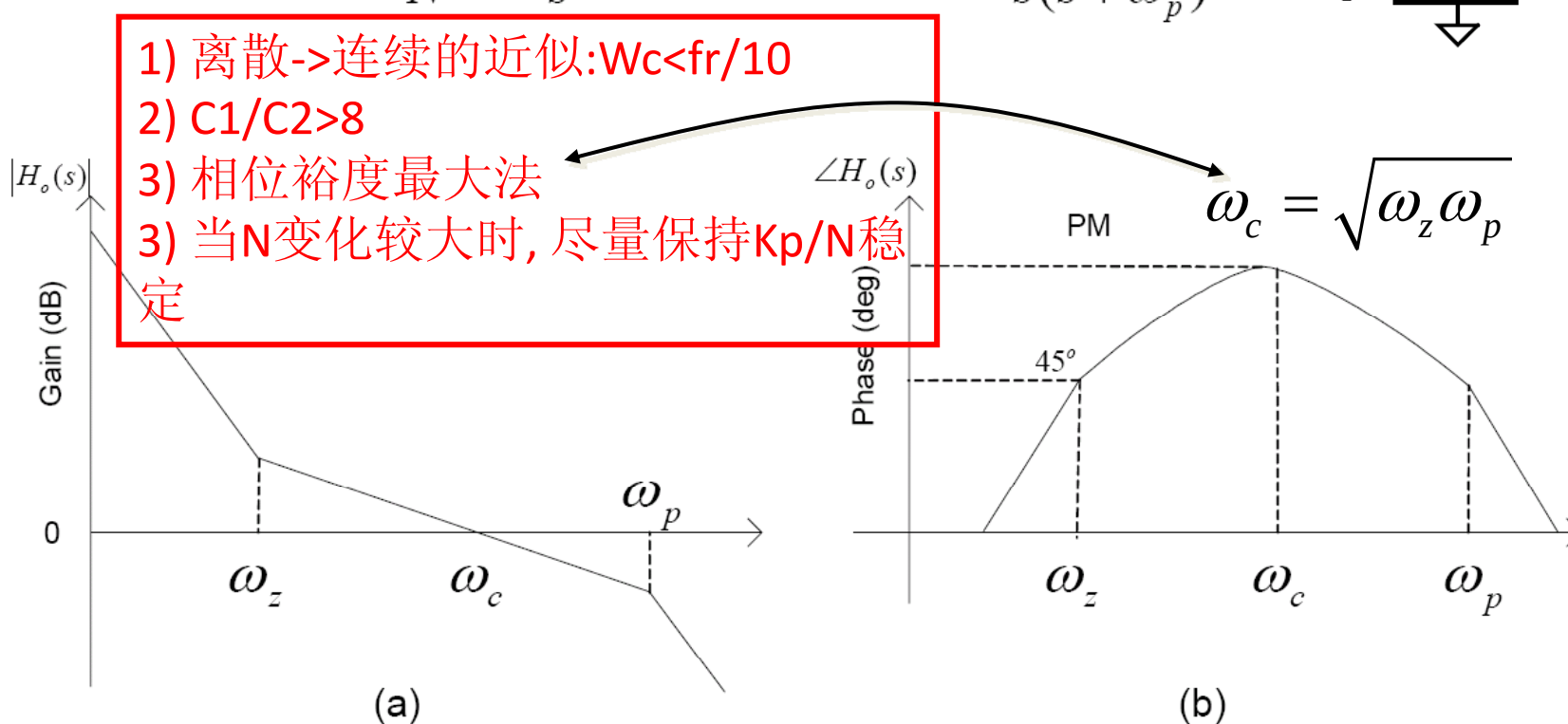
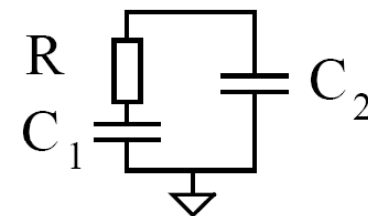
稳定性

开环传递函数

$$H_o(s) = \frac{K_p K_v}{N} \frac{H(s)}{s}$$

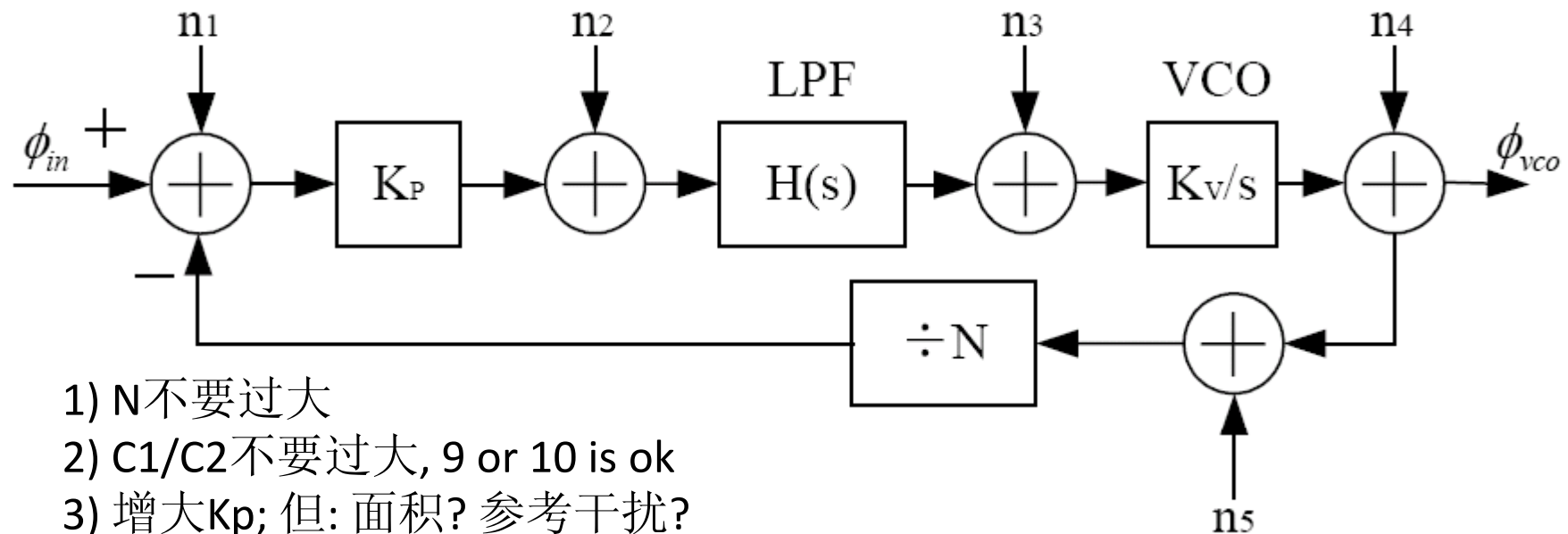
滤波器传递函数

$$H(s) = \frac{s + \omega_z}{s(s + \omega_p)}$$

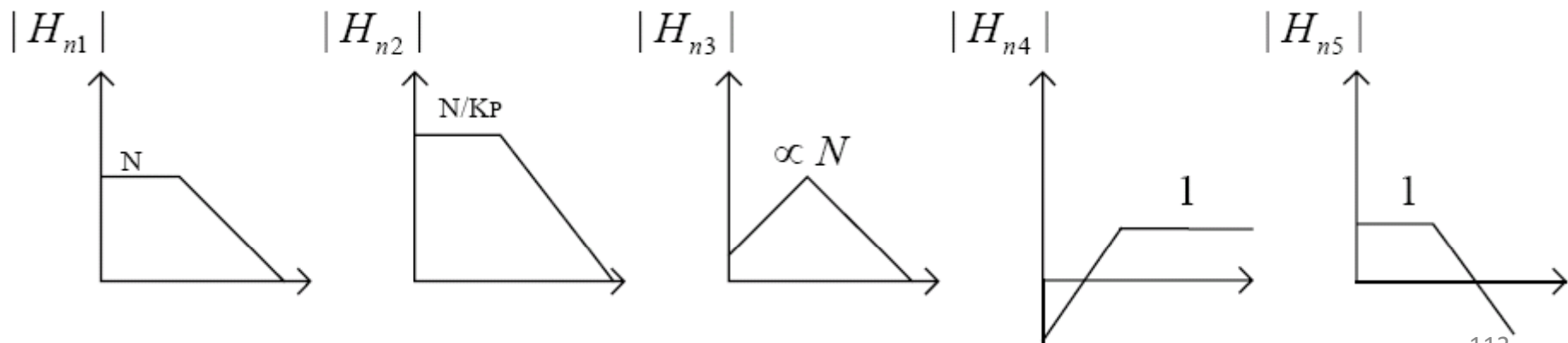


开环传输函数的 Bode 图 (a) 幅度特性 (b) 相位特性

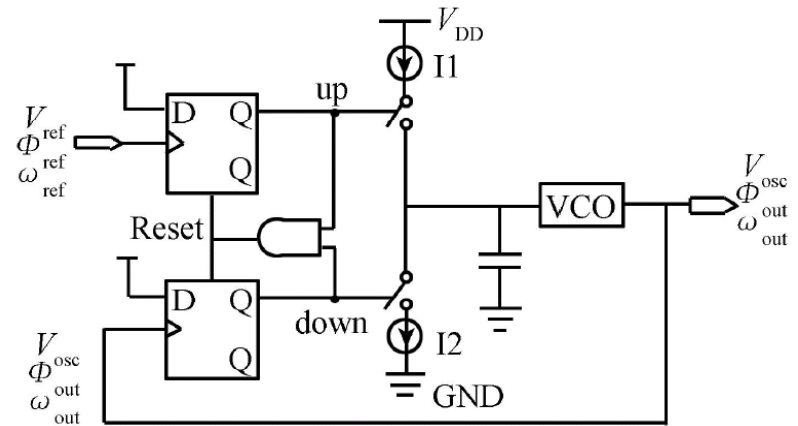
Phase noise



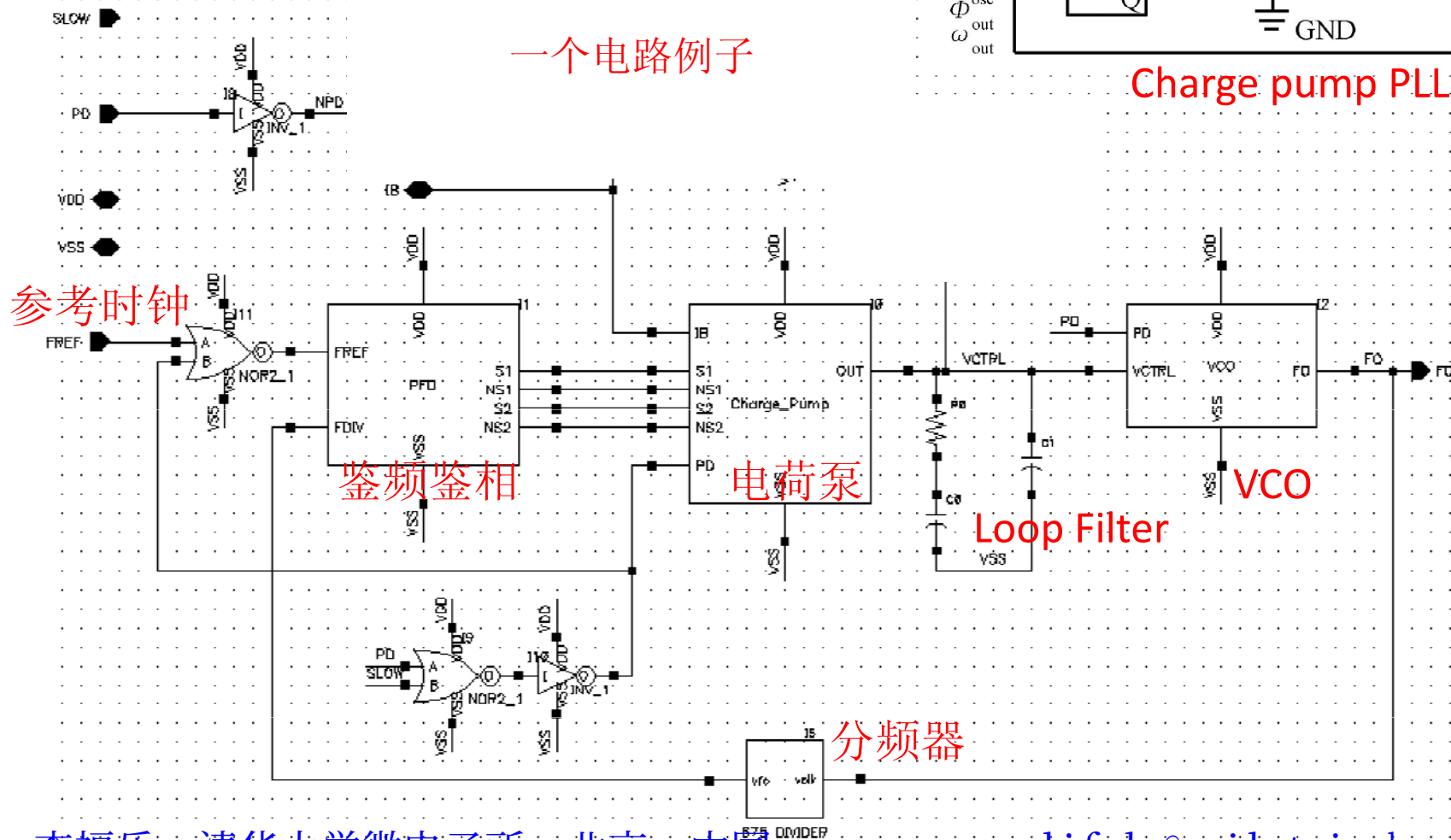
- 1) N 不要过大
- 2) $C1/C2$ 不要过大, 9 or 10 is ok
- 3) 增大 K_P ; 但: 面积? 参考干扰?
- 3) 减小 K_V , 但: 频率范围?
- 4) 环路带宽? $N1, n2, n5$: 低通; $n4$: 高通. tradeoff



PLL电路



Charge pump PLL



PLL CAD

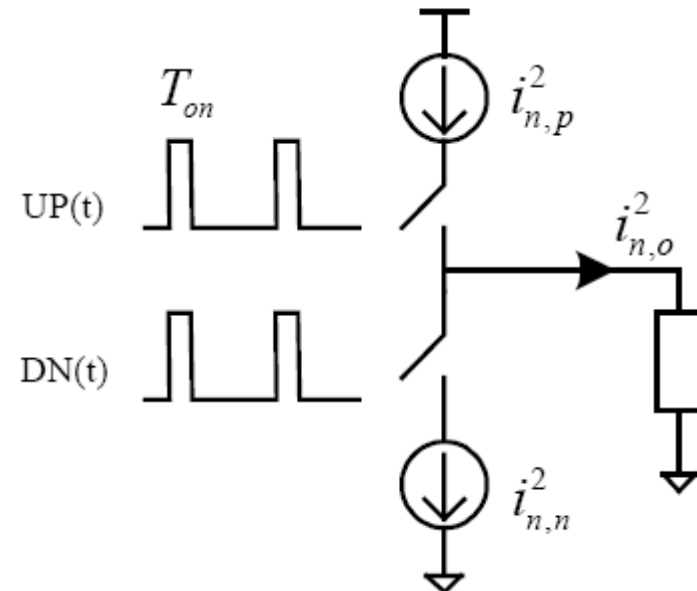
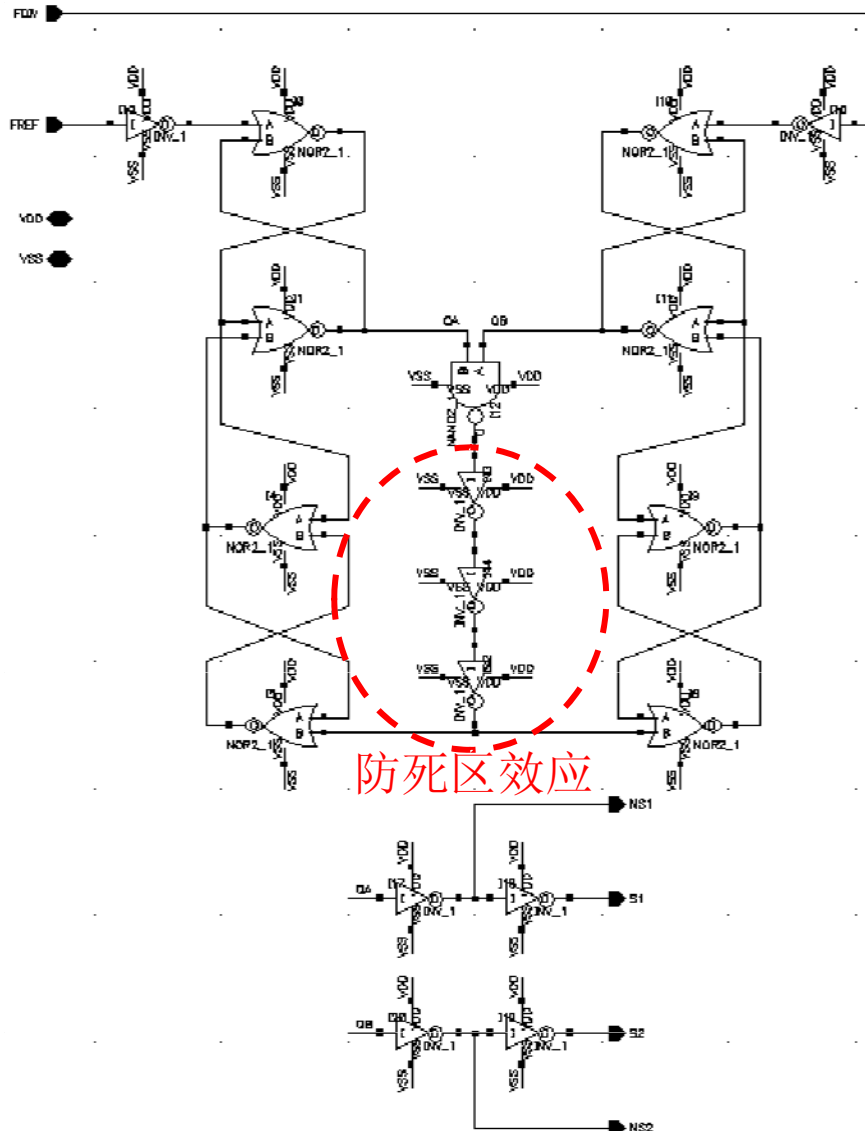
LOOP FILTER DESIGN SCREEN

Kphi		KUCO	Output
0.00		31.30	22.118 MHz
mA/2pi		MHz/U	
rad			
	C1	C2	
	GND	R2	
		GND	
32.768000 KHz		N Counter	
		1/675	

SPECIFY/INQUIRE *	FILTER PARAMETERS	COMPONENTS
<1> Charge Pump Gain	Phi 55.000 deg.	C1 0.077 nF
<2> UCO Gain	wc 2.200 KHz	C2 0.693 nF
<3> N	Atten 0.000 db	
<4> Comparison Frequency	<make Atten = 0 for	R2 330.964 Kohm
<5> Phase Margin	2nd Order filter >	
<6> Loop Bandwidth		
<7> Atten	* For Help, select the parameter, and	
<0> EXIT	read the instructions	

用easy_pll软件来设计Loop Filter

PFD

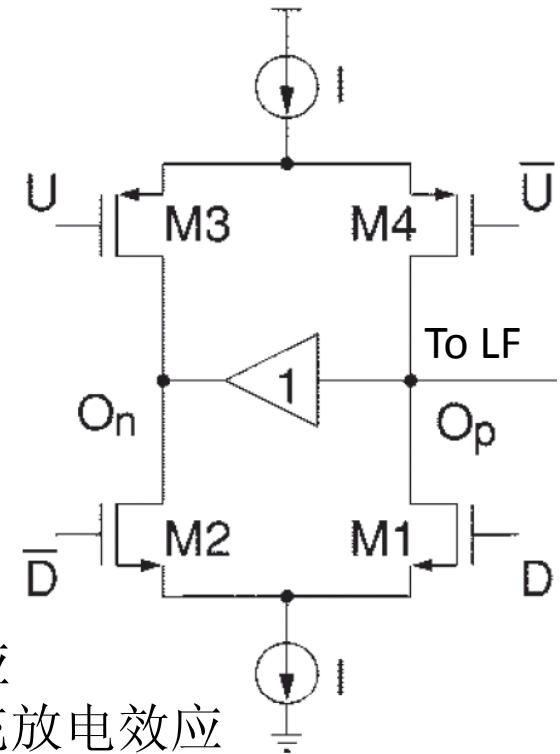
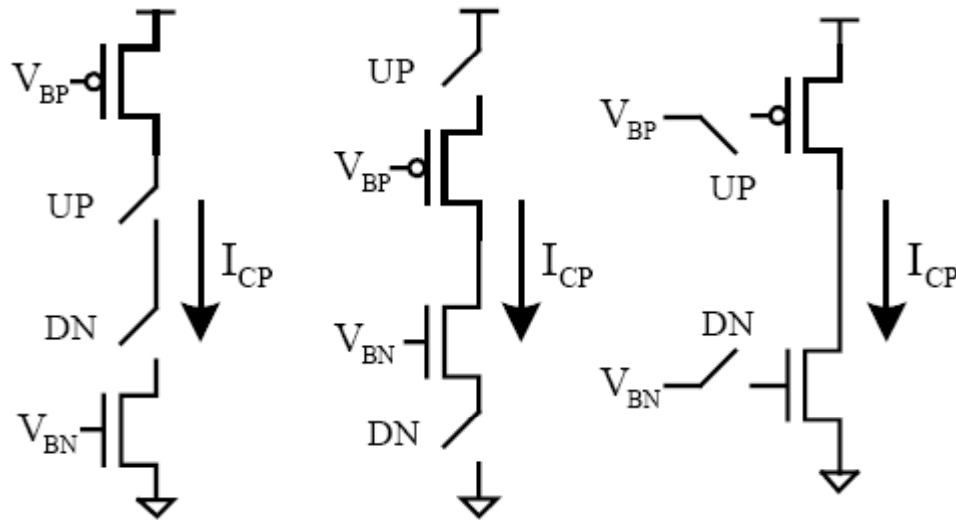


$$S_{UP}(f) = \sum_{n=-\infty}^{+\infty} a_n^2 \delta(f - nf_{ref})$$

$$a_n = \frac{1}{T} \int_{-\frac{T_{on}}{2}}^{\frac{T_{on}}{2}} UP(t) e^{-jn\omega_{ref}t} dt = \frac{\sin \frac{n\omega_{ref}T_{on}}{2}}{n\pi}$$

- 1) T_{on} 不能太小，防死区效应
- 2) T_{on} 增大，噪声增大
- 3) T_{on} 应取死区出现前的最小值

Charge Pump



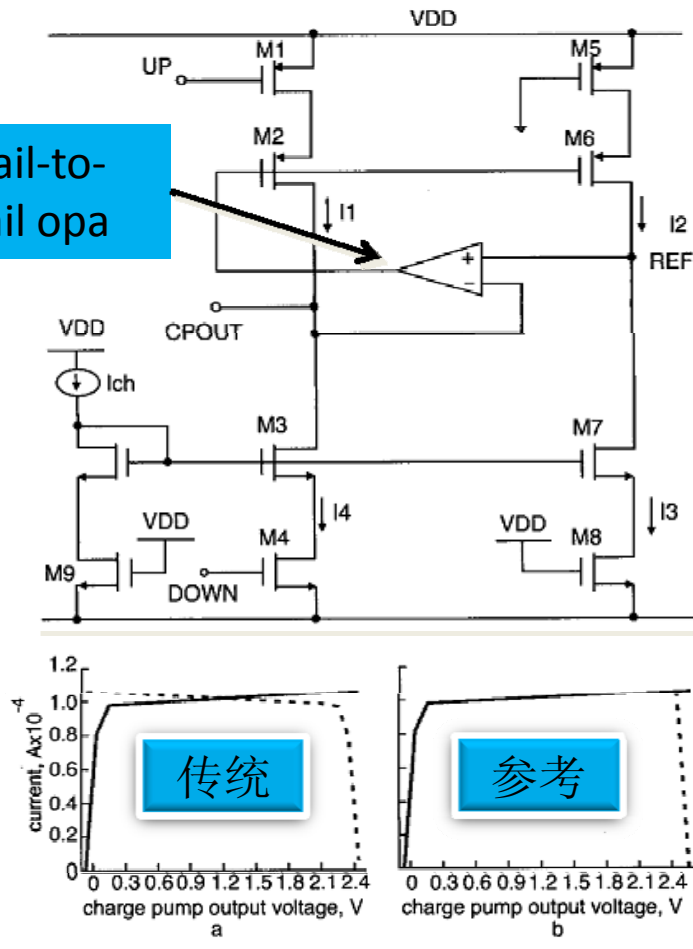
- 1) 尽量保证上下偏置电流匹配，减小沟道调制效应
- 2) 尽量保持电流源MOS的工作点，防止寄生电容充放电效应
- 3) 提升电路本身的开关速度，减小Ton
- 4) 尽量减小开关的时钟馈通与电荷注入效应
- 5) 减小current noise

$$S_{i,thermo} \propto \frac{8}{3} kT(1+\eta) \frac{I_{CP}}{V_{gs} - V_{th}}$$

$$S_{i,flicker} \propto \frac{K_f \cdot I_{CP}}{C_{ox} \cdot L_{eff}^2 \cdot f^\beta}$$

Charge pump

Rail-to-rail opa



Phase error due to current mismatch:

$$\Phi_{\epsilon} = 2\pi \frac{\Delta t_{on}}{T_{ref}} \frac{\Delta i}{I_{cp}} \quad [\text{rad}]$$

三阶PLL

Ref. spur due to current mismatch:

$$P_r = 20 \log \left[\frac{\sqrt{2}(I_{CP}R/2\pi)\Phi_{\epsilon}K_{VCO}}{2f_{ref}} \right] - 20 \log \frac{f_{ref}}{f_{pl}} \quad [\text{dBc}]$$

fpl: the pole of the loop filter

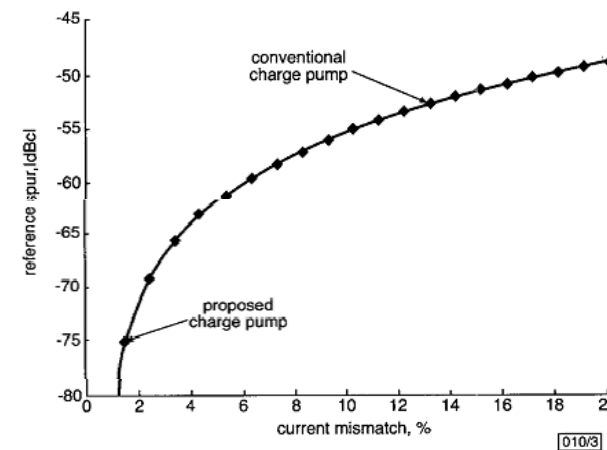


Fig. 3 Reference spur against current mismatch

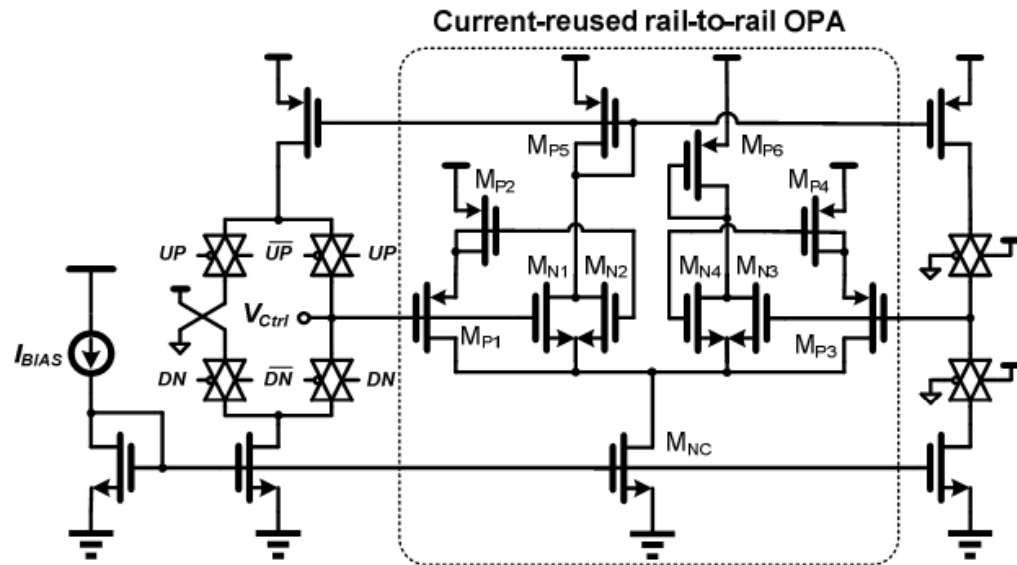
Conditions: $I_{cp} = 100\mu\text{A}$, $f_{ref} = 5\text{MHz}$, $\Delta t_{on} = 50\text{ns}$, $K_{VCO} = 200\text{MHz/V}$, $f_{pl} = 16.5\text{kHz}$, $R = 7.5\text{k}\Omega$

Ref: J. -S. Lee, et al., Charge pump with perfect current matching characteristics in phase-locked loops, Electronics Letters, Nov., 2000.

李福乐，清华大学微电子所，北京，中国

lifule@mail.tsinghua.edu.cn

Charge Pump

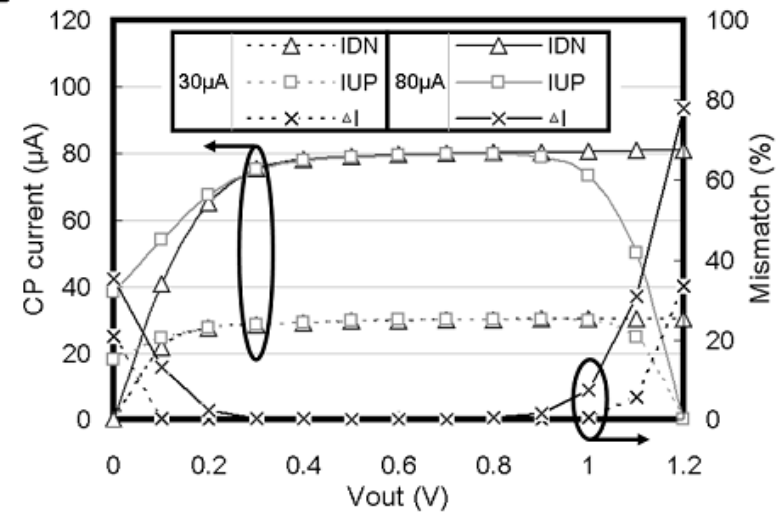


MN1~MN4: input stage

Mp1~Mp4: level shifter

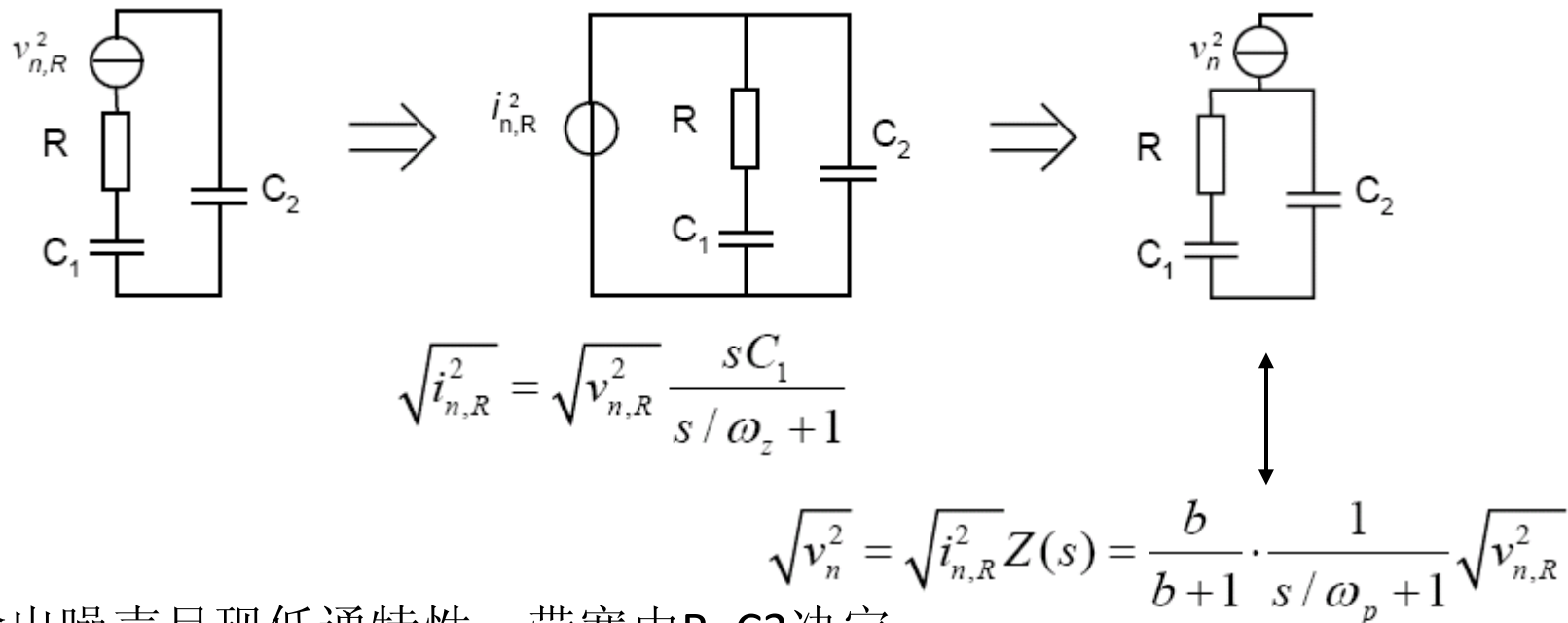
OPA: 与传统rail-to-rail电路相比节约一半功耗

Ref: Meng-Ting Tsai, et, al. A 65nm CMOS Adaptive Bandwidth Clock Generator with Clock Multiplexer Embedded. ASSC2010投稿, 暂时保密!



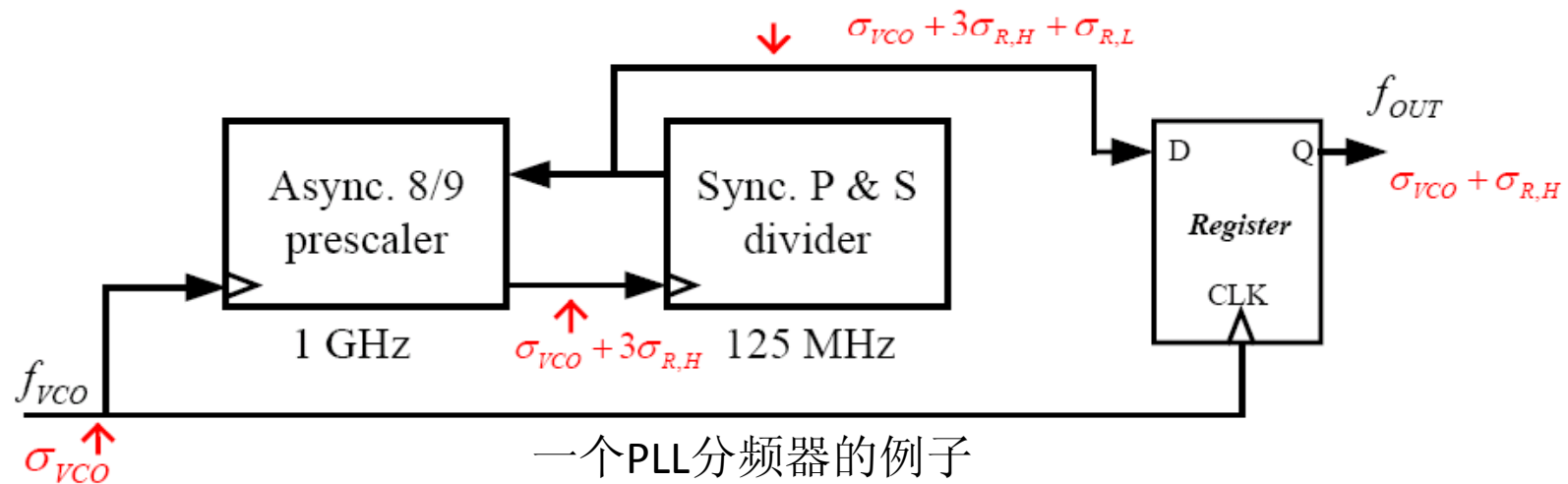
Charge pump output current and mismatch simulation

Loop filter



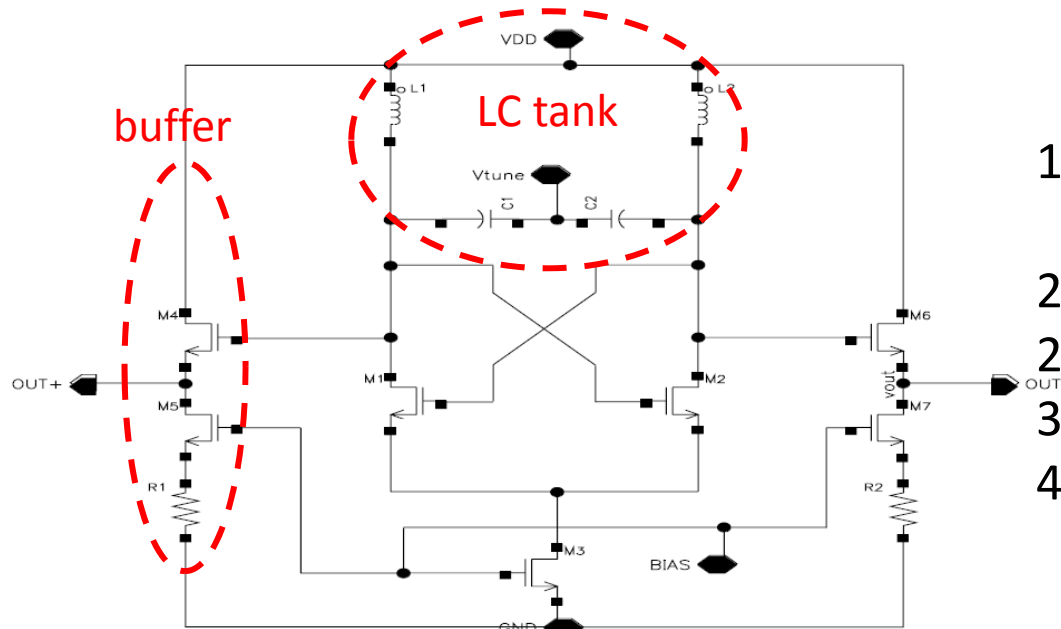
- 1) 输出噪声呈现低通特性，带宽由R, C2决定
- 2) 在闭环系统中，低频噪声受到高通滤波而衰减，极点为 ω_c
- 3) 在闭环系统中，高频噪声受到 k_v/s 滤波而进一步衰减
- 4) Loop filter到输出的噪声为带通特性
- 5) 噪声优化：加大 ω_c ，降低 k_v
- 6) 还可采用有源滤波器，优点是电荷泵没有沟道调制效应问题，缺点是增加了运放的噪声，使其只适用于 ω_c 较大的宽带系统

分频器



- 1) 寄存器会带来额外的时钟抖动
- 2) 异步分频器电路简单，功耗小，但时钟抖动会逐步累加
- 3) 同步分频器逻辑复杂，功耗大，但抖动小
- 4) 通常分频器由高频异步前置分频器和低频同步分频器级联而成，最后由一个寄存器进行同步
- 5) 条件：中间过程中总积累的抖动 $< T_{VCO}$

VCO



1) Ring OSC or LC OSC?

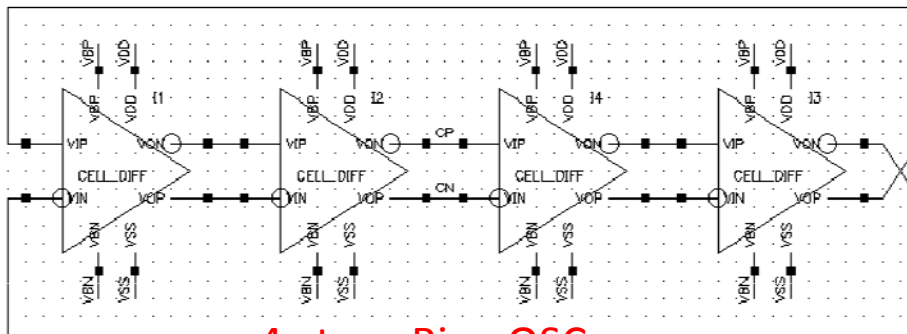
前者调谐范围大; 后者噪声小

2) 噪声来源: 电路噪声+干扰

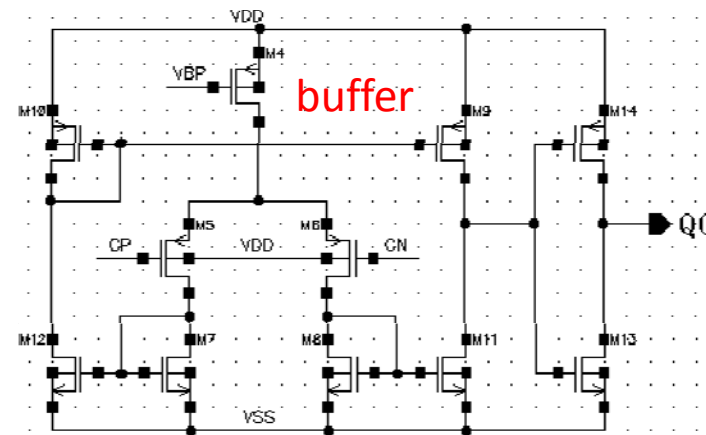
2) 增大WL减1/f noise;

3) 增大功耗->减少噪声

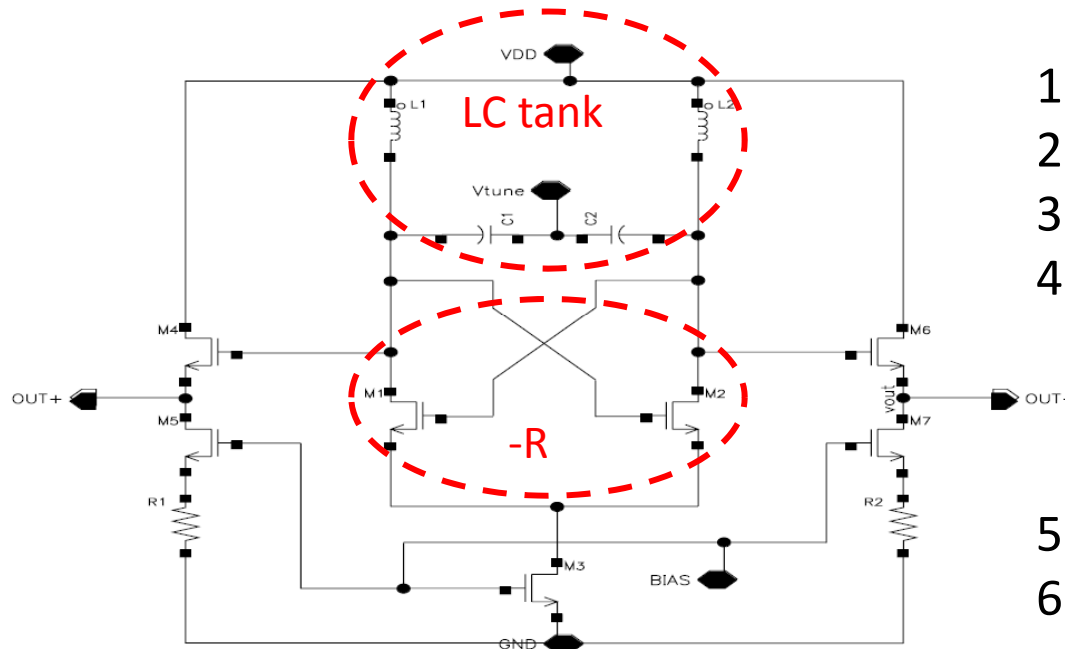
4) 采用差分结构



4-stage Ring OSC

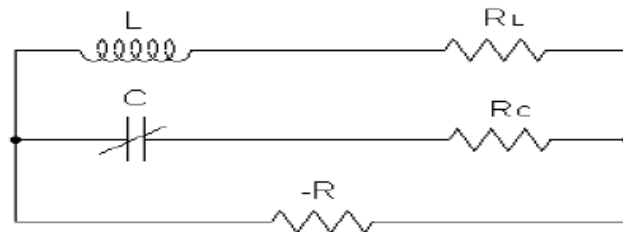


LC VCO



$$\omega_{osc} = \frac{1}{\sqrt{LC}}$$

$$Q \approx \frac{\omega L}{R_p}$$



Leeson equation:

$$L\{\Delta\omega\} = 10 \log \left\{ \frac{FkT}{2P_{avs}} \left[1 + \frac{\omega_c}{\Delta\omega} + \left(\frac{\omega_{osc}}{2\Delta\omega Q_L} \right)^2 \left(1 + \frac{\omega_c}{\Delta\omega} \right) \right] \right\}$$

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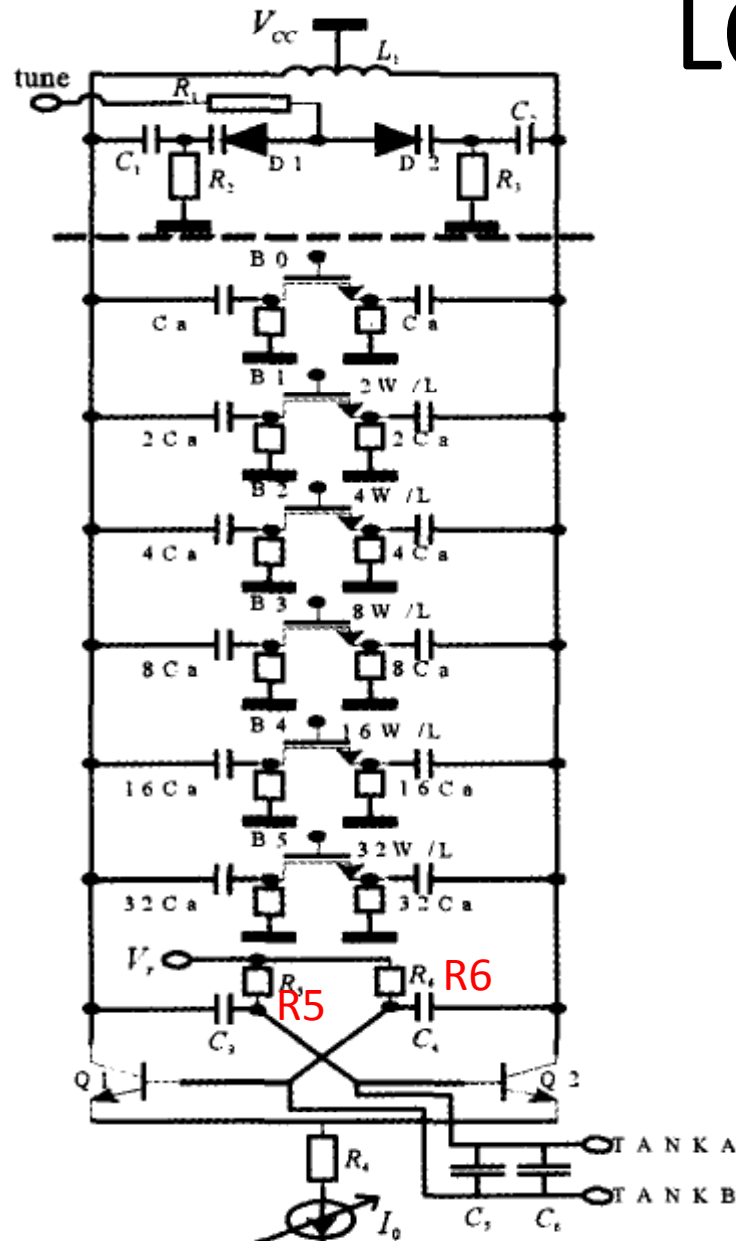
- 1) 带通特性，Q值高，噪声小
- 2) 尽量减小LC TANK寄生电阻
- 3) 电路Q值主要由电感决定；
- 4) 电感可为spiral inductor or bonding wire

前者值易控制，但Q值较低
后者Q值高，但值不易控制

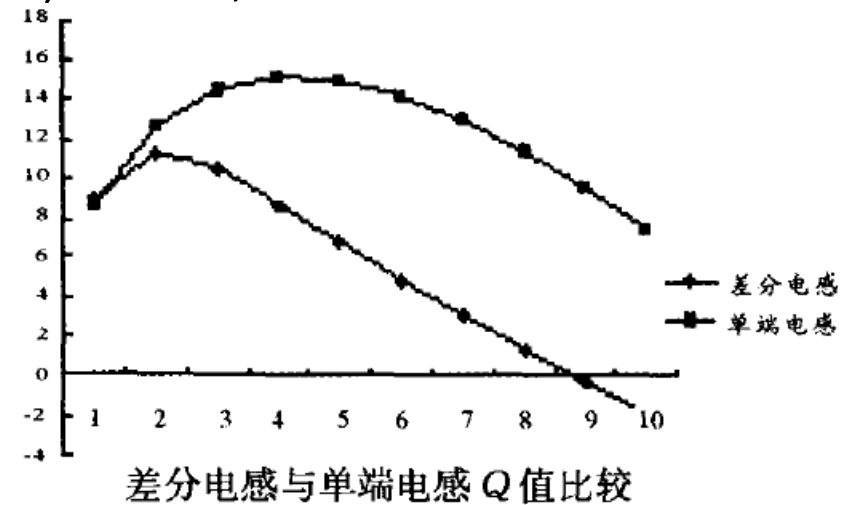
- 5) 频率调整通过nmos varactor
- 6) 可通过programable cap array+varactor来增加调谐范围

- 7) 电感占主要面积，且不随工艺scale down，可采用差分结构来优化面积

LC VCO Example

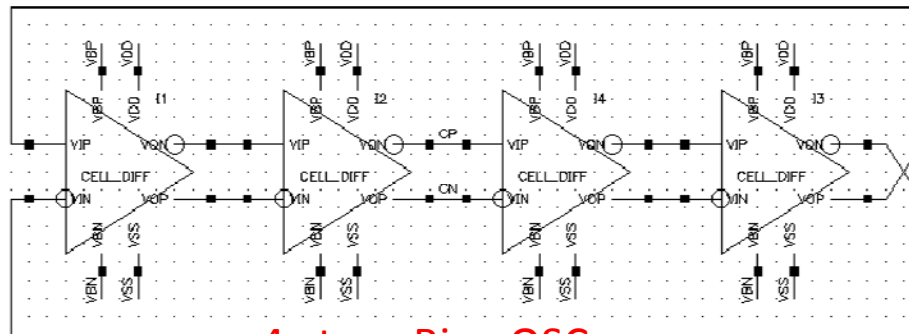


- 1) 频率调整通过变容二极管
- 2) 可通过programable cap array来增加调谐范围
- 3) 采用差分电感来优化面积和Q值
- 4) 采用金属电容提高电容Q值
- 5) 注意开关导通电阻对Q值的影响
- 6) 注意R5, R6要优化

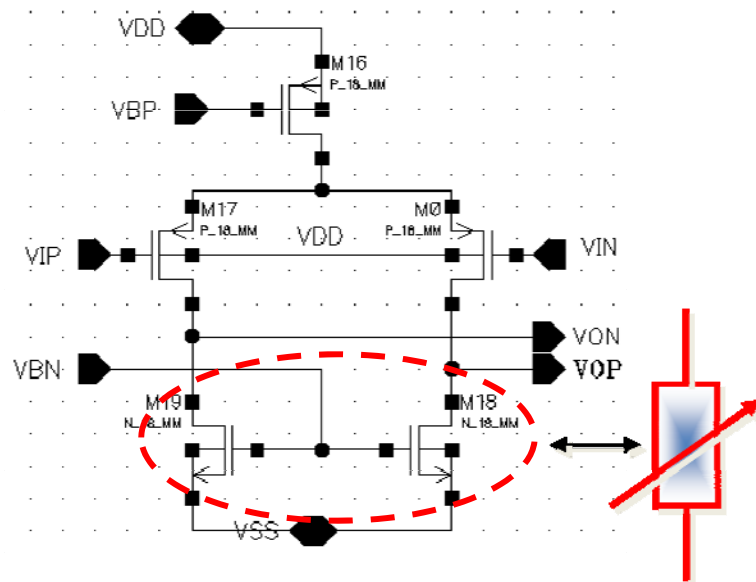


Ref: 李明剑等, 单片集成的高频宽带LC VCO设计

Ring VCO



4-stage Ring OSC



- 1) Ring OSC结构灵活，频率调整范围大，面积小，但噪声大
- 2) 由多级反相放大器级联成环
- 3) 结构: 3级 or 4级 or更多?
- 4) 4级更易起振, 幅度受工艺变化影响小, 但功耗大一些; 在某些场合需要多级, 如需输出多相时钟
- 3) 注意起振条件! $A_{dc}=g_m R$ 应比起振条件稍大, 以对付工艺角问题, 但也不能过大, 否则jitter将会增大

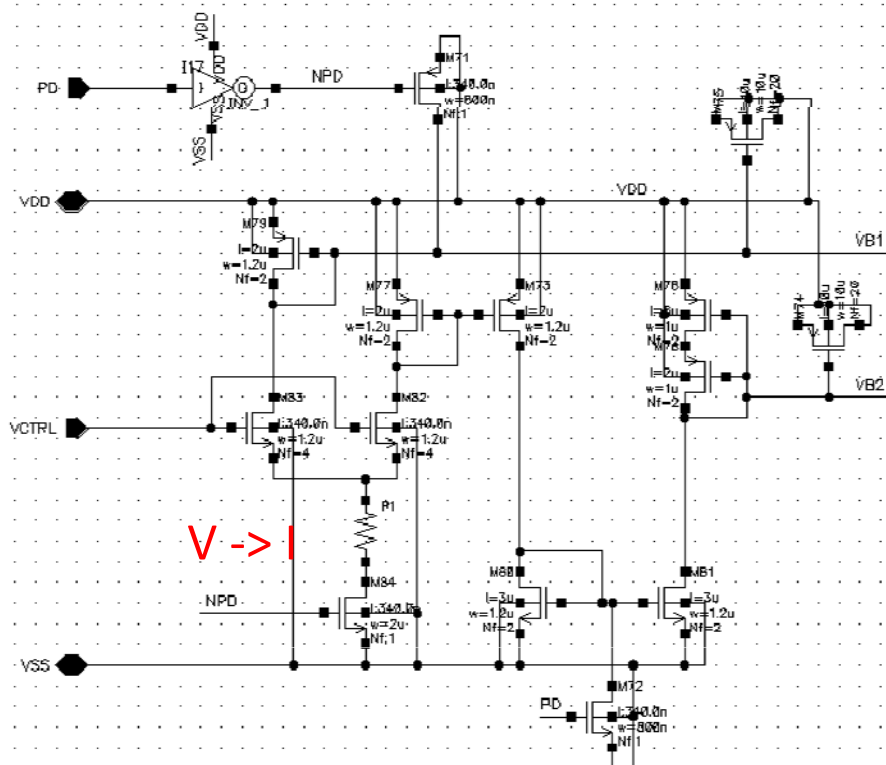
$$\arg[H(j\omega_{osc})] = \pi/N$$

$$|H(j\omega_{osc})| = 1$$

$$A_{dc} = g_m R \geq \sqrt{1 + \left(\tan\left(\frac{\pi}{N}\right) \right)^2}$$

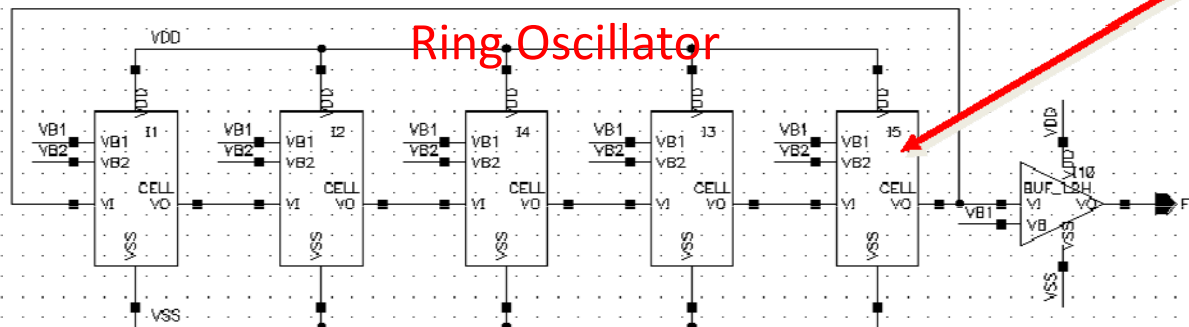
124

Ring VCO Example

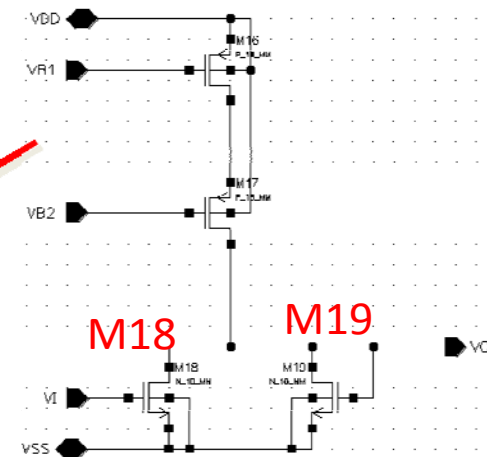


V -> I

- 1) $V/I + I_{CO} = V_{CO}$
- 2) I_{CO} 采用单端结构，电路简单
- 3) 需采用奇数级级联
- 4) $A_{dc} = g_{m18}/g_{m19}$
- 4) 采用固定偏置电流(A类)设计，具有较高的PSRR
- 5) 速度取决于单级带宽



Ring Oscillator



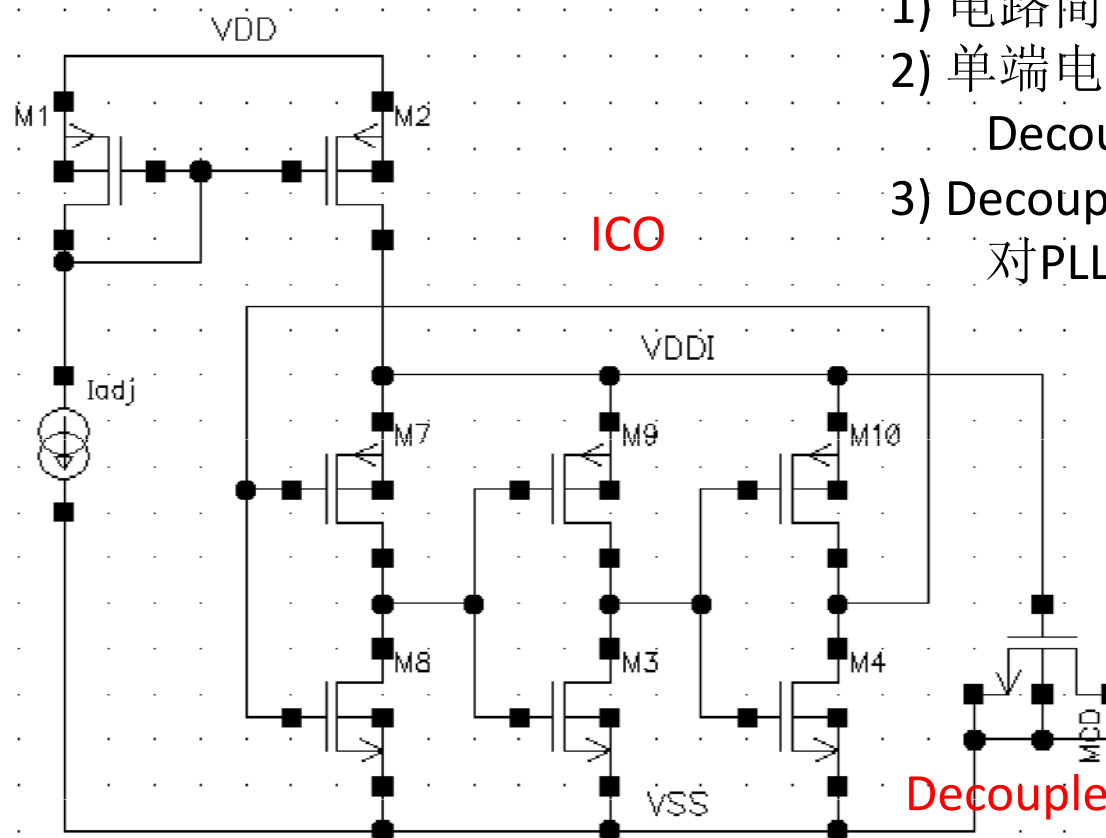
M18

M19

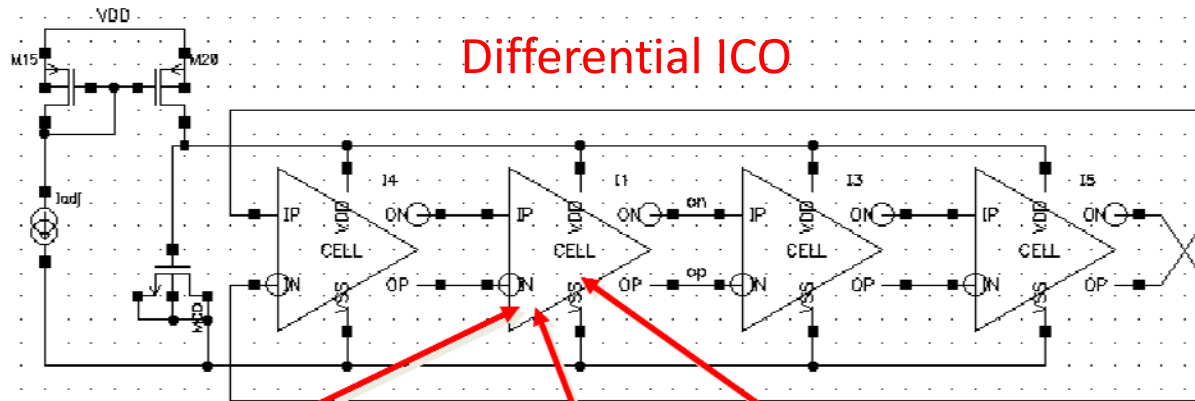
Ring VCO Example

GHz电路中常用的Ring VCO电路

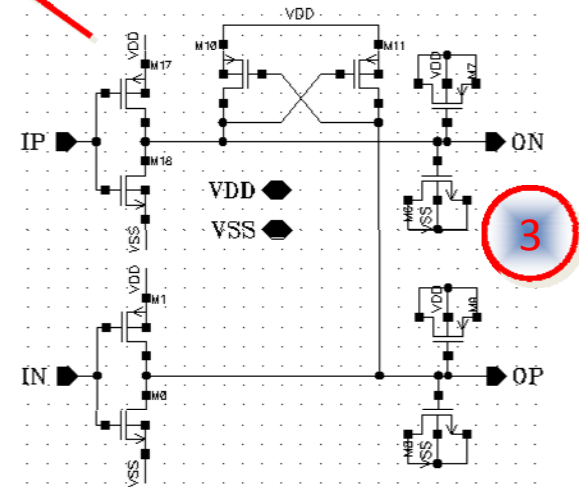
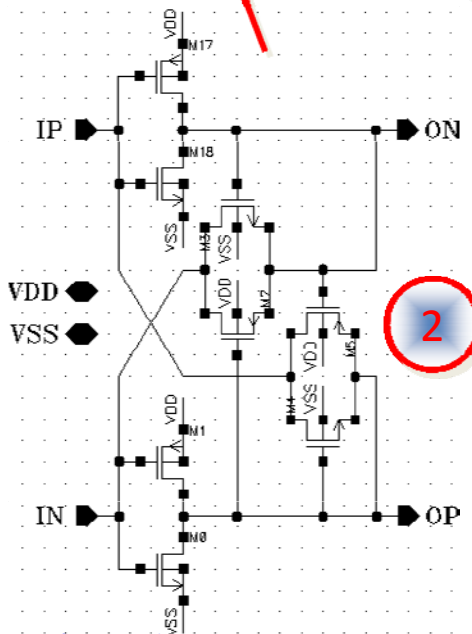
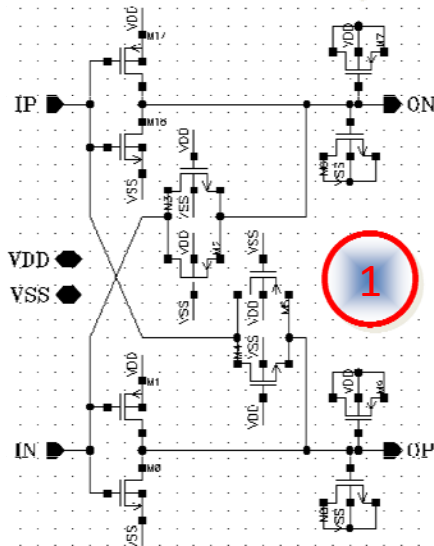
- 1) 电路简单，频率高
- 2) 单端电路，易受干扰，VDDI上要加Decouple
- 3) Decouple cap不能过大，要注意其对PLL稳定性的影响



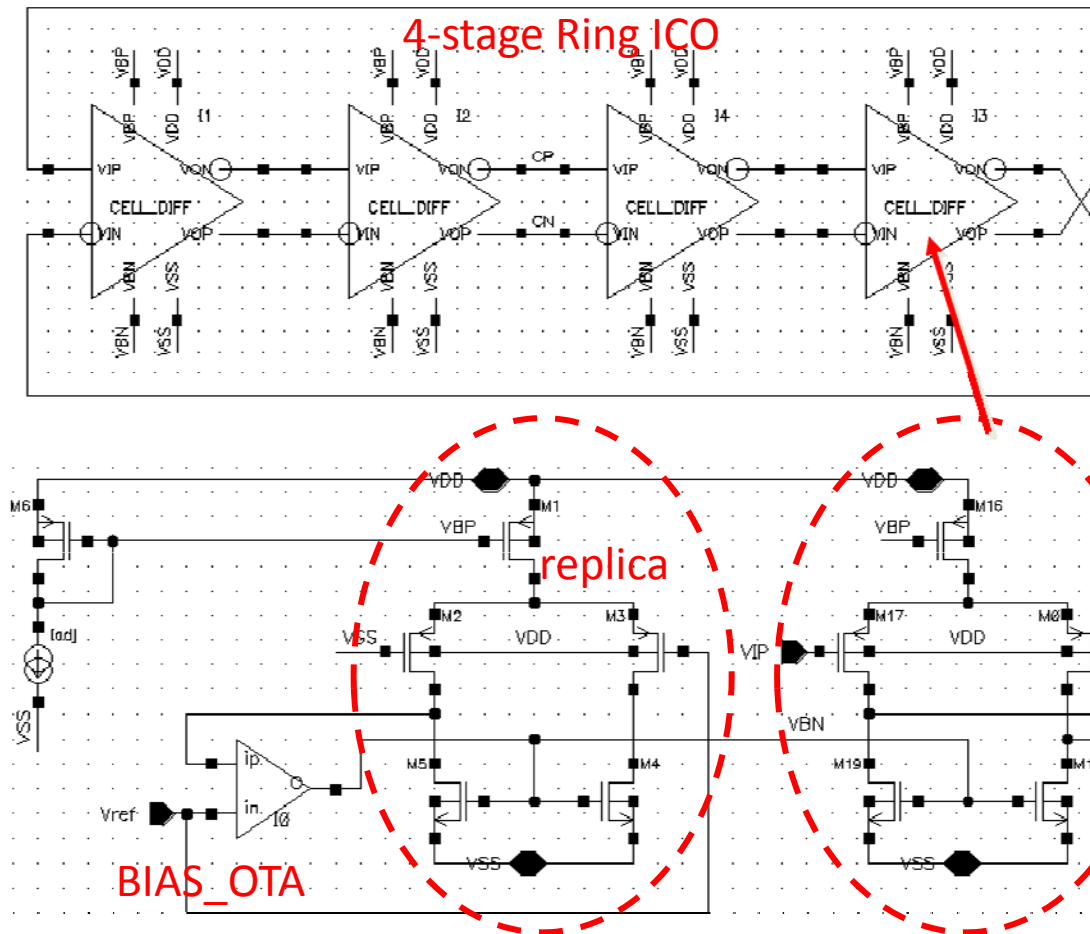
Ring VCO Example



差分cell的各种实现形式



Ring VCO Example

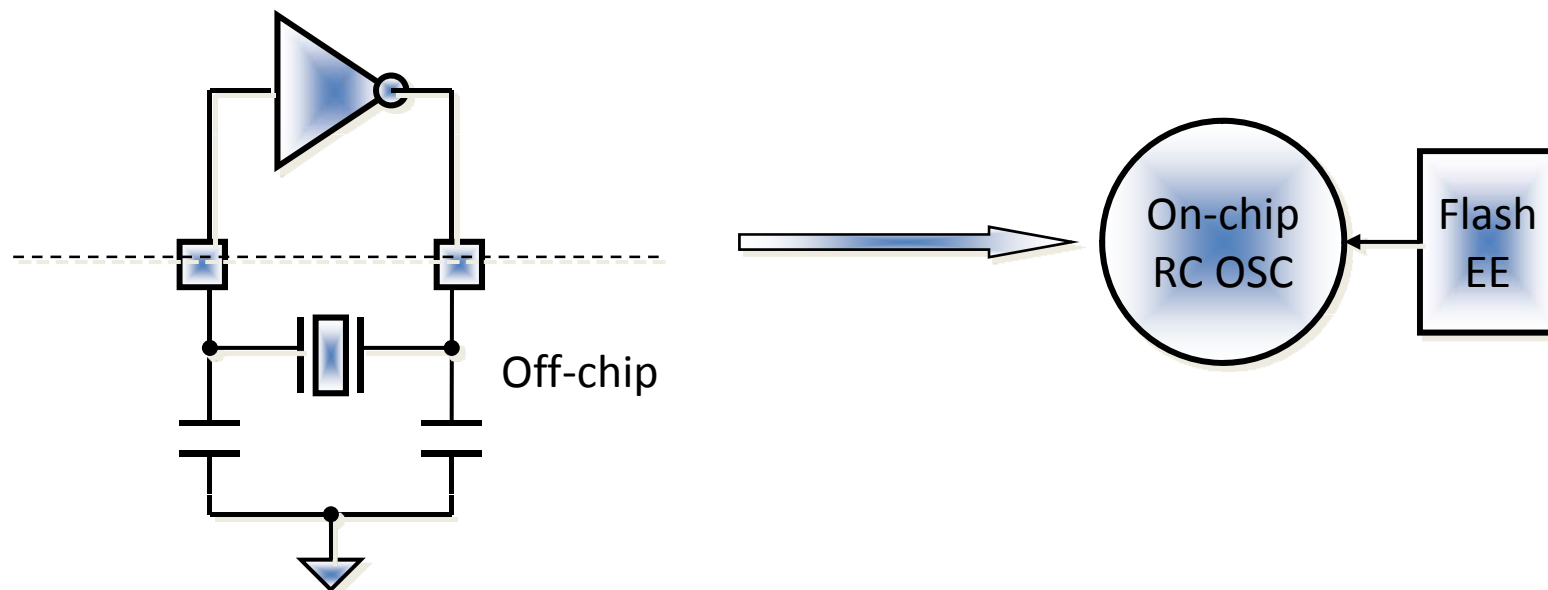


常用的模拟差分Ring VCO电路

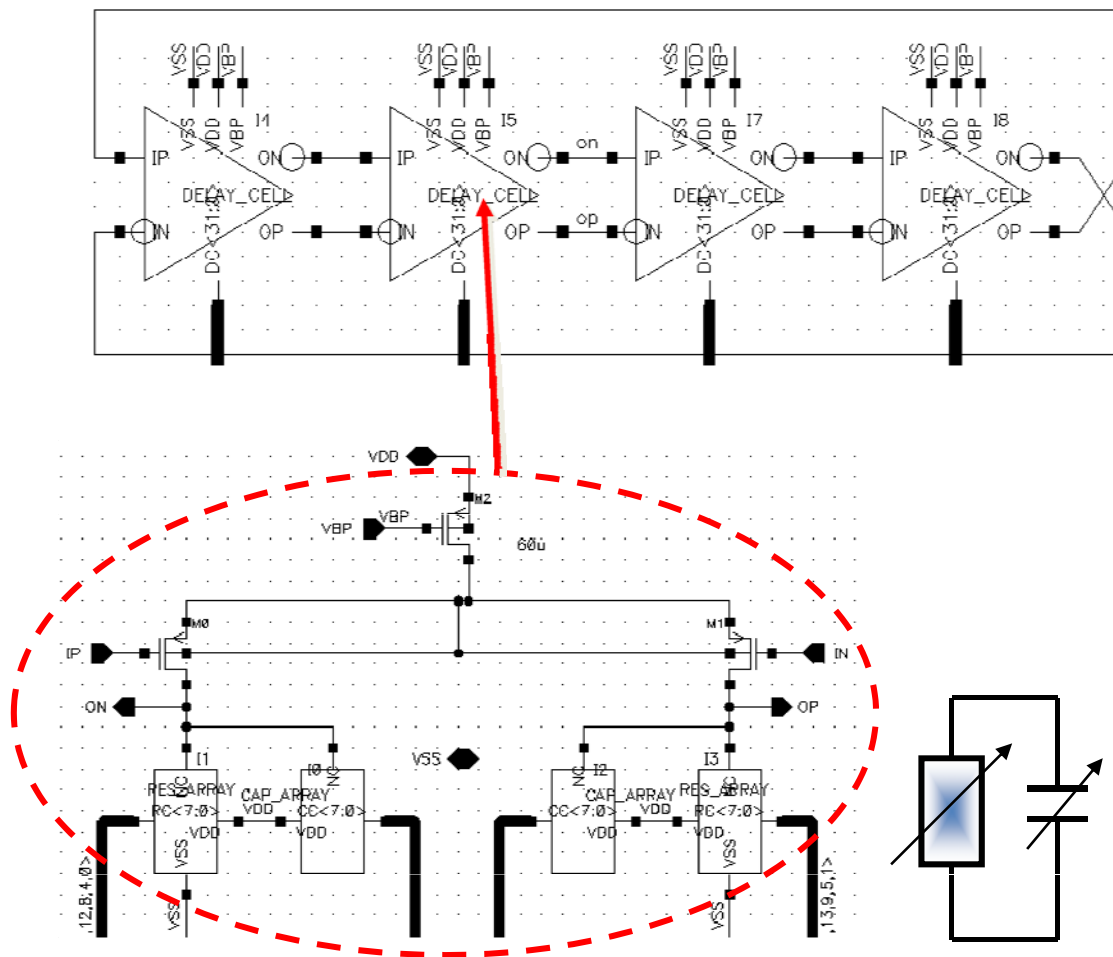
- 1) 差分结构，带尾电流偏置，抵御共模干扰性能好
- 2) 输入管饱和区工作，负载管线性区工作
- 3) 通过replica来控制振荡幅度
- 4) $I_{ss} \cdot R_n = V_{ref}$ ，调节 I_{ss} ，就是调节 R_n ，就是调节振荡频率
- 5) BIAS_OTA的输出范围应尽可能大，故一般采用两级放大器结构

RC OSC

- 实现全集成的比较精确的时钟源(如频率偏差5%之内)
- 晶体振荡器
 - 很精确($10^{-4} \sim 10^{-5}$); 成本高; 无法全集成
- RC振荡器
 - 校准后频偏可控制在x%之内; 成本低; 全集成, 无片外元件

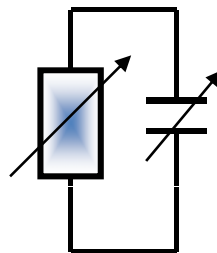


RC OSC

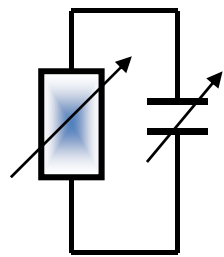


频率精度控制

- 1) 频率调节: R, C 均为可编程的阵列设计
- 2) 频率校准: 芯片出厂后进行一次, 校准结果存放在Flash, EE等存储器中, 正常使用时只读设置, 不再校准
- 3) 频率稳定: 与电源, 温度等环境变化不敏感!
- 4) 设计要点: 尽量使频率只取决于 RC , R, C 需选有温度系数低的元件



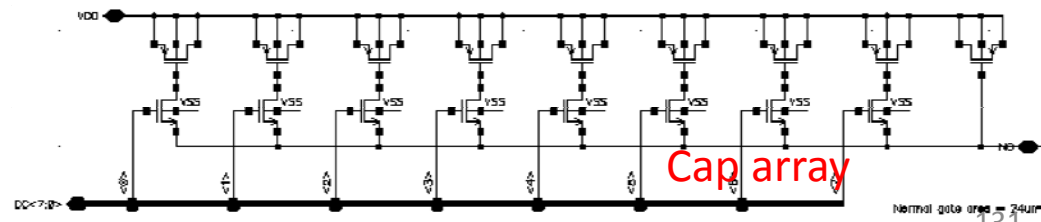
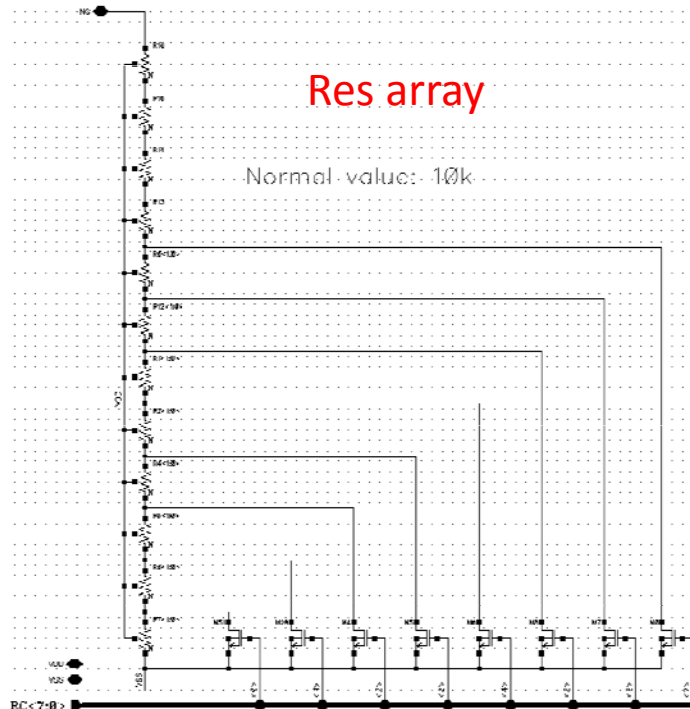
RC OSC



$$f_{osc} \propto \frac{1}{RC}$$

R, C阵列设计

- 集成R, C的值 $\pm 20\%$ ，对应频率范围为：(69.4%, 156.3%) f_0
- 2) 频率分辨率： $\Delta f = (156.3 - 69.4) / N \%$
- 3) 根据频率精度的要求来设计N，即总的档数，如电容分为M档，电阻分为K档，采用组合法，则： $N = M * K$ 若采用叠加法，则： $N = M + K$
- 4) 组合法控制简单，但可能会不单调的情况；**叠加法控制复杂，但可保证单调性**

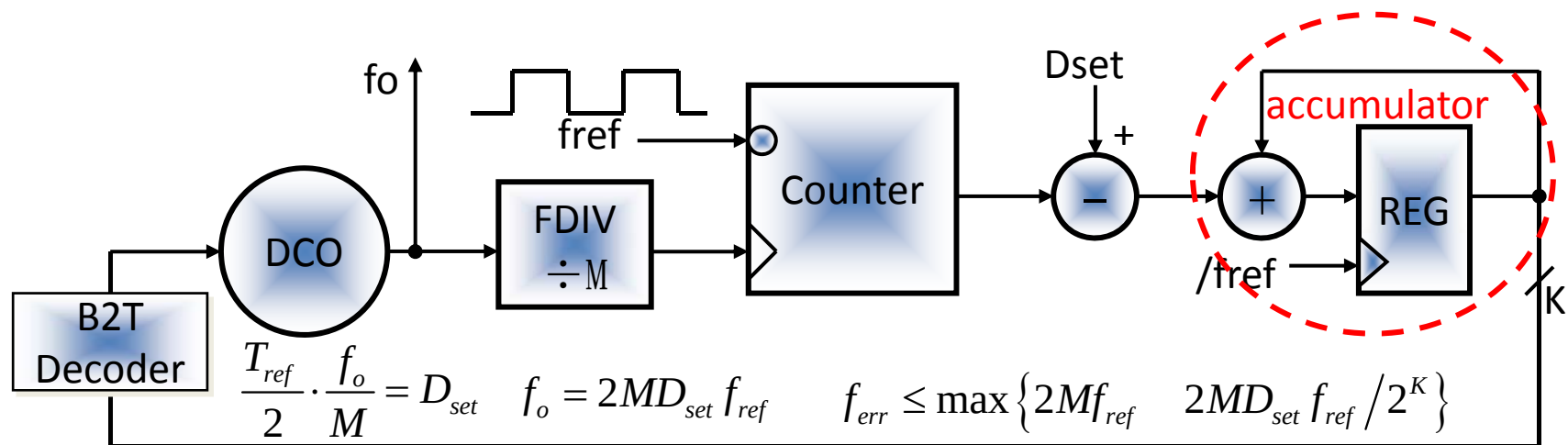


Freq. multiplier based RC OSC

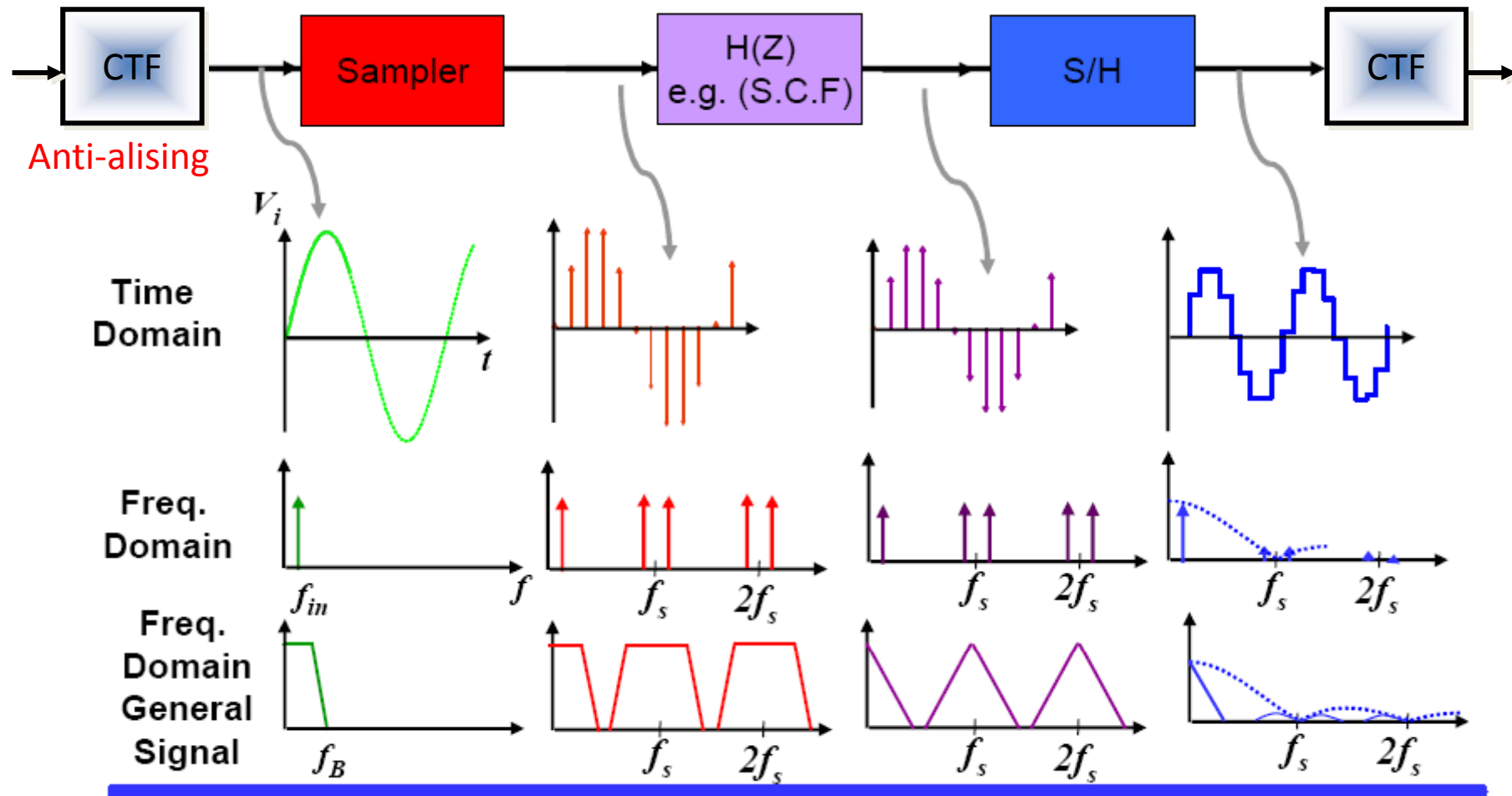
应用：需要一个高频信号，如915MHz，频率精度要求较高，但低于晶振精度，**对成本很关心**

解决：可用一个低成本的**32768Hz晶振**作为基准，进行倍频

- 若采用PLL锁相倍频环路，参考频率太低，面积太大，倍频比太高
- 采用下图的数控振荡器环路，环路最慢部分在数字部分实现，节约芯片面积
- **DCO须保证单调性，否则环路控制会有问题**



SC Filter architecture



Ref: http://www-inst.eecs.berkeley.edu/~ee247/fa05/lectures/L9_22_f05.pdf

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设计流程

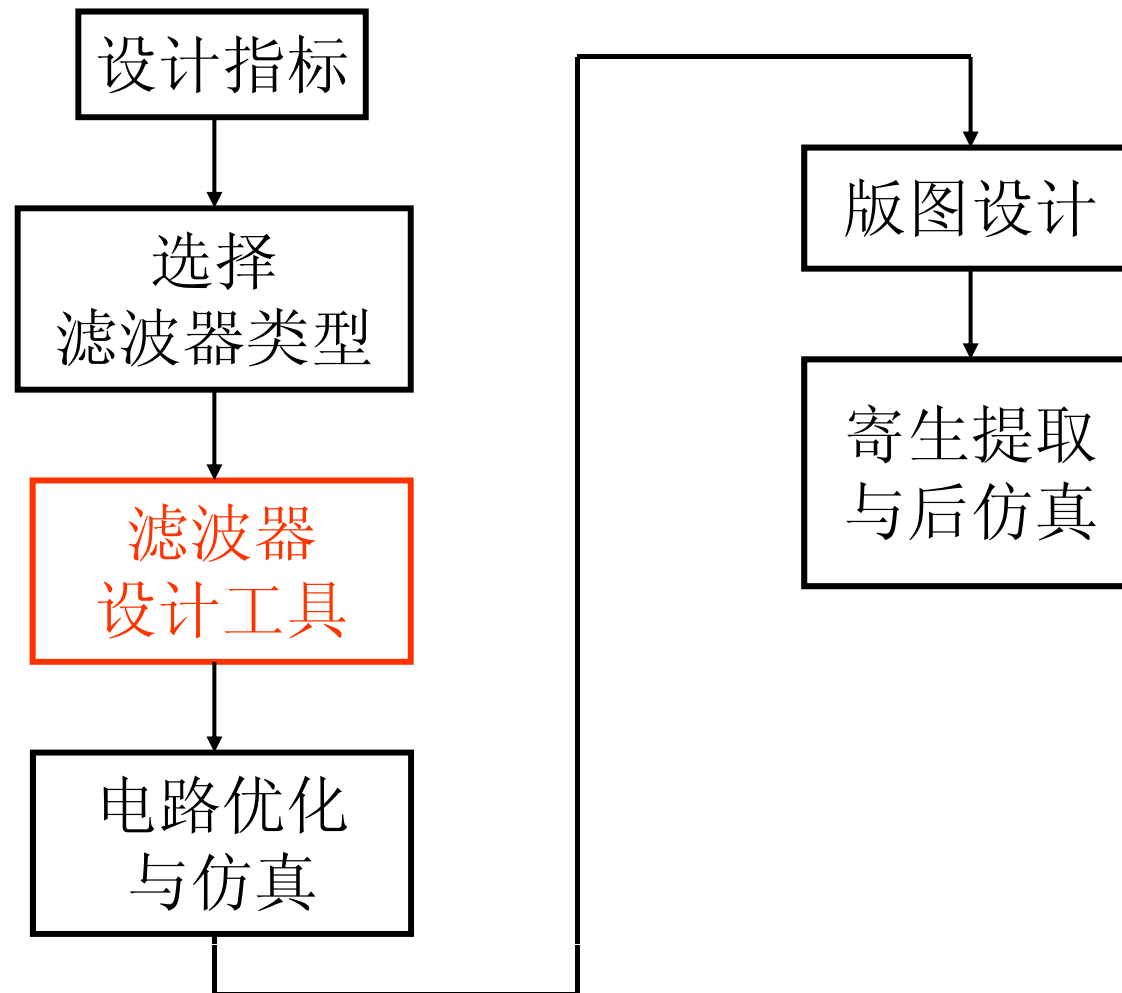
滤波器是非常成熟的设计，有成熟的设计工具

Matlab

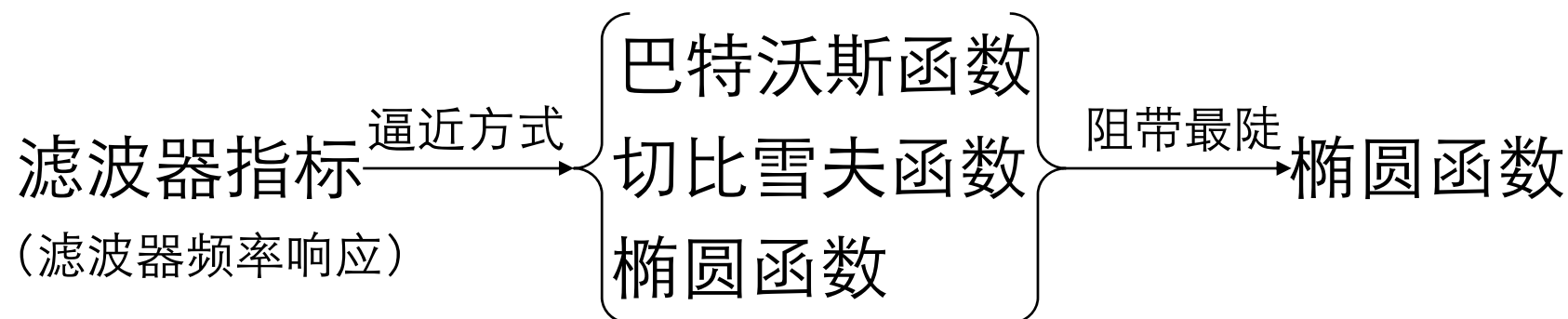
Filter lab

Filter Solution

...



选择类型

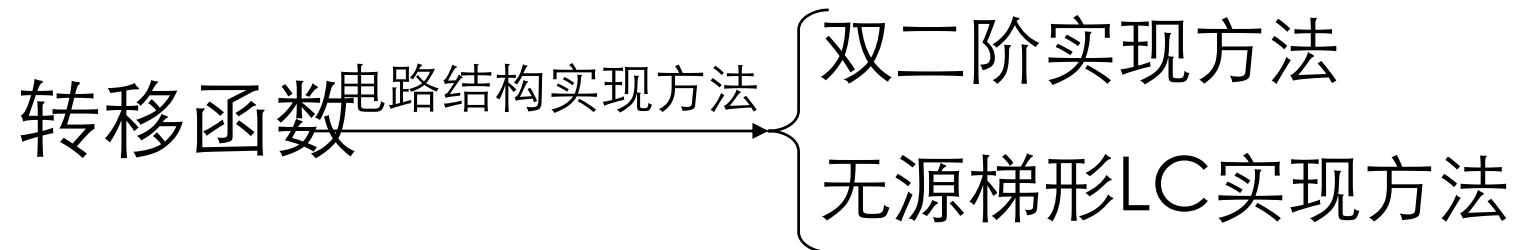


椭圆函数 $\xrightarrow{\text{软件迭代}}$ 滤波器转移函数



$$H(Z) = \frac{8.513 \times 10^{-3} (Z^2 - 1.929Z + 1)(Z^2 - 1.65Z + 1)}{(Z^2 - 1.834Z + 0.8448)(Z^2 - 1.927Z + 0.9473)}$$

电路结构实现方法

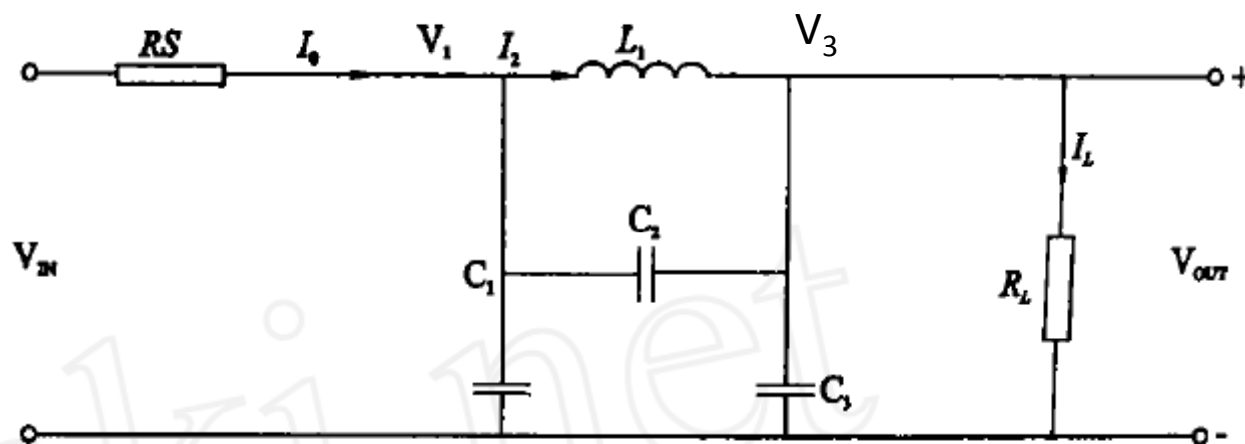


- 双二阶实现方法 | 单元之间相互独立，便于调试，采用该实现方法

利用二阶滤波器单元级联组成有源RC, 开关电容滤波器

- 无源梯形RLC 实现方法
信号流图变换法

信号变换法



$$I_0 = (V_N - V_1) / R_s$$

$$V_1 = (I_0 - I_2) / S(C_1 + C_2) + V_{OUT} C_2 / (C_1 + C_2)$$

$$I_2 = (V_1 - V_{OUT}) / S L_2$$

$$V_3 = (I_2 - I_L) / S(C_3 + C_2) + V_1 C_2 / (C_3 + C_2)$$

$$I_L = V_{OUT} / R_L$$

$$V_0' = R(V_N - V_1) / R_s$$

$$V_1 = (V_0' - V_2') / S(C_1 + C_2) R + V_3 C_2 / (C_1 + C_2)$$

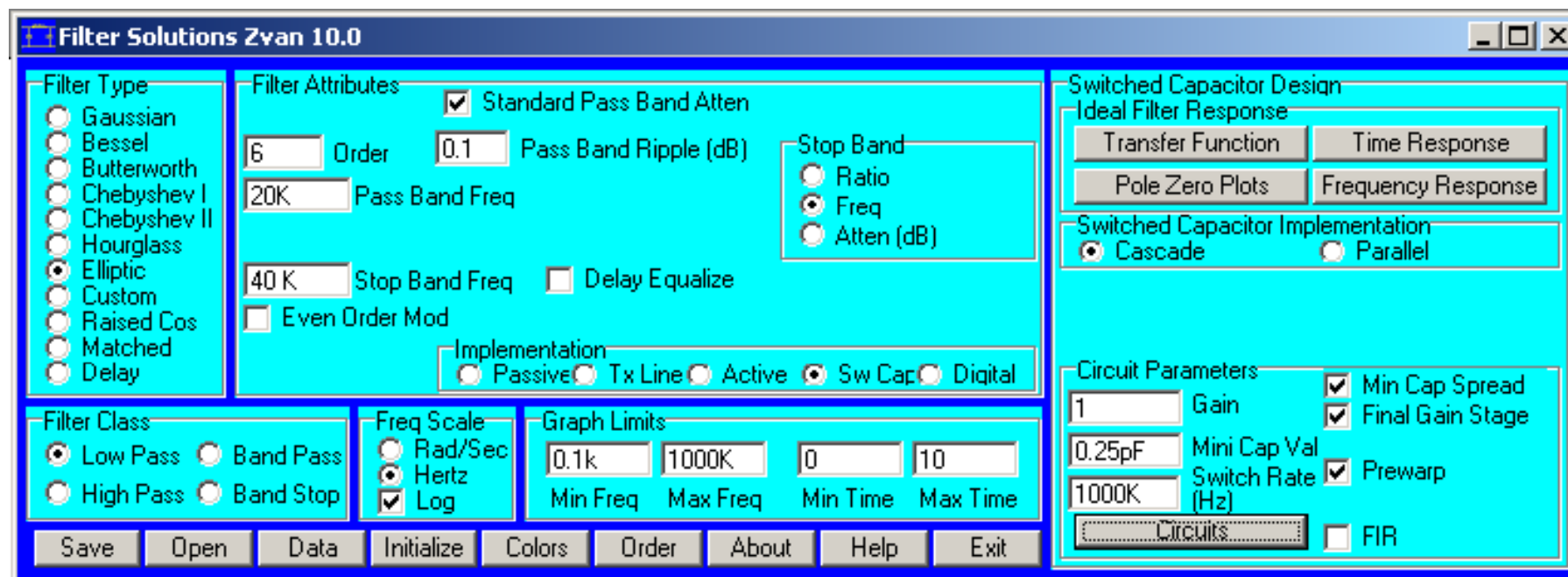
$$V_2' = (V_1 - V_3) R / S L_2$$

$$V_3 = (V_2' - V_L') / S(C_3 + C_2) R + V_1 C_2 / (C_3 + C_2)$$

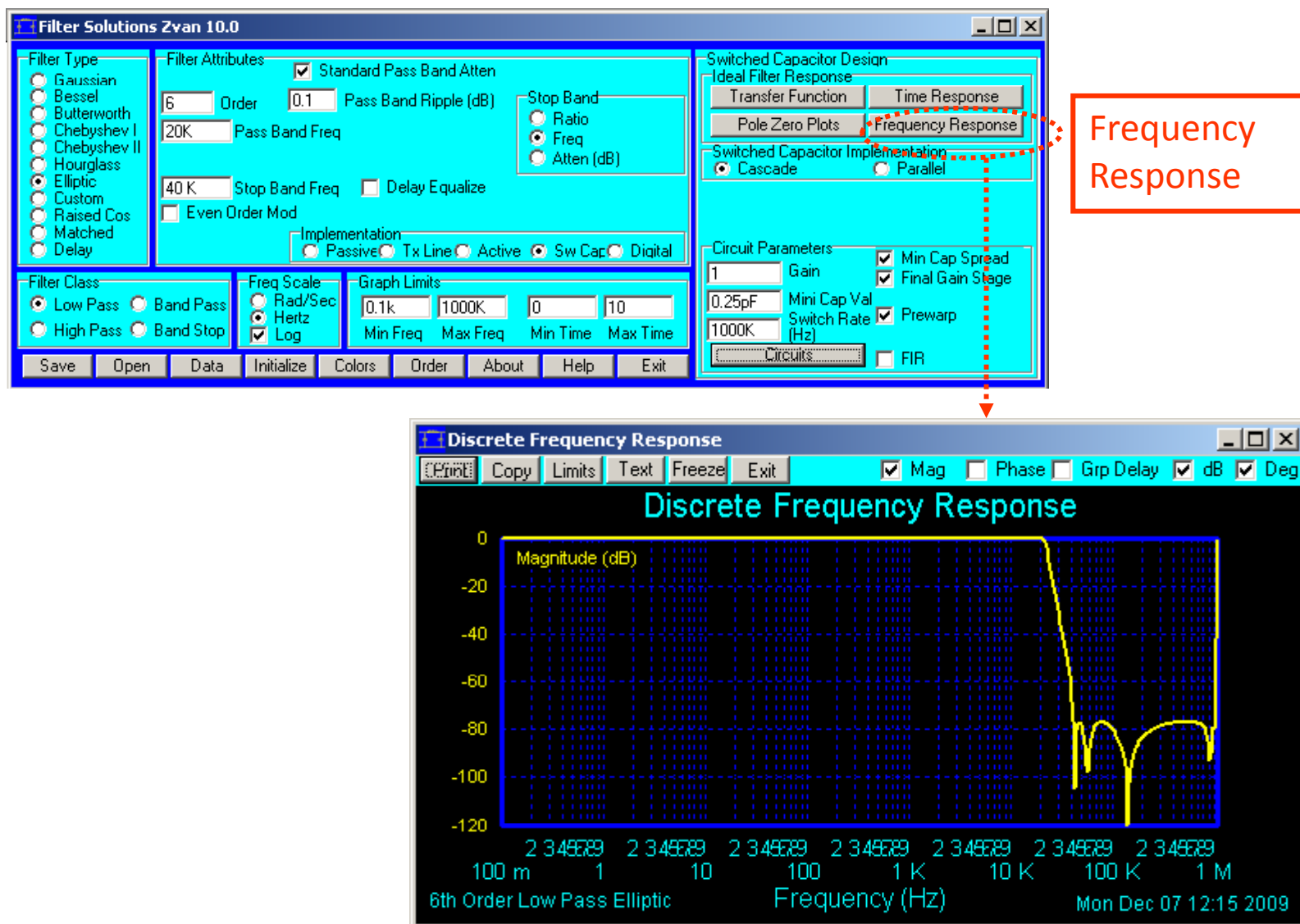
$$V_L = R V_{OUT} / R_L$$

设计工具

- Matlab FDA toolbox
 - Matlab界面: start → Toolboxes → Filter Design
→ Filter Design & Analysis Tool (FDA Tool)
- **Filter Solution**



设计工具



设计工具

Filter Solutions Zvan 10.0

Filter Type

- ☐ Gaussian
- ☐ Bessel
- ☐ Butterworth
- ☐ Chebyshev I
- ☐ Chebyshev II
- ☐ Hourglass
- ☒ Elliptic

Filter Attributes

☒ Standard Pass Band Atten

Order: 6, Pass Band Ripple (dB): 0.1, Pass Band Freq: 20K, Stop Band Freq: 40 K, Delay Equalize: ☐

Stop Band

- ☐ Ratio
- ☒ Freq
- ☐ Atten (dB)

Switched Capacitor Design

☒ Ideal Filter Response

☐ Transfer Function, ☐ Time Response, ☐ Pole Zero Plots, ☐ Frequency Response

Switched Capacitor Implementation

☒ Cascade, ☐ Parallel

Discrete Transfer Function

Print Copy Vec Freeze Exit Stan Casc Para 4 Sig Dig Fit

Bilinear Transformation with Prewarping
Sample Frequency = 1.000 MHz

$$\frac{1.737e-04 \cdot Z^6 - 8.916e-04 \cdot Z^5 + 2.026e-03 \cdot Z^4 - 2.615e-03 \cdot Z^3 + 2.026e-03 \cdot Z^2 - 8.916e-04 \cdot Z + 1.737e-04}{Z^6 - 5.762 \cdot Z^5 + 13.86 \cdot Z^4 - 17.82 \cdot Z^3 + 12.91 \cdot Z^2 - 4.998 \cdot Z + .8076}$$

6th Order Low Pass Elliptic

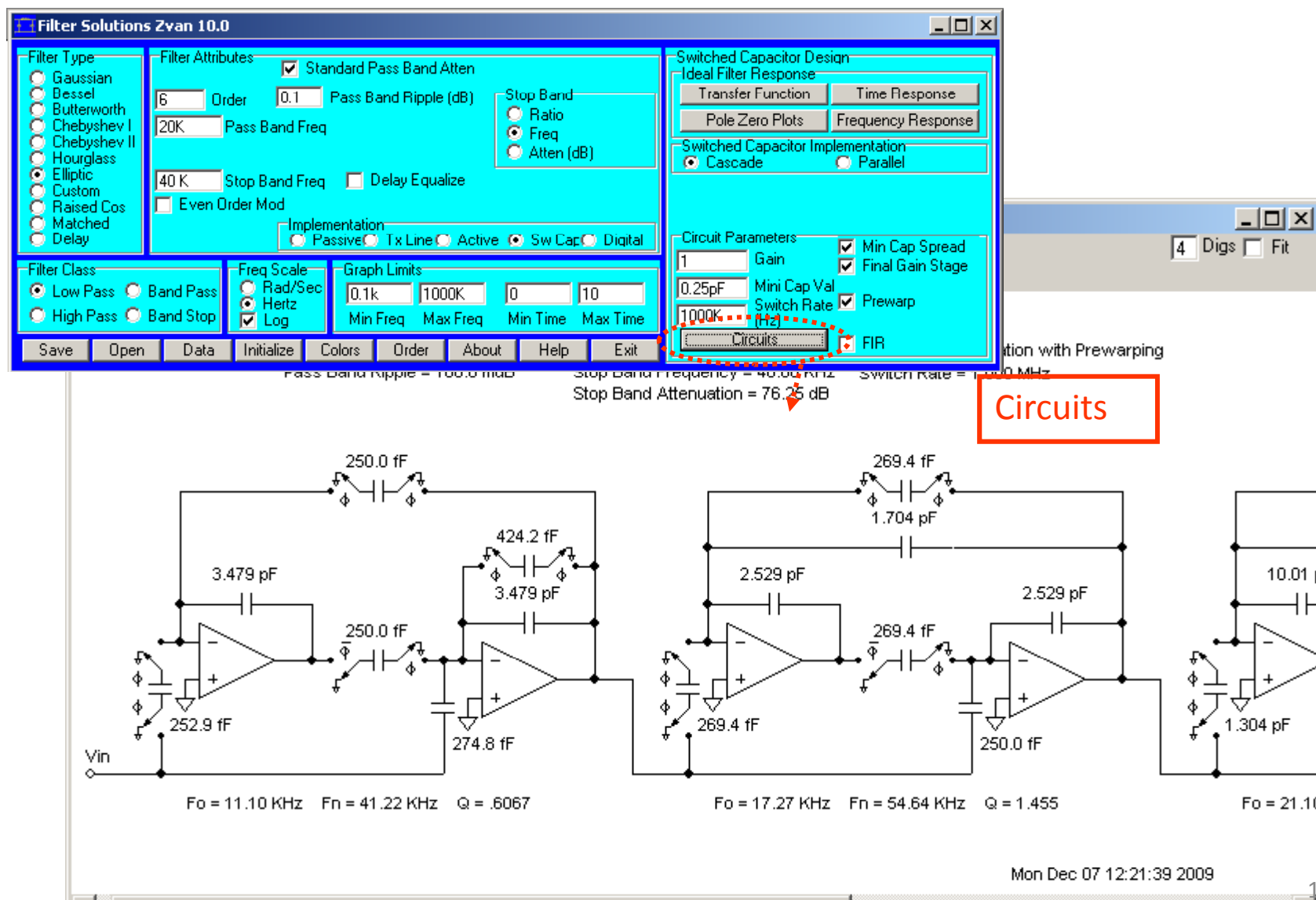
Pass Band Frequency = 20.00 KHz, Stop Band Ratio = 2, Pass Band Ripple = 100.0 mdB, Stop Band Frequency = 40.00 KHz, Stop Band Attenuation = 76.25 dB

Hint: Filter Accuracy and Stability are Affected by Significant Digits and Form Selection

Mon Dec 07 12:20 2009

Transfer Function

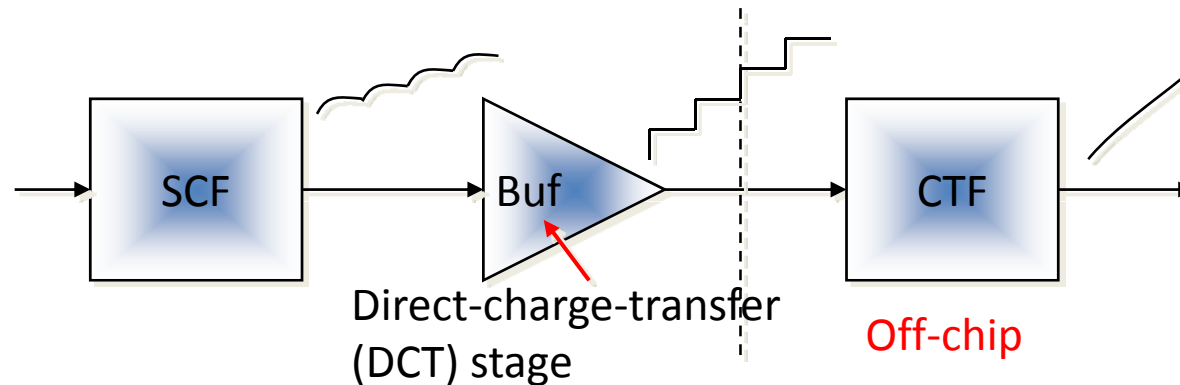
设计工具



后续工作

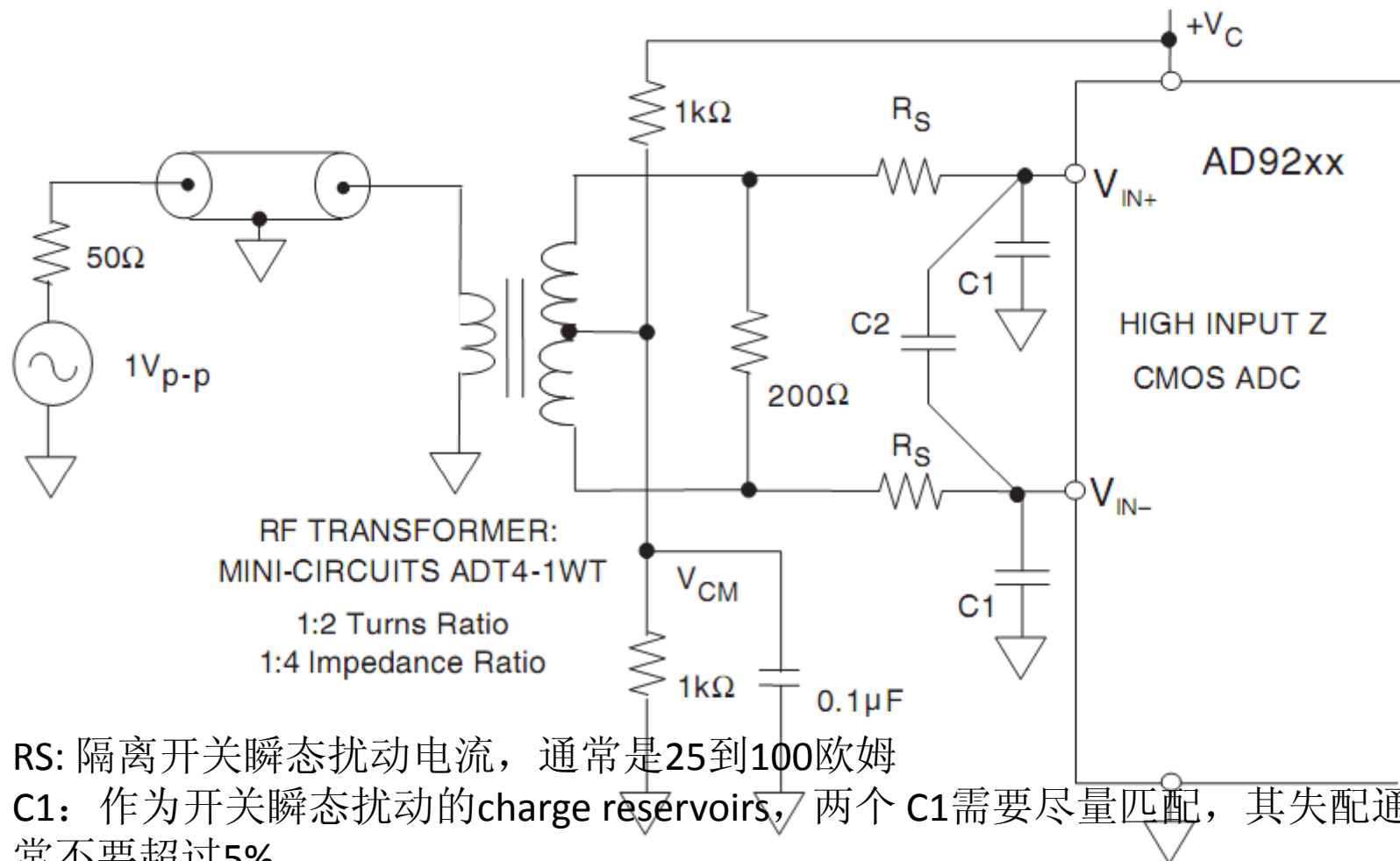
- 从单端→差分转换
- 各积分器动态范围标定
- 电容匹配设计方案
- 运放设计
- SCF信号输出片外前要加buf整形

注意：对连续时间滤波器也可用**filter solution**，在工具的支持下，有类似的设计过程



DCT: J.A.C. Bingham, "Applications of a direct-transfer SC integrator".

Input buffer for CMOS ADC



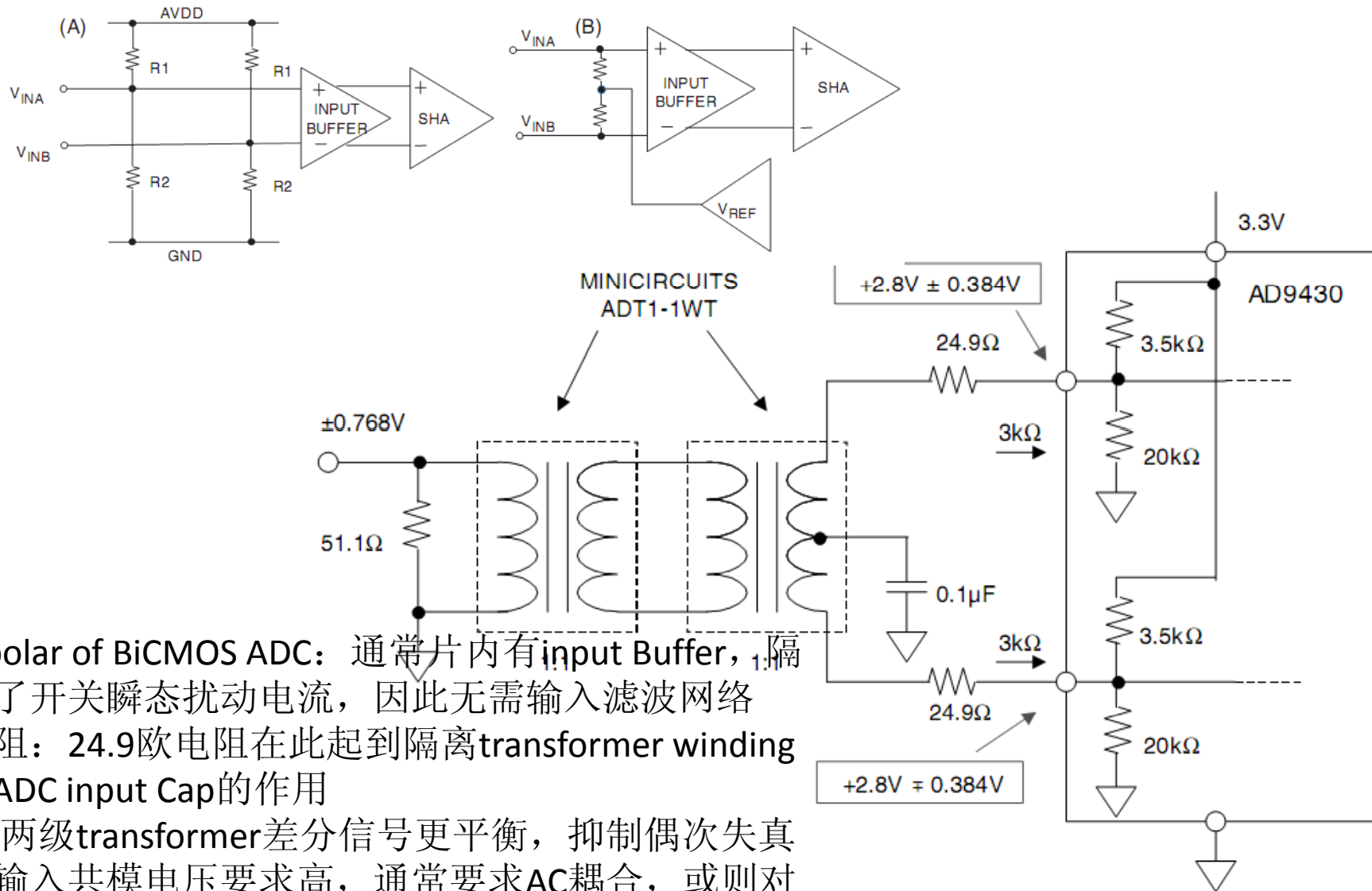
R_S: 隔离开关瞬态扰动电流，通常是25到100欧姆

C1: 作为开关瞬态扰动的charge reservoirs，两个 C1需要尽量匹配，其失配通常不要超过5%

C2: 专门作为差分charge reservoirs

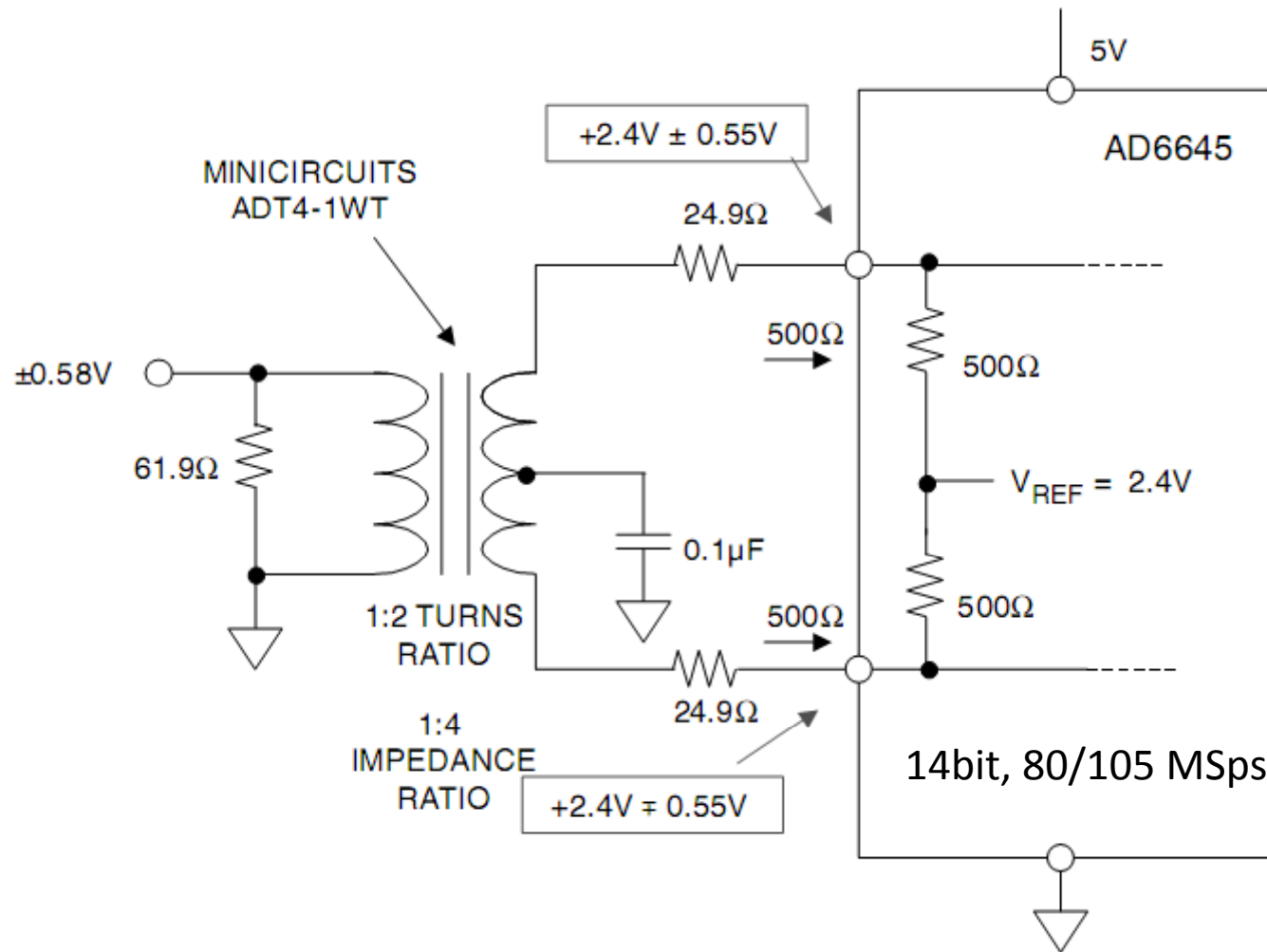
同时C1, C2还构成对输入的共模和差模低通滤波

Input Buffer for BiCMOS ADC



Bipolar of BiCMOS ADC: 通常片内有input Buffer, 隔离了开关瞬态扰动电流, 因此无需输入滤波网络
 电阻: 24.9Ω 电阻在此起到隔离transformer winding和ADC input Cap的作用
 用两级transformer差分信号更平衡, 抑制偶次失真
 因输入共模电压要求高, 通常要求AC耦合, 或则对input buffer单独提供一个更高的电源

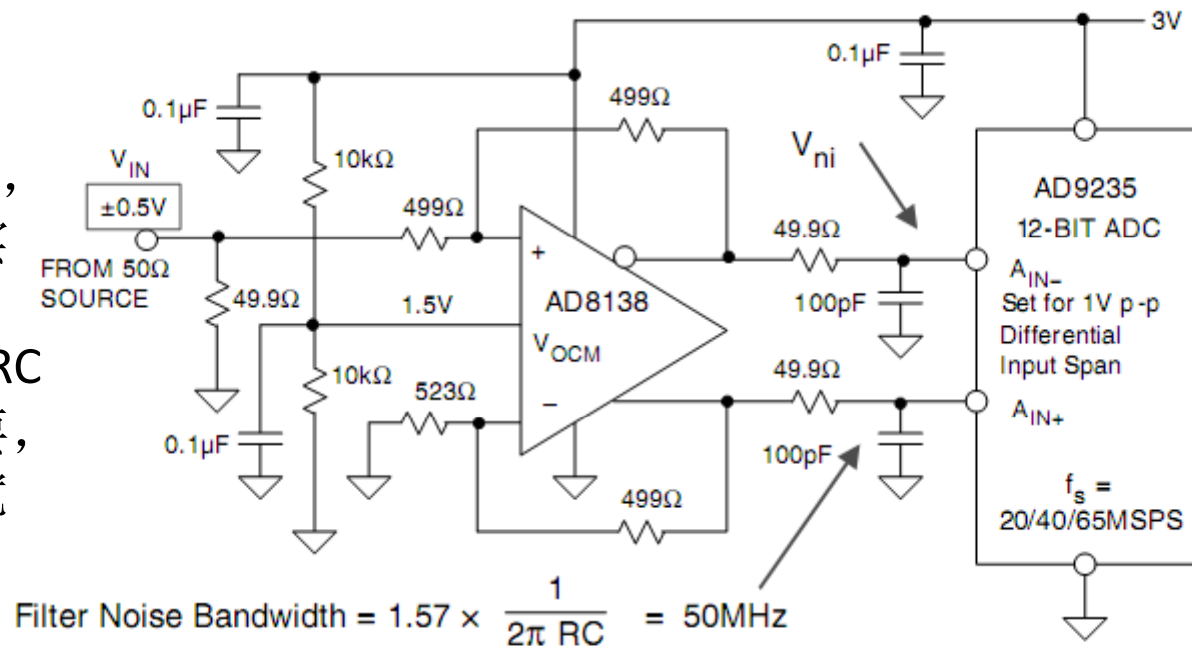
Input Buffer for BiCMOS ADC



Input Buffer for ADC

➤输入低频信号，
用全差分运放实现单转差

➤ADC输入端的RC
低通滤波很重要，
可限制噪声带宽



AD8038 DIFF. AMP SPECIFICATIONS

- Output Voltage Noise = $11.6\text{nV}/\sqrt{\text{Hz}}$
- Closed-Loop BW = 300MHz
- Closed-Loop Noise BW = $1.57 \times 300\text{MHz} = 471\text{MHz}$

AD9235 ADC SPECIFICATIONS

- Effective Input Noise = $132\mu\text{V rms}$
- Small Signal Input BW = 500MHz
- Input Noise BW = $1.57 \times 500\text{MHz} = 785\text{MHz}$

AD8038 Output Noise Spectral Density = $11.6\text{nV}/\sqrt{\text{Hz}}$ (Including Resistors)

$$V_{ni} = 11.6\text{nV}/\sqrt{\text{Hz}} \times \sqrt{50\text{MHz}} = 78.2\mu\text{V rms}$$

Lecture 3

END